Geometric Computation: Twin Helices and the Algebra of Flow

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Abstract

This work constructs a manifold consisting of twin helices to investigate the existence of computational geometric bordisms, as part of the broader program of the Cobordism Hypothesis. We use these structures to define a class of deterministic computational substrates, termed *Geometrically Computable Manifolds*, and study their geometric, algebraic, and computational properties. We show that, in a discrete regime, these manifolds encode arbitrary cyclic groups and support universal computation in the sense of Kleene's partial recursive functions. This leads to a theorem describing the structural trade-offs in physical computation across distinct computational phases. We then analyze a regime in which abelian gauge computation is modeled with ordinary differential equations, linking the construction to continuous geometric dynamics. This signals a new research direction in physical mathematics at the intersection of geometry, gauge theory, and computability theory, termed *Geometric Computability Theory*, in which computation is treated as a property of geometric compositions. This perspective opens avenues for future investigation, motivating the development of a constructive computational gauge theory with applications to Yang–Mills theory. As of writing, no algebraic systems yet exist to formalize the behavior of these manifolds; future work will develop such research.

Keywords: geometric computation, higher gauge theory, prequantum geometry, Yang-Mills theory, cobordism hypothesis, helices, topological quantum field theory.

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Contents

Introduction

> *"In the long run, we hope that some theory of language expressiveness develops into a formal theory of the programming language design space."*
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> — Mattias Felleisen, *On the Expressive Power of Programming Languages* (1991)

> *"A twisted version of four-dimensional supersymmetric gauge theory is formulated..."*
>
> — Edward Witten, *Topological Quantum Field Theory* (1988)

The following work constructs a geometric model of computation.

The gap between computer science and physics has been a long-standing one. While various attempts to bridge the divide have been made over the years, a full resolution remains out of reach. On the one hand, familiar models of computation are fundamentally discrete, while much of the symmetries observed in fundamental physics are continuous. So, are the laws of physics truly computable? Penrose and others have argued that certain key physical (and cognitive) phenomena may elude algorithmic description; by contrast, mainstream practice treats numerical approximation on a continuum as acceptable. Here we bridge this divide.

Section 1 constructs a deterministic geometric substrate from manifolds built out of joined helices. This leads naturally to coherence conditions imposed by chirality, torsion, and gluing, and from these conditions emerges a class of symbolic flow manifolds with non-cancellative monoidal structure. We then analyze their basic group-theoretic, geometric, and algebraic behavior.

Section 2 introduces the main computational objects of the paper: the *Geometrically Computable Manifold* and its associated state object, the *geobit*. The goal is to reconcile semantic expressiveness in computation with the structural stability and reversibility of geometric processes. Instead of treating semantics syntactically, we recast it geometrically in terms of moduli spaces of helices and their categorical behavior under symbolic flow. This leads to a framework in which symbolic computation is expressed through geometry itself, and from this analysis, we arrive at the central result of the paper: a conservation law governing computational behavior.

Section 3 places the theory on a computability-theoretic footing by relating it to Kleene's partial recursive functions. It also marks the shift from finite group actions to groupoid-based computation, where admissible operations are local, context-sensitive, and in general only partially defined.

Section 4 studies the algebraic behavior of these manifolds as computation evolves toward increasingly gauge-like regimes. We examine foundational issues in continuous computation, establish formal bounds, and give explicit models for computational behavior in this setting. An extended homotopy-theoretic framework is then outlined, together with an abelian gauge-theoretic phase formalism and a discussion of the relevant moduli spaces.

Section 5 situates the framework within a broader categorical and physical setting. There, we examine how computational randomness may arise as a quotient of geometric complexity, through coarse-graining of torsion and chirality across moduli. Holonomy-based semantics are introduced as the theory passes into non-abelian gauge-theoretic behavior, and we analyze the behavior of the resulting moduli space under unrestricted analog flow. We conclude by outlining possible connections to LQG, string-theoretic, and twistor-based programs, together with a broader interpretation of quantum theory in light of our results.

In the end, we remain hopeful that a new form of constructive computational gauge theory may provide a way forward in the study of non-perturbative field theory.

Our task here, by its nature, is a work of synthesis as much as rigor. If its scope sometimes appears unruly, or its exposition occasionally ventures beyond traditional boundaries, it is only because the questions at hand resist treatment by standard means. I have aimed for clarity without sacrificing depth. So, if arguments sometimes feel unconventional?

I hope that patient reading and engagement will reward the effort.

Ultimately, my intent is not to assert authority, but to invite the reader to explore together an unexpected landscape that may, in time, reshape our understanding of computation itself.

1 Twin Helical Manifolds

1.1 Volumetric Helices

For a concrete realization, fix parameters

$$R > 0, \qquad h \neq 0, \qquad \theta_{\min} < \theta_{\max},$$

and define the helical spine

$$\gamma(\theta) = (R\cos\theta,\ R\sin\theta,\ h\theta), \qquad \theta \in [\theta_{\min}, \theta_{\max}].$$

Let $(T(\theta), N(\theta), B(\theta))$ denote the Frenet–Serret frame of γ.

Definition 1.1 (Volumetric Helix)**.** Let ρ_{base} and ρ_{apex} satisfy

$$0 < \rho_{\text{apex}} < \rho_{\text{base}},$$

and define the normalized parameter

$$u = \frac{\theta - \theta_{\min}}{\theta_{\max} - \theta_{\min}}.$$

The cross-sectional radius profile is

$$\rho(\theta) = \rho_{\text{apex}} + \big(\rho_{\text{base}} - \rho_{\text{apex}}\big)\cos^2\left(\frac{\pi}{2}u\right).$$

The corresponding tubular body is the image of

$$\Phi(\theta, r', \phi) = \gamma(\theta) + r'\big(\cos\phi\, N(\theta) + \sin\phi\, B(\theta)\big),$$

for

$$\theta \in [\theta_{\min}, \theta_{\max}], \qquad 0 \le r' \le \rho(\theta), \qquad 0 \le \phi < 2\pi.$$

The *base* is the distinguished disk at $\theta = \theta_{\min}$, and the *apex* is the terminal end at $\theta = \theta_{\max}$.

The profile $\rho(\theta)$ is monotone decreasing, attains its maximal value ρ_{base} at the base, attains its minimal value ρ_{apex} at the apex, and satisfies

$$\rho'(\theta_{\min}) = \rho'(\theta_{\max}) = 0.$$

Thus the tube tapers smoothly from base to apex.

To remove the terminal boundary disk at the apex and obtain a smooth capped body, we attach a dome-shaped cap.

Definition 1.2 (Apical Cap)**.** Let $(T_{\max}, N_{\max}, B_{\max})$ be the Frenet–Serret frame of γ evaluated at $\theta = \theta_{\max}$. The *apical cap* is the surface

$$S_{\text{cap}}(u, v) = \gamma(\theta_{\max}) + \rho_{\text{apex}}\sin(u)\, T_{\max} + \rho_{\text{apex}}\cos(u)\big(\cos v\, N_{\max} + \sin v\, B_{\max}\big),$$

for

$$u \in [0, \pi/2], \qquad v \in [0, 2\pi).$$

At $u = 0$ this cap matches the terminal circle of the tube, and because $\rho'(\theta_{\max}) = 0$, the tangent plane of the cap agrees with the tangent plane of the tubular body along the junction. Hence the capped volumetric helix has C^1 boundary.

Remark 1.1. We impose

$$\rho_{\text{base}} < R \qquad \text{and} \qquad \rho_{\text{base}} < \pi|h|$$

to ensure that the thickened tube is embedded and does not intersect either the helical axis or adjacent turns. We separate the *abstract topology* of the object from its *embedded geometric realization.* Abstractly, a capped finite helix is a compact smooth 3–manifold diffeomorphic to the 3–ball D^3, equipped with a distinguished embedded base disk and a distinguished core arc. The helical geometry enters through the embedding in $\mathbb{R}^3$.

1.2 Chirality and Alignment

Definition 1.3 (Chirality)**.** A subset $O \subset \mathbb{R}^n$ is said to have *chirality* if it is not properly congruent to its mirror image. Equivalently, O is chiral if for every reflection r across a plane in $\mathbb{R}^n$, there exists no proper isometry, i.e. orientation-preserving Euclidean motion (translations or rotations), g such that $g(O) = r(O)$. We say that helices may possess either left-handed or right-handed chirality. A helix is left-handed if $h < 0$ and right-handed if $h > 0$.

Lemma 1.1 (Spiral Matching Lemma)**.** *Two Archimedean spirals, S_1 and S_2, with distinct centers and the same pitch, can intersect to form a C^0 curve with a matching phase if and only if they have opposite chirality.*

Proof. Let the two spirals be defined in $\mathbb{R}^2$ by the vector equations:

$$S_i(\theta_i) = C_i + a_i\theta_i \begin{pmatrix} \cos\theta_i \\ \sin\theta_i \end{pmatrix}, \quad i = 1, 2$$

We are given that the centers are distinct ($C_1 \neq C_2$) and the pitches are equal ($|a_1| = |a_2| = a > 0$). The lemma requires two conditions to be met simultaneously:

(a) **C^0 Intersection.** The spirals meet at a common point. $S_1(\theta_1) = S_2(\theta_2)$

(b) **Phase Matching.** The local polar angles are the same. $\theta_1 = \theta_2 =: \theta$

Substitute the phase-matching condition into the intersection equation:

$$C_1 + a_1\theta \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} = C_2 + a_2\theta \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

Rearranging the terms to isolate the vector between the centers gives our primary relation:

$$C_2 - C_1 = (a_1 - a_2)\theta \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

We analyze this relation for both chirality cases.

(a) **Same Chirality.** ($a_1 = a_2$) If the spirals have the same chirality, the term $(a_1 - a_2)$ is zero. The relation becomes

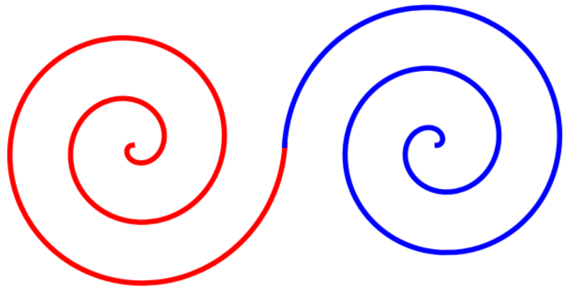
$$C_2 - C_1 = 0 \implies C_1 = C_2$$

This contradicts the given condition that the centers are distinct. Therefore, an intersection with matching phase is impossible for spirals of the same chirality.

(b) **Opposite Chirality.** ($a_2 = -a_1$) If the spirals have opposite chirality, the term $(a_1 - a_2)$ becomes $(a_1 - (-a_1)) = 2a_1$. The relation becomes

$$C_2 - C_1 = 2a_1\theta \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

Thus, an intersection with matching phase is possible if and only if the spirals have opposite chirality. □



Remark 1.2. We impose the phase-matching condition $\theta_1 = \theta_2$ so that the radial directions—and hence the unit tangent directions—agree at the intersection, removing any kink, the necessary first step toward any C^1 or C^2 splice.

Corollary 1.2 (Coaxial Spiral Chirality Constraint)**.** *Let S_1 and S_2 be Archimedean spirals centered at the same point. Then, a C^0 bounded path connecting them with matching phase at a common point can exist only if their chiralities are identical.*

Proof. This is a direct consequence of the Spiral Matching Lemma. As proven above, with $C_1 = C_2$, equation:

$$(C_2 - C_1) = (a_1 - a_2)\theta(\cos\theta, \sin\theta)$$

which gives

$$(a_1 - a_2)\theta\,(\cos\theta, \sin\theta) = 0$$

for $\theta \neq 0$, this forces $a_1 = a_2$, which in turn requires the spirals to have the same chirality. $\square$

Remark 1.3. This illustrates a fundamental principle in geometric design: adding a simple local constraint, such as matching phase, can enforce a relationship between the curves' global properties (chirality). In this case, the seemingly mild phase-matching condition is sufficient to rule out a connection entirely for same-chirality spirals with offset centers.

1.3 Twin Helical Manifolds

We now form pairs of volumetric helices by gluing them along their base disks.

Definition 1.4 (Binary Orientation of a Helix)**.** Let H be a helix in $\mathbb{R}^3$ with axis of symmetry parallel to the z-axis. The orientation of H is defined by the monotonicity of its z-coordinate as a function of its intrinsic parameter u, increasing from base to apex:

(a) **Upwards-oriented.** $z(u)$ is strictly increasing from base to apex.

(b) **Downwards-oriented.** $z(u)$ is strictly decreasing from base to apex.

Definition 1.5 (Type 1 Twin Helix)**.** Fix $R > 0$ and $h > 0$. Let the first spine be

$$\gamma_A(\theta) = (R\cos\theta,\ R\sin\theta,\ h\theta), \qquad \theta \in \mathbb{R},$$

and define the second spine by rotation through π about the x-axis:

$$\gamma_B(\theta) = (R\cos\theta,\ -R\sin\theta,\ -h\theta).$$

This produces the basic *Type 1* twin configuration. The remaining types are obtained from this model by applying the Euclidean isometries recorded in the final column of table 1.

Definition 1.6 (Twin Helical Manifold)**.** Let H_A and H_B be volumetric helices, and let

$$B_A \subset \partial H_A, \qquad B_B \subset \partial H_B$$

be their distinguished base disks. Given a diffeomorphism

$$f : B_A \to B_B,$$

the *Twin Helical Manifold* associated to (H_A, H_B, f) is the quotient

$$H_{AB} = (H_A \sqcup H_B)/\sim, \qquad p \sim f(p) \quad (p \in B_A).$$

This quotient defines the abstract manifold obtained by gluing the two helices along their bases. In the present paper we typically take f to be the identity in the natural cylindrical coordinates on the base disks. To realize this quotient as an embedded C^1 manifold in $\mathbb{R}^3$, we may interpolate between the two bases by an explicit bridge.

Definition 1.7 (Embedded Bridge Realization)**.** Let P_A and P_B be the centers of the two base disks, and let $\mathbf{t}_A, \mathbf{t}_B$ be the corresponding unit tangent vectors of the spines at the base. Fix a tension parameter $\ell > 0$ and define

$$\mathbf{m}_A = \ell\,\mathbf{t}_A, \qquad \mathbf{m}_B = \ell\,\mathbf{t}_B.$$

The bridge spine is the cubic Hermite curve

$$\Gamma(s) = h_{00}(s)P_A + h_{01}(s)P_B + h_{10}(s)\mathbf{m}_A + h_{11}(s)\mathbf{m}_B, \qquad s \in [0,1],$$

where

$$\begin{aligned} h_{00}(s) &= 2s^3 - 3s^2 + 1, \\ h_{01}(s) &= -2s^3 + 3s^2, \\ h_{10}(s) &= s^3 - 2s^2 + s, \\ h_{11}(s) &= s^3 - s^2. \end{aligned}$$

By construction,

$$\Gamma(0) = P_A, \qquad \Gamma(1) = P_B, \qquad \Gamma'(0) = \mathbf{m}_A, \qquad \Gamma'(1) = \mathbf{m}_B,$$

so the bridge matches the endpoint positions and tangent data.

For Types 2 and 4, where the naive Hermite curve may develop a small hair-pin, we replace Γ by the bumped curve

$$\widetilde{\Gamma}(s) = \Gamma(s) + \alpha s^2(1-s)^2\,\mathbf{n},$$

where $\mathbf{n}$ is a unit vector orthogonal to the relevant tangent directions and

$$\alpha > 2\rho_{\text{base}}.$$

This perturbation preserves the endpoint position and tangent data while moving the bridge clear of the parent tubes. A tubular neighborhood of Γ (or $\widetilde{\Gamma}$) of radius at most ρ_{base} then gives an explicit C^1 bridge between the two volumetric helices. This produces an embedded realization of the abstract quotient H_{AB} inside $\mathbb{R}^3$.

Remark 1.4. The spline bridge is *not* part of the abstract definition of a Twin Helical Manifold. It is only one convenient geometric realization used to exhibit a C^1 embedding in $\mathbb{R}^3$. The underlying manifold is determined by the gluing of the base disks, while the particular bridge affects only the chosen embedded representative.

We call any such glued object a *Twin Helical Manifold* (THM), or more informally, a *Twin Helix.*

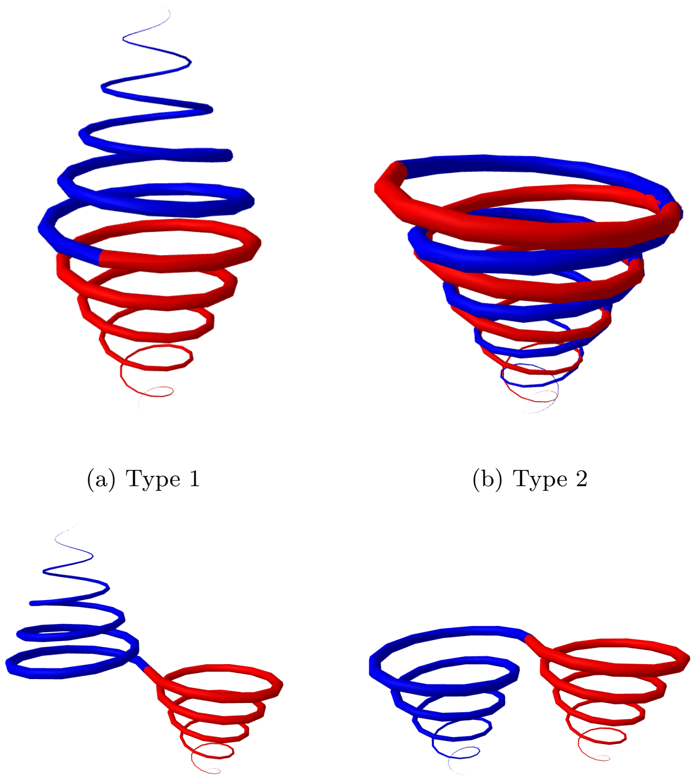
Definition 1.8 (Twin Helical Manifold and Unit Helices)**.** A THM consists of two embedded helical surfaces $\Sigma_\pm \subset M$ whose boundary circles are identified along a smooth annulus $A \simeq S^1 \times I$, with orientations matched so that the resulting subset $\Sigma_+ \cup_A \Sigma_-$ is a smooth submanifold (no cone angle or cusp). A connection $A \in \Omega^1(M;\mathfrak{g})$ restricts smoothly across A; its curvature F is bounded (no measure concentration). The two helices composing a single THM are termed *unit helices*, denoted H_A and H_B.

By combining different chiralities and orientations, we define a geometric family of four canonical manifolds, denoted THM_4. This yields a total of 4 distinct invariant configuration states. However, our Type 2 twin helix has a small problem. A simple drawing shows that constructing a twin helix with the specified attributes becomes problematic in the previously defined $\mathbb{R}^3$ space. A closer investigation leads to the following conclusion.

Lemma 1.3 (Twin Helix Intersection Lemma)**.** *No twin helical manifold whose unit helices with turns $N \geq 1/2$, sharing both axial alignment, orientation, and opposing chirality, can exist without intersection of its respective unit helices in $\mathbb{R}^3$.*

Table 1: Essential type chart for twin helical manifolds

Type	**Chirality**	**Orientation**	**Axiality**	**Mnemonic**	**Map** $\Rightarrow H_B$
1	Opposite	Opposite	Coaxial	Helical Orb	$R_x(\pi)$
2	Opposite	Same	Coaxial	Twisted Pair	Σ_y
3	Same	Opposite	Offset	Symmetrical Flow	$R_y(\pi), T$
4	Same	Same	Offset	Co-propagating Twins	$R_z(\pi), T$



(a) Type 1 (b) Type 2

(c) Type 3 (d) Type 4

Figure 1: The THM_4 geometric family.

Proof. Let H_A and H_B be two coaxial helices parameterized by $t \in [0,1]$. Both helices share the same radial function $r(t)$ and the same axial function $z(t)$. Their distinction lies in chirality and a possible phase offset.

The angular position of a point on helix H_x at parameter t is given by

$$\alpha_x(t) = 2\pi N \chi_x t + \phi_x,$$

where:

(a) N is the number of turns in one unit helix,

(b) χ_x is the chirality of H_x, with $\chi_A = -1$ and $\chi_B = 1$ for Type 2,

(c) ϕ_x is the phase offset. Without loss of generality, we may set $\phi_A = 0$.

The center lines of the two helices intersect if, for some $t \in [0,1]$, their angular positions coincide modulo 2π. That is

$$\alpha_B(t) - \alpha_A(t) = 2k\pi$$

for some integer k. Substituting the expressions for $\alpha_A(t)$ and $\alpha_B(t)$, we obtain:

$$\begin{aligned}(2\pi N\chi_B t + \phi_B) - (2\pi N\chi_A t) &= 2k\pi\\ 2\pi N t(\chi_B - \chi_A) + \phi_B &= 2k\pi\\ 4\pi N t + \phi_B &= 2k\pi.\end{aligned}$$

Define the function $f(t) = 4\pi N t + \phi_B$. An intersection occurs if there exists a $t \in [0,1]$ such that $f(t) = 2k\pi$ for some integer k.

The function $f(t)$ is continuous and monotonic on $[0,1]$ (assuming $N > 0$). The values it takes range from

$$f(0) = \phi_B, \quad f(1) = 4\pi N + \phi_B.$$

This interval has length $4\pi N$. For $f(t)$ to contain at least one multiple of 2π, it suffices that

$$4\pi N \geq 2\pi \quad \Rightarrow \quad N \geq \frac{1}{2}.$$

By the Intermediate Value Theorem, since $f(t)$ is continuous on $[0,1]$, it must attain all values in the interval $[\phi_B, \phi_B + 4\pi N]$. If this interval has length at least 2π, it must contain at least one value $2k\pi$. Therefore, for $N \geq \frac{1}{2}$, there always exists a $t \in [0,1]$ for which $\alpha_B(t) \equiv \alpha_A(t) \mod 2\pi$, and thus the helices intersect.

Hence, for any $N \geq \frac{1}{2}$, it is mathematically impossible to choose a constant angular phase offset ϕ_B that avoids the intersection of the center lines of the Type 2 twin helices on $[0,1]$. □

Twin helices and the Whitney paradigm. In $\mathbb{R}^3$, a closed orientable surface has a line normal bundle; demanding a global 2-frame (to preserve both chirality and phase across the join) overconstrains codimension 1. Equivalently, any such immersion develops transverse self-intersections at the Type–2 junction, so the framed join cannot be realized without crossings. Passing to $\mathbb{R}^4$ removes this extrinsic obstruction: the normal bundle is rank 2 and can be trivialized, so one obtains a V_4–equivariant, self–intersection–free embedding that preserves the twist data. This matches the general Whitney [1] paradigm: every smooth 2–manifold embeds in $\mathbb{R}^4$, and the extra normal direction carries the required framing.

Chirality rule-reversal between 2- and 3-dimensional gluing. A curious but systematic reversal occurs when one passes from planar curves to spatial curves concerning the handedness required for a C^2 splice. In the planar setting (spirals), the data to be matched at an intersection consists of position, unit tangent, and signed curvature. Because curvature is the only first-order invariant beyond the tangent, one free phase parameter suffices to adjust its magnitude but not its sign. Consequently, two offset spirals can be C^0-glued iff their curvature signs are opposite.

In the spatial setting (helices), a space curve adds a second scalar invariant (torsion), whose sign records right-handed versus left-handed screw motion. The gluing constraints now depend on the relative placement of the helical axes:

Proposition 1.4 (Rules for Smooth Gluing of Helices)**.** *Let*

$$\gamma_i(t) \;=\; c_i + R_i\big(\cos t\,\mathbf{u}_i + \sin t\,\mathbf{v}_i\big) + p_i\,t\,\mathbf{e} \quad (i = 1,2)$$

be two circular helices in $\mathbb{R}^3$ whose axes are parallel to the same unit vector $\mathbf{e}$.
Define curvature and torsion in the usual way:

$$\kappa_i = \frac{R_i}{R_i^2 + p_i^2}, \qquad \tau_i = \frac{p_i}{R_i^2 + p_i^2}\ ,$$

so $\mathrm{sign}(\tau_i)$ records right- or left-handedness.

(a) ***Coaxial case.*** *If the centres c_1, c_2 differ only along the common axis $(c_2 - c_1) \parallel \mathbf{e}$, then one can splice γ_1 to γ_2 by a C^2 curve through an intersection point P* iff

$$\tau_1 + \tau_2 = 0, \quad \textit{equivalently} \quad \mathrm{sign}(\tau_1) = -\,\mathrm{sign}(\tau_2).$$

(b) ***Offset-axis case.*** *If the axis directions coincide but the centres are displaced sideways, $(c_2 - c_1) \perp \mathbf{e}$, then a C^2 splice exists* iff

$$\tau_1 = \tau_2, \quad \textit{equivalently} \quad \mathrm{sign}(\tau_1) = \mathrm{sign}(\tau_2).$$

Proof. Identify $P = \gamma_1(t_1) = \gamma_2(t_2)$. Matching a 2–jet at P requires:

$$\mathrm{J}^0(\gamma_1) = \mathrm{J}^0(\gamma_2), \quad \mathrm{J}^1(\gamma_1) = \mathrm{J}^1(\gamma_2), \quad \mathrm{J}^2(\gamma_1) = \mathrm{J}^2(\gamma_2).$$

Explicitly this means (position) P coincides, (tangent) $\dot{\gamma}_1(t_1)$ is a positive multiple of $\dot{\gamma}_2(t_2)$, and (curvature + torsion) the Frenet frames agree.

Coaxial case. The equality of normals forces $t_2 = t_1 + \pi$ up to 2π, leaving a single phase parameter. The curvature equality $\kappa_1 = \kappa_2$ fixes R_2 once R_1 is given. The remaining torsion constraint is $\tau_1 + \tau_2 = 0$, hence opposite signs.

Offset case. Because $d \neq 0$ there are now *two* independent phase angles (t_1, t_2) available. These can be chosen so that the binormals (and hence normals) align, provided $\tau_1 = \tau_2$. If the torsion signs differ, the binormals rotate in opposite senses and cannot be made parallel.

Details are an elementary exercise in solving the linear constraints on $\dot{\gamma}_i$ and $\ddot{\gamma}_i$. □

Remark 1.5. Passing from $\mathbb{R}^2$ to $\mathbb{R}^3$ promotes the matching problem from the 1-jet to the 2-jet bundle: curvature sign is now unsigned (since $\kappa > 0$), while torsion sign becomes the relevant orientation datum. Whether this new constraint binds or is absorbed by an extra phase degree of freedom hinges on the axial configuration. In effect, the added dimension introduces a framing (normal-binormal plane) whose sign can obstruct or permit the splice depending on how many phase parameters survive the geometric symmetry.

Proposition 1.5 ($THM_4 \cong V_4$)**.** *The group of transformations generated by the Chirality Swap (X) and Orientation Swap (Y) operators, which acts on the set of four canonical THM types, is isomorphic to the Klein four-group, V_4.*

Proof. Let $H_{\text{types}} = \{\text{Type 1}, \text{Type 2}, \text{Type 3}, \text{Type 4}\}$, representing the four configurations of unit helices determined by combinations of chirality (Same vs Opposite) and orientation (Same vs Opposite).

Define the following transformations acting on H_{types}:

- Identity I: Trivial transformation. $I(T_i) = T_i$ for all $T_i \in H_{\text{types}}$.
- Chirality Swap X: Swap the chirality relationship (Same $\leftrightarrow$ Opposite), while preserving orientation.
- Orientation Swap Y: Swap the orientation relationship (Same $\leftrightarrow$ Opposite), while preserving chirality.
- Composite $Z = X \circ Y$: Acts by applying both swaps.

The action on the four manifold types proceeds as follows:

Type	(Chirality, Orientation)	X	Y
H_1	(O, O)	H_3	H_2
H_2	(O, S)	H_4	H_1
H_3	(S, O)	H_1	H_4
H_4	(S, S)	H_2	H_3

Composition rules:

$$\begin{aligned} Z &= X \circ Y = Y \circ X \\ Z(H_1) &= X(Y(H_1)) = X(H_2) = H_4 \\ Z(H_2) &= X(Y(H_2)) = X(H_1) = H_3 \\ Z(H_3) &= X(Y(H_3)) = X(H_4) = H_2 \\ Z(H_4) &= X(Y(H_4)) = X(H_3) = H_1 \end{aligned}$$

All non-identity elements have order 2:

$$X^2 = I, \quad Y^2 = I, \quad Z^2 = (X \circ Y)^2 = X^2 \circ Y^2 = I$$

Pairwise compositions:

$$X \circ Y = Z, \quad Y \circ X = Z \quad \Rightarrow \quad X \circ Y = Y \circ X$$

$$X \circ Z = X \circ (X \circ Y) = (X \circ X) \circ Y = I \circ Y = Y$$

$$Y \circ Z = Y \circ (X \circ Y) = (Y \circ Y) \circ X = I \circ X = X$$

The set of transformations $G = \{I, X, Y, Z\}$ forms a group under composition. All elements have order 1 or 2, and the group is abelian. Thus, $G \cong V_4$. □

Table table 2 of the calculated FSG progressions via the GAP for reference.

Table 2: Group progression of the THM_4

Group Structure	**Order**	**Defined by...**	**Type**
$V_4 \cong C_2 \times C_2$	4	Chirality and orientation combinations of the 4 canonical THM_4 types	abelian
$D_4 \cong (C_4) \rtimes C_2$	8	Chirality flips + swapping identity of the two helices	non-abelian
C_2^4	16	Independent flips of (c_A, c_B, d_A, d_B) for each unit helix	abelian
$C_2^4 \rtimes C_2$	32	Above + helix identity swapping symmetry	non-abelian
$(C_2^4 \rtimes C_2) \times C_4$	128	Adds 90° global rotation symmetry around z-axis	non-abelian
$(C_2^4 \rtimes C_2) \times C_{4h}$ $\cong (C_2^4 \rtimes C_2) \times (C_4 \times C_2)$	256	Full group including xy-plane reflection symmetry (σ_h)	non-abelian

Table 3: Generators of the THM_{256} symmetry group

Generator	**Action**
c_1	Flip chirality of Helix A
c_2	Flip chirality of Helix B
d_1	Flip orientation of Helix A (up/down)
d_2	Flip orientation of Helix B (up/down)
S	Swap identity of Helix A and Helix B
R_{90}	Global 90° rotation of the twin helix around the z-axis
σ_h	Reflect the entire twin helix across the xy-plane

In addition, the generators of THM_{256} and lower orders are shown in table 3. These symmetries exhibit a progression toward non-abelian behavior as more generators and state permutations are introduced. We note that this group structure may not be unique to the present construction. Similar generator sets arise in various contexts of symmetry and braid groups. For our purposes here, we do not require the full symmetry group, so the explicit relations and whole structure of THM_4, including its precise order, are left open for further investigation.

Definition 1.9 (Regularity Principle for Geometric Computation)**.** A geometric computational state consists of an admissible manifold configuration together with a specified local evolution law (flow, pulse update, or hybrid dynamics). A necessary condition for well-posed computation is that the geometric data and the evolution law possess sufficient regularity to prevent singular breakdown and to make local updates unambiguous.

At minimum, the state manifold must admit the geometric structures used by the model (e.g. tangent directions, phase variables, curvature), and the associated vector field or update operator must be at least locally Lipschitz on the relevant state space. This ensures local existence and uniqueness of the induced ordinary differential evolution whenever the model is expressed in ODE form.

Definition 1.10 (Minimal Regularity by Computational Regime)**.** The required differentiability depends on which geometric invariants actively participate in the computation:

- **Abelian regimes** (symbolic and quasi-fluidic). It is generally sufficient that the geometric state be at least C^1 and that the local evolution law be locally Lipschitz (or C^1) in the active variables. This guarantees well-defined tangent/phase transport and deterministic local evolution. Higher regularity is often convenient but not essential.

- **Non-abelian regimes** (fully fluidic and gauge). When the computation depends explicitly on curvature, torsion, holonomy, or other higher-order geometric data, one typically requires at least C^2 regularity of the underlying geometric configuration, together with compatible regularity of the evolution law. In practice, C^k regularity for $k \geq 2$ (and often smoother) is preferable for stable geometric evolution.

Remark 1.6. Geometric computation must be regular enough for its update rules to be well-defined. In the symbolic and quasi-fluidic bands, this usually means controlling tangent and phase data; in the higher regimes, curvature and higher-order structure enter directly, forcing stronger regularity assumptions. Thus, singularities are not merely geometric defects: they mark a breakdown of the computational model itself.

2 Geometrically Computable Manifolds

We now begin the construction of a Turing Machine ("TM") from a lattice of THM. The feasibility of this approach rests on the fact that THM, unlike most manifolds, provides a rich but controlled symmetry space. It is precisely because most well-known shapes in this metric (e.g., circles, tori, spheres) possess too much symmetry—not too little—that they fail to offer the structural asymmetry needed for computational behavior. The THM construction breaks this impasse.

2.1 Geometrically Computable Manifolds

Definition 2.1 (Geometrically Computable Manifold)**.** A **GCM** on a space is:

$$\mathcal{G} \;=\; \big(\mathcal{M},\; \{R_i\}_{i\in I},\; \pi : E \to \mathcal{M},\; \mathcal{I},\; \mathsf{Ev}\big)$$

consisting of:

(a) **Smooth Manifold $\mathcal{M}$.** A smooth manifold (often a finite working region, possibly with boundary). A finite family of compact submanifolds $\{R_i\}_{i\in I}$ (cells) covers the working region and provides locality (finite overlap pattern).

(b) **State bundle $\pi : E \to \mathcal{M}$.** A smooth fiber bundle with finite dimensional fibers. A *global state* is a section $\sigma \in \Gamma(E)$. Discrete/symbolic, toroidal/phase, and integer-valued components may all live in the fiber.

(c) **Interfaces $\mathcal{I}$.** For each nonempty overlap $R_i \cap R_j$, an interface map:

$$I_{ij} :\; \big(\sigma|_{R_i},\, \sigma|_{R_j},\, \text{local data}\big) \longmapsto \big(\sigma'|_{R_i},\, \sigma'|_{R_j}\big),$$

attached to a portion of $\partial R_i \cap \partial R_j$. Interfaces couple neighboring cells and enforce locality: updates at R_i depend only on σ over R_i and finitely many overlaps $R_i \cap R_j$.

(d) **Evolution Ev.** Either:

- discrete time: an update operator $U : \Gamma(E) \to \Gamma(E)$ that is local with respect to $\{R_i\}$, or
- continuous time: a flow of sections $(\sigma_t)_{t\in\mathbb{R}}$ solving $\dot{\sigma}_t = F(\sigma_t)$ for a smooth, fiber-wise vector field F over $\mathcal{M}$;

possibly accompanied by a base diffeomorphism $\Phi_t : \mathcal{M} \to \mathcal{M}$. In either case, evolution is non-singular; it exists for all steps/times on admissible initial states and remains in a fixed finite-energy subset of E.

What a GCM is. A GCM is a manifold that can encode its computable properties. A single THM can qualify as a GCM. A lattice of THM is also a GCM—just with more computing capacity. We call $\mathcal{G}$ a GCM when its evolution is local, interface-driven, and closed under composition: finite products of such systems and gluing along matched interfaces yield new systems of the same kind.

2.2 Geobits—Bits, Geometric Bits, and State Space

Definition 2.2 (Geometric Bits — Symbolic and Enriched)**.** Fix an ambient manifold region $\mathcal{M} \subset \mathbb{R}^n$ and a twin helix template H (a pair of embedded tubular helices) together with a framing: a preferred axial direction $\hat{z}$ and a reference scale r_0. Let $\mathrm{Emb}^{\mathrm{fr}}(H, \mathcal{M})$ denote the set of framed embeddings $e : H \hookrightarrow \mathcal{M}$ modulo framed ambient isotopy that preserves the axial framing and the scale r_0 so that radius and orientation remain part of the state.

Let $H_{\mathrm{ext}} \leq O(n)$ be a chosen ambient symmetry group (allowing reflections so that chirality changes are representable) and let H_{int} be a finite internal symmetry group acting on the labels of the twin (e.g. the swap of the two helices). Set $G := H_{\mathrm{ext}} \times H_{\mathrm{int}}$, acting on $\mathrm{Emb}^{\mathrm{fr}}(H, \mathcal{M})$ by pre/post–composition.

(a) **Symbolic geobit.** Define the symbolic geobit state set:

$$\mathbf{S}_{\mathrm{sym}} \;:=\; \Big\{ [e; \chi, \omega] \;\Big|\; e \in \mathrm{Emb}^{\mathrm{fr}}(H, \mathcal{M}),\ \chi,\ \omega \Big\} \Big/ G,$$

where $[e; \chi, \omega]$ denotes a framed embedding decorated by chirality (χ) and axial orientation (ω) relative to $\hat{z}$, and the quotient is by the G–action, so states are defined up to global symmetry. A *symbolic geobit* is an element $s \in \mathbf{S}_{\mathrm{sym}}$. The finite subgroup generated by "flip chirality" C, "flip orientation" D, and (optionally) "swap" $S \in H_{\mathrm{int}}$ acts on $\mathbf{S}_{\mathrm{sym}}$; in the canonical twin helix unit this induces a $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ action on the labels (χ, ω).

(b) **Enriched (quasi–fluidic) geobit.** To allow compact phase and quantized charges, extend the label by a finite–type fiber:

$$P \;\cong\; (S^1)^m \times \mathbb{Z}^p \times F,$$

where the S^1–coordinates encode bounded phases (e.g. axial angle), the $\mathbb{Z}$–coordinates encode twist/winding/linking numbers, and F is a finite set of discrete tags (e.g. mode, parity). Set

$$\mathbf{S}_{\mathrm{enr}} \;:=\; \Big\{ [e; \chi, \omega, p] \;\Big|\; e \in \mathrm{Emb}^{\mathrm{fr}}(H, \mathcal{M}),\ \chi,\ \omega,\ p \in P \Big\} \Big/ G,$$

with G acting equivariantly on the S^1–factors (by rotation) and preserving the $\mathbb{Z}$–charges. An *enriched geobit* is an element of $\mathbf{S}_{\mathrm{enr}}$. The G–action need not be transitive; its orbits in $\mathbf{S}_{\mathrm{enr}}$ are the *geobit types*. The symbolic case is recovered by taking $m = p = 0$ and F trivial.

In both cases, the geobit is a G–set (not necessarily a group): the finite operations generated by C, D (and S when present) act on states, yielding the canonical V_4 action in the symbolic unit. The enrichment by P provides the quasi–fluidic degrees of freedom (compact phases and integer counters) used elsewhere in the text.

For the time, we work only with S_{sym}. Now, let:

$$(H_1, H_2, H_3, H_4) \mapsto (00, 01, 10, 11)$$

where each H_n is a configuration from the THM_4 symmetry schema defined in table 1, preserving the order of the original type chart. We assign bit values based on symmetry: if the chirality and orientation of the unit helices are the same, assign bit value 1; if they differ, assign 0. This yields a total of 2 bits per THM_4 configuration.

Definition 2.3 (Geobit Capacity)**.** Let L be the finite set of *internal labels* for a GCM cell (e.g. chirality/orientation choices before identifying symmetries), and let the *encoding symmetry* O act on L by intrinsic transformations (excluding extrinsic rigid motions). The set of *observable* states is the orbit set L/O, and the *geobit capacity* is

$$\mathrm{Cap}(L, O) \;:=\; \log_2 |L/O|.$$

When $L = O$ with the left–regular action, this reduces to $\mathrm{Cap} = \log_2 |O|$.

Example 2.1 (Embedding Relationships for Helices). Let G act on a configuration space and let $O \subseteq G$ be the intrinsic encoding subgroup; put L for the raw label set and compute observable states via L/O.

(a) THM_{16}. $O \cong C_2^4$, $L = O$ with the left–regular action. Then $|L/O| = |O| = 16$, so

$$\mathrm{Cap} = \log_2 16 = 4 \text{ bits} \quad (\text{"2 bits from } \chi\text{, 2 from } \omega\text{"}).$$

(b) THM_4. Impose identifications (e.g., identity–swap/pairing symmetries) modeled by a normal subgroup $H \trianglelefteq O$ of order 4; the effective encoding collapses to $O_{\text{eff}} \cong O/H \cong C_2^2$. With $L = O_{\text{eff}}$ (regular action),

$$\mathrm{Cap} = \log_2 |O_{\text{eff}}| = \log_2 4 = 2 \text{ bits}.$$

(c) THM_2. If only orientation is swappable, the effective group is $O_{\text{eff}} \cong C_2$, hence

$$\mathrm{Cap} = \log_2 2 = 1 \text{ bit}.$$

(d) THM_8. Let the raw labels be $L = C_2 \times C_4$ with elements (χ, ω) (chirality $\chi \in \{\pm 1\}$, orientation $\omega \in \mathbb{Z}/4$). Let $O = D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle$ act by

$$r \cdot (\chi, \omega) = (\chi, \omega + 1), \qquad s \cdot (\chi, \omega) = (-\chi, -\omega).$$

Then $|L| = 8$, but the action has nontrivial stabilizers (reflections identify (χ, ω) with $(-\chi, -\omega)$), yielding $|L/O| = 4$ orbits. Hence

$$\mathrm{Cap} = \log_2 4 = 2 \text{ bits} \quad \text{while} \quad \log_2 |O| = 3,$$

illustrating the reduction caused by non–commuting symmetries (orbit collisions).

This leads to a key structural pattern. Let O act on the finite label set L:

- **Abelian Case.** If $O \cong (C_2)^n$ acts by independent flips on L with a free action (e.g. the regular action on $L = O$), then $|L/O| = |O| = 2^n$ and the capacity is n bits.
- **Non–abelian Case.** When O is non–abelian, typical elements introduce larger stabilizers and merge labels across conjugate directions; consequently, $|L/O| < |L|$ and often $\mathrm{Cap} < \log_2 |O|$ (as in the D_4 example).

Table 4: Relationship between geobits and their groups

Group	Order	Dimension	# Bits	Comments
C_2	2	2	1	Classical bit
$(C_2)^2 = V_4$	4	4	2	Base level 2-geobit
$(C_2)^4$	16	4	4	Canonical abelian 4-geobit
D_4	8	3	≤ 3	Non-abelian, simple global swap
$D_4 \ltimes (C_2)^4$	128	≥ 5	< 8	Non-abelian, global rotations

Why the focus on abelian vs non-abelian groups? In an abelian group, every state can be constructed as the direct sum of independent generators. That is, every geobit state is uniquely addressable by an n-tuple over $\mathbb{C}_2^n$. There are no collisions. The total number of distinguishable geobit states is exactly 2^n, and the encoding is non-redundant.

In contrast, a non-abelian group acts less cleanly. Its action "mixes" states: some group elements act non-trivially, creating identifications among configurations. Orbits form under conjugation or subgroup stabilizers, and the full state space becomes quotient-like. Redundancy becomes intrinsic—some geobit states are indistinguishable under the group action. As a result, the number of distinct encodable states in the orbit space may be strictly less than the order of the group itself.

This distinction becomes crucial when interpreting a GCM as a computational substrate: *abelian structure maximizes symbolic resolution, while non-abelian structure imposes constraints that must be managed or exploited.*

Remark 2.1. The embedding dimension for a THM symmetry group is the minimal integer n such that all distinct geometric configurations corresponding to the elements of the group can be realized as non-self-intersecting embedded submanifolds in Euclidean space. It is essential to note that global reflections or entire rotational symmetry groups do not contribute to the intrinsic geobit capacity in $\mathbb{R}^4$. These operations require action on the global geometry and do not alter the intrinsic state of the unit helices. Thus, group orders beyond 16 that arise from external symmetry do not increase state space in a computable sense. Table 5 compares geobits with their close cousin.

Table 5: Qubit vs. Geobit

Property	**Quantum qubit**	**Geometric geobit**
Pure state space	$\mathbb{C}P^1 \cong S^2$ (Bloch sphere)	Finite labels $\times$ continuous geometric parameters (e.g. chirality/orientation, phase)
Dynamics	Unitaries (e.g. $SU(2)$ rotations)	Geometric transformations (flips, rotations, deformations)
Measurement/Readout	Born rule (probabilistic)	Geometric probe/partition (deterministic given microstate)
Accessible classical info	≤ 1 bit per use[1]	Depends on chosen encoding and symmetries
Elementary operations	Single-/two-qubit gates; entanglement via tensor product	Local geometric updates; composition by connecting/gluing components
Key feature	Coherence and entanglement central	Robustness tied to analog stability

Definition 2.4 (Geometric Kets)**.** We refer to an individual geobit ket array as "n-geobit," directly paralleling the usage of qubit arrays with ket-style notation. Following the same notation, we employ a modified bra-ket notation.

Let the microstate of a twin helix cell be

$$\alpha = (C_A, O_A, C_B, O_B) \in C_2^4.$$

Where C is chirality and O is orientation. We write the *state ket* $|\alpha\rangle := |C_A\, O_A\, C_B\, O_B\rangle$. Intrinsic flips act by the Klein-four group

$$V_4 = \{e,\, s_A,\, s_B,\, s_A s_B\},$$

where s_A flips (C_A, O_A) and s_B flips (C_B, O_B) coordinatewise. A 1-bit readout is any fixed function $r : C_2^4 \to C_2$ (e.g. parity $r = o_A \oplus o_B$ or equality $r = [o_A = o_B]$).

Example 2.2 (Update Step with V_4). Take parity readout $r = o_A \oplus o_B$ and the microstate

$$|\alpha\rangle = |1\,1\,0\,1\rangle.$$

Then $b = r(x) = 1 \oplus 1 = 0$. Suppose the next bit should be $b' = 1$. Apply any toggler in V_4, e.g. s_A:

$$s_A \cdot |\alpha\rangle \;=\; |1\,0\,0\,1\rangle, \qquad r(s_A \cdot \alpha) = 0 \oplus 1 = 1 = b'.$$

(If instead $b' = b$, use a parity-preserver e or $s_A s_B$.) Optionally, a coarse projection $\pi : C_2^4 \to C_2^2$ such as

$$\pi(C_A, O_A, C_B, O_B) = (C_A \oplus C_B,\; O_A \oplus O_B)$$

can be displayed as $|\pi(\alpha)\rangle$; note that s_A or s_B need not descend to operators on the quotient, so actions are applied on C_2^4 before projecting. The resulting ket would then become $|11\rangle$.

2.3 The GCM Lattice

Definition 2.5 (GCM Lattice)**.** A *GCM Lattice* or *GL* is a finite or locally finite assembly of GCMs arranged over a graph. Formally, it is the data:

$$GL \;=\; \big(\mathcal{L},\; G=(I,E),\; \{\mathcal{G}_i\}_{i\in I},\; \{\iota_i\},\; \mathcal{I}^{\mathrm{lat}},\; \mathsf{Ev},\; \mu\big)$$

with:

(a) **Ambient region $\mathcal{L}$.** Smooth manifold (often a bounded working region) into which cells will be embedded.

(b) **Lattice graph $G = (I, E)$.** A finite or locally–finite graph (e.g. a box of $\mathbb{Z}^d$) with uniformly bounded degree; vertices index the cells, edges encode adjacency.

(c) **Cells $\{\mathcal{G}_i\}$.** For each $i \in I$, a GCM $\mathcal{G}_i = (\mathcal{L}_i, \{R^i_\alpha\}, \pi_i : E_i \to \mathcal{L}_i, \mathcal{I}_i, \mathsf{Ev}_i)$ as in definition 2.1.

(d) **Placements $\{\iota_i\}$.** Smooth embeddings $\iota_i : \mathcal{L}_i \hookrightarrow \mathcal{L}$ with pairwise disjoint interiors. The support of cell i is $\mathrm{supp}(i) := \iota_i(\mathcal{L}_i)$.

(e) **Lattice interfaces $\mathcal{I}^{\mathrm{lat}}$.** For each edge $(i, j) \in E$, a family of interface maps attached to portions of $\partial\mathrm{supp}(i) \cap \partial\mathrm{supp}(j)$,

$$I_{ij} :\; \big(\sigma|_{\mathrm{supp}(i)},\, \sigma|_{\mathrm{supp}(j)},\, \text{local flow data}\big) \;\longmapsto\; \big(\sigma'|_{\mathrm{supp}(i)},\, \sigma'|_{\mathrm{supp}(j)}\big),$$

which are local (depend only on a finite stencil around i, j) and equivariant with respect to the chosen cell symmetries.

(f) **Evolution Ev.** Either (a) a discrete, synchronous update $U : \Gamma(E) \to \Gamma(E)$ on the global state bundle $E := \bigsqcup_{i\in I} \iota_i E_i$, or (b) a continuous flow of sections $(\sigma_t)_{t\in\mathbb{R}}$ solving $\dot\sigma_t = F(\sigma_t)$ and compatible with all interfaces. Evolution is local and non-singular on admissible initial states.

(g) **Symbolic observation μ.** A readout map $\mu : \Gamma(E) \to \Sigma^N$ that, at each cell i, selects a coding $q_i : E_i \to \Sigma^{n_i}$ of its geobits; the total capacity is $N = \sum_{i\in I} n_i$.

We say a *GL* is *computationally coherent* if the symbolic dynamics induced by the geometric evolution commutes with readout, i.e., there exists $\Phi_s : \Sigma^N \to \Sigma^N$ with:

$$\mu\big(U(\sigma)\big) \;=\; \Phi_s\big(\mu(\sigma)\big) \quad \text{(discrete time)}, \qquad \text{or} \qquad \frac{d}{dt}\,\mu(\sigma_t) \;=\; F_s\big(\mu(\sigma_t)\big) \quad \text{(continuous time)}.$$

Remark 2.2. The architecture of geometric computability is decidedly a non–Von Neumann one. A GL is necessary to compute programs that are not trivial, since one can't discretely read out results on a single GCM with symbolic or quasi-fluidic computation, not without a projection layer or a symbolic "slice." In a GCM there is no separation of program and data, no instruction pointer, or fetch–decode–execute cycle, the geometry *is* the computation. Computation is intrinsic, continuous, and spatially distributed. While one can clearly see the similarity to cellular automata in the discrete case, parallels can be drawn to neural fields or reaction-diffusion systems if an outside example is required.

2.4 Achieving Turing Completeness in R4

Finite state machines (FSMs) are the building blocks of classical computation. A FSM consists of a quintuple $(S, \Sigma, \delta, s_0, F)$: a finite set of states S, a finite input alphabet Σ, a transition function $\delta : S \times \Sigma \to S$, an initial state $s_0 \in S$, and a designated accepting set $F \subseteq S$ (or an output map). Given an input word over Σ, the machine updates its state by iterating δ; acceptance (or output) is determined by the terminal state.

Proposition 2.1 (FSM Embedding in Symbolic Mode)**.** *In the symbolic regime, any finite-state machine $(S, \Sigma, \delta, s_0, F)$ embeds into a GCM tile with finite internal labels and symbolic interfaces.*

Table 6: Classical vs. Geometric Computation

Feature	Classical Computation	Geometric Computation
Basic substrate	Strings, tapes, registers, graphs of discrete symbols	Embedded geometric cells/manifolds arranged in a lattice or interface network
Primitive state	Symbol/bit / finite register value	Local geometric state (e.g. chirality, orientation/phase, and optional topological counters)
State space	Finite alphabet or discrete configuration space	Structured local fiber, typically discrete or hybrid: symbolic labels and, in richer regimes, continuous phase data and quantized invariants
Elementary operations	Rewrite rules, gate updates, transition functions	Admissible geometric transformations, local interface updates, and symmetry-respecting deformations
Dynamics	Discrete step semantics (FSM/TM/CA update)	Local evolution on geometric states: symbolic updates, driven pulses, or continuous flows with induced return maps
Composition	Product of machines, circuit wiring, cellular adjacency	Gluing/coupling of cells through interfaces, adjacency graphs, and local geometric compatibility conditions
Input	External symbol stream or initial tape/register contents	Boundary conditions, perturbations, forcing pulses, or initialized geometric configurations
Output	Tape contents, accepting state, register readout	Readout of geometric observables via a chosen projection/partition of the state space
Symmetry	Usually implicit or combinatorial	Explicit: finite groups in the symbolic regime, abelian phase symmetries and richer geometric symmetries beyond
Constraint structure	Logical consistency and finite update rules	Smoothness constraints, and admissibility of local geometric moves
Computational viewpoint	Computation is manipulation of discrete symbols	Computation is evolution of structured geometry, with symbolic data recovered as a readout of the substrate

Sketch. Let a cell carry a finite internal label set $L \cong S$ (e.g. $L = C_2^k$ with a fixed encoding of S). Inputs are a finite set of discrete interface signals Σ (boundary toggles). Specify a local update table $\Delta : L \times \Sigma \to L$ implementing δ. Fix an initial label $\ell_0 \in L$ encoding s_0. Choose a readout predicate $R : L \to \{0,1\}$ and define the accepting set $F = \{\ell \in L : R(\ell) = 1\}$.

A single tile thus realizes the FSM; a single-dimensional lattice with nearest-neighbor symbolic communication realizes synchronous products. □

Proposition 2.2. *A one-dimensional GL whose cells are THM_4 tiles with symbolic nearest–neighbor communication can simulate Rule 110.*

Proof. Each cell contains a twin helix with two binary orientation labels

$$(o_A, o_B) \in C_2 \times C_2,$$

and intrinsic orientation flips act by the Klein four group

$$V_4 \;\cong\; C_2 \times C_2 \;=\; \{e,\, s_A,\, s_B,\, s_A s_B\},$$

where s_A flips o_A, s_B flips o_B, and s_As_B flips both.

Readout/encoding. Fix the bit readout as the parity (XOR)

$$r(o_A, o_B) \;:=\; o_A \oplus o_B \;\in\; \{0,1\}.$$

Any fixed local encoding is acceptable; parity is chosen because s_A or s_B toggles r, while s_As_B preserves r.

Cellular update. Let $b_i^t := r(o_{A,i}^t, o_{B,i}^t)$ be the bit at site i and time t. Compute the Rule 110 local rule

$$b_i^{t+1} \;=\; \delta(b_{i-1}^t, b_i^t, b_{i+1}^t), \qquad \delta = 110 \to 1,\ 101 \to 1,\ 011 \to 1,\ 010 \to 1,\ 001 \to 1,\ \text{else } 0.$$

Now choose a group element $g_i^t \in V_4$ to enforce the new readout:

$$g_i^t \;=\; \begin{cases} e \text{ or } s_As_B, & \text{if } b_i^{t+1} = b_i^t \quad \text{(preserve parity)}; \\ s_A \text{ or } s_B, & \text{if } b_i^{t+1} \neq b_i^t \quad \text{(toggle parity)}. \end{cases}$$

Apply the intrinsic update

$$(o_{A,i}^{t+1}, o_{B,i}^{t+1}) \;=\; g_i^t \cdot (o_{A,i}^t, o_{B,i}^t).$$

This uses only nearest–neighbor bits to compute b_i^{t+1} and a local V_4 action to realize it; hence, the induced dynamics on the readout $\{b_i^t\}_{i\in\mathbb{Z}}$ is exactly Rule 110.

By Cook's [2] theorem, such a lattice simulates a cyclic tag system and therefore a Turing machine. Consequently, the GL with THM_4 cells and the above encoding is Turing complete. □

2.5 The Shape of Computation

A classical bit is typically modeled as a two-point discrete state space, while a qubit has a canonical continuous pure-state space, namely the Bloch sphere. A geobit, by contrast, does not come equipped with a unique ambient state manifold of this kind. Its state space is determined only up to a chosen *encoding*: one may represent the same symbolic transition structure by different Cayley graphs, polytopes, or orthogonal symmetry realizations.

In the symbolic regime, the relevant object is therefore not a canonical "geobit sphere," but a finite transition system equipped with a symmetry action. For THM_4, this structure is governed by the Klein four group and may be visualized by a 3-dimensional Cayley graph. For the full symbolic model, one may ask for the smallest orthogonal representation in which the entire symmetry group acts faithfully.

Proposition 2.3 (Minimal Orthogonal Embedding Dimension for $(C_2)^4$)**.** *A faithful embedding of the symbolic symmetry group $(C_2)^4$ into an orthogonal rotation group requires at least $SO(5)$.*

Proof. Let $E \subset SO(n)$ be an elementary abelian 2-subgroup. Since every element of E has order 2, each element is an orthogonal involution and hence diagonalizable over $\mathbb{R}$ with eigenvalues in $\{\pm 1\}$. Because the elements of E commute, they are simultaneously diagonalizable. Therefore, after conjugation, E is contained in the subgroup of diagonal sign matrices

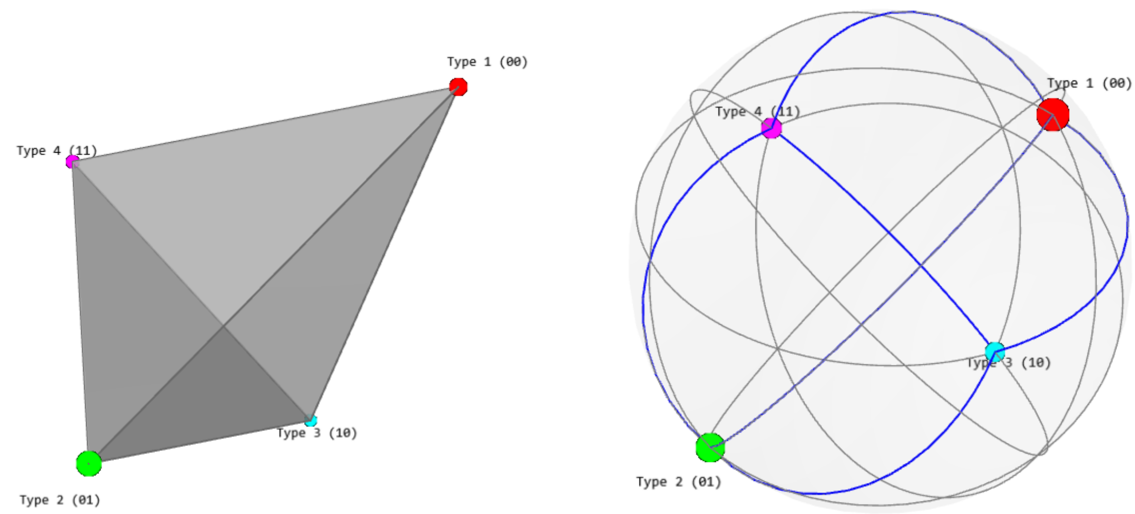
$$\Big\{\mathrm{diag}(\varepsilon_1, \ldots, \varepsilon_n) : \varepsilon_i \in \{\pm 1\},\ \varepsilon_1 \cdots \varepsilon_n = 1\Big\} \subset SO(n),$$

which is isomorphic to $(C_2)^{n-1}$.

It follows that every elementary abelian 2-subgroup of $SO(n)$ has rank at most $n-1$. In particular, $(C_2)^4$ cannot embed in $SO(4)$, since the largest such subgroup there has rank 3. On the other hand, in $SO(5)$ the determinant-1 diagonal sign subgroup is exactly

$$\Big\{\mathrm{diag}(\varepsilon_1, \ldots, \varepsilon_5) : \varepsilon_i \in \{\pm 1\},\ \varepsilon_1 \cdots \varepsilon_5 = 1\Big\} \cong (C_2)^4,$$

so the bound is sharp. □



(a) V_4 as a 3D Cayley graph (b) An analogous spherical representation

Figure 2: Encoding-dependent realizations of the V_4 symbolic state space.

Thus, although the underlying twin-helix geometry is naturally constructed in low dimension, a faithful orthogonal realization of its full symbolic symmetry requires a higher-dimensional ambient representation. This should not be interpreted as an intrinsic physical dimension of the manifold itself, but rather as a statement about the geometry of *state-space realization*. In particular, the "shape" of a geobit is not canonical: it depends on which observables, generators, and equivalence relations one chooses to encode. For visualization, we therefore restrict attention to the smallest symbolic cases, such as V_4, whose Cayley graph admits a simple 3-dimensional rendering.

Remark 2.3. The tetrahedral appearance of the V_4 state space is notable in light of Baez's discussion of the quantum tetrahedron [3] where similar state collapses were noted. In the present setting, however, the tetrahedral structure arises from a purely combinatorial symbolic transition system rather than from spin-network quantization. We return to issues of state space comparison in section 5.3.

2.6 The Logic of Geometry

In a GCM, logic may be encoded geometrically rather than symbolically. The most basic mechanism is *orientation*: once a preferred axis has been fixed, the phase of a unit helix around that axis determines a logical value. This enables a passage from binary logic to finite n–ary logic.

Orientation as a geometric alphabet. Fix a helical axis and let $\theta \in S^1 \cong \mathbb{R}/2\pi\mathbb{Z}$ denote the corresponding orientation phase. For each integer $n \geq 2$, partition the circle into n equal sectors and define the discrete readout

$$b(\theta) \;:=\; \left\lfloor \frac{n\,\theta_{\text{norm}}}{2\pi} \right\rfloor \in \{0, 1, \ldots, n-1\}, \qquad \theta_{\text{norm}} \in [0, 2\pi).$$

This induces an n–valued logic on the orientation channel, which we call *ω–logic*. Equivalently, one may regard the underlying continuous phase as

$$s(\theta) \;:=\; \frac{\theta_{\text{norm}}}{2\pi} \in [0, 1),$$

with the discrete alphabet obtained by thresholding s into n bins. In this way, binary, ternary, quaternary, and higher n–ary logics all arise from the same geometric mechanism: a cyclic phase together with a chosen readout.

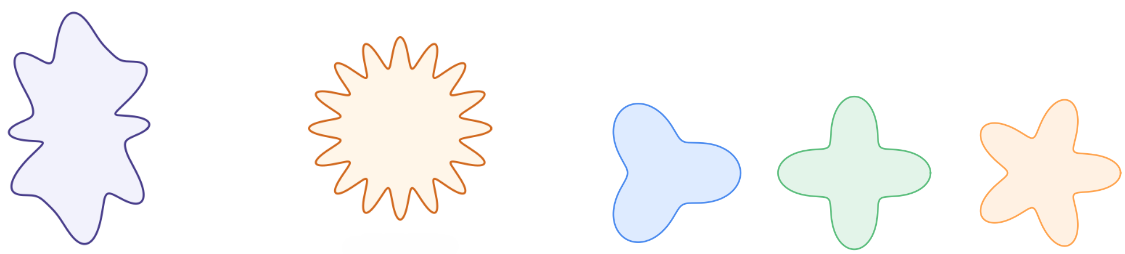
Definition 2.6 (ω–logic)**.** Let $n \geq 2$. The *n–ary orientation logic* of a unit helix is the map

$$\omega_n : S^1 \longrightarrow \{0, 1, \ldots, n-1\}, \qquad \omega_n(\theta) = \left\lfloor \frac{n\,\theta_{\text{norm}}}{2\pi} \right\rfloor .$$

When $n = 2$ this recovers binary logic; when $n > 2$ it gives finite n–ary logic; and if one retains θ itself rather than thresholding, one obtains the continuous $U(1)$ phase corridor. Table 7 organizes a table of models.

Table 7: The chart of n–logics

Geometric Expressiveness	**Join Symmetry**	**Algebraic Closure Group**	**Result**
Extreme	—	$\mathrm{Diff}(M)$ (Maximal)	Computational gauge field
High	1-fold (360°)	C_1	Geometric chaotic analog flow
Medium	2-fold (180°)	C_2	Classical Boolean logic
Low	n-fold ($2\pi/n$), finite	C_n	Finite n–ary logic
Minimal	Full rotational symmetry	$U(1)$ (Lie)	Algebraic fuzzy analog flow



(a) Expressiveness in binary vs. hexadecimal (b) n–logical gluing maps for 3, 4, and 5–ary logic

Figure 3: Cross-sectional shape families and the n–logic tradeoff.

A spectrum of physical logic. From the computational point of view, low-order cyclic logics such as C_2 or C_4 occupy a particularly stable region: they retain enough discrete algebraic structure to support verification and symbolic composition, while still allowing genuinely geometric realizations. At one extreme, the continuous $U(1)$ limit becomes an abelian phase medium; at the other, the loss of usable discrete closure drives the system toward geometric or chaotic analog behavior. In this sense, Boolean and low-order finite logic form a kind of *logical habitable zone* for physically stable symbolic computation.

Hence, even in an idealized, noiseless world, n-ary logical computers are "sandwiched" between two analog extremes. For n-ary logic, the cross-section of the THM can be any smooth closed manifold with at least n distinguished points on its boundary that realize the requisite "logical angularities." The primary family consists of sinusoidal perturbations ("clovers"), that can be globally smooth and are easily visualized in polar coordinates—see fig. 3b. However, one can readily imagine richer forms that realize the required angular structure. Figure 3a illustrates such alternatives and compares expressiveness across different logical regimes.

A geometric tradeoff. The passage from C_1 to larger cyclic groups and finally to $U(1)$ exhibits a *systematic tradeoff.* Increasing rotational symmetry enlarges the continuity of the orientation channel, but at the same time restricts the available shape space and weakens finite algebraic closure. Conversely, low-order discrete symmetry yields fewer phase values but stronger symbolic verifiability. This motivates the following lemma.

Lemma 2.4 (Geometric Logic Conservation Lemma). *Fix a THM G together with a chosen readout. Let $\mathcal{C}(G)$ denote the complexity of the induced algebra of local symbolic operations (for example, the size of the effective gate clone or finite symmetry algebra visible under the readout), and let $\mathcal{E}(G)$*

denote the corresponding geometric expressiveness (for example, the number of orbit-distinguishable cross-sectional shapes or phase patterns available at fixed regularity). Then one expects a bounded tradeoff of the schematic form

$$\mathcal{C}(G) + \mathcal{E}(G) \lesssim \Lambda(G),$$

where $\Lambda(G)$ depends on the ambient geometric constraints and symmetry class of the model.

Remark 2.4. The essential point is that the symmetry of the system's configuration space directly induces an invariant quantity in the system's dynamics. Every symmetry break in topology introduces a logical ambiguity, and vice versa. Although we are only in the initial regime of computation, we will eventually encounter higher-order manifolds that devolve into pure fluidic computability, leaving discrete logic altogether and entering increasingly abstract and chaotic forms of analog computation.

Definition 2.7 (Geometric Expressiveness at Finite Cutoff)**.** Let $\mathcal{F}_K$ denote the space of real trigonometric polynomials of degree at most K,

$$r(\theta) = a_0 + \sum_{k=1}^{K} \big(a_k \cos(k\theta) + b_k \sin(k\theta)\big),$$

interpreted as smooth 2π–periodic radial functions. Let $\mathcal{F}_{n,K} \subseteq \mathcal{F}_K$ be the subspace of functions invariant under rotation by $2\pi/n$:

$$r(\theta + 2\pi/n) = r(\theta) \qquad \forall\, \theta.$$

The *relative geometric expressiveness* at symmetry level n and cutoff K is

$$E_K(n) \;:=\; \frac{\dim(\mathcal{F}_{n,K})}{\dim(\mathcal{F}_K)}.$$

Proposition 2.5 (Symmetry Collapses Cross-sectional Expressiveness)**.** *For fixed cutoff K,*

$$\dim(\mathcal{F}_{n,K}) \;=\; 2\Big\lfloor \frac{K}{n} \Big\rfloor + 1, \qquad E_K(n) = \frac{2\lfloor K/n \rfloor + 1}{2K+1}.$$

In particular, for fixed K, $E_K(n)$ decreases monotonically with n, and

$$E_K(n) \to \frac{1}{2K+1} \qquad \text{as } n \to \infty.$$

At the level of smooth functions (without cutoff), full rotational symmetry forces the radial function to be constant, hence the cross-section is a circle.

Proof. If

$$r(\theta) = a_0 + \sum_{k=1}^{K} \big(a_k \cos(k\theta) + b_k \sin(k\theta)\big)$$

satisfies $r(\theta + 2\pi/n) = r(\theta)$ for all θ, then only those Fourier modes with frequency divisible by n may remain. Hence the allowed modes are $k = 0, n, 2n, \ldots, \lfloor K/n \rfloor n$, giving

$$\dim(\mathcal{F}_{n,K}) = 1 + 2\Big\lfloor \frac{K}{n} \Big\rfloor.$$

Since $\dim(\mathcal{F}_K) = 2K + 1$, the formula for $E_K(n)$ follows immediately. In the full smooth category, invariance under all rotations implies that only the constant Fourier mode survives, so the radial profile is constant, and the cross-section is a circle. □

Progression to algebraic analog flow. This is the essential phenomenon as the size of the rotational symmetry group grows. Even as the symmetry group becomes continuous, the system's logic remains identical to its geometry. This is the core idea behind geometric computation, and why such systems are not limited to discrete operations.

Definition 2.8 (χ–logic)**.** Let $\mathcal{C}$ be a class of framed geometric configurations equipped with a continuous *chiral observable*

$$\kappa : \mathcal{C} \longrightarrow \mathbb{R},$$

where κ may be realized, for example, by signed torsion, signed twist density, or another orientation-sensitive local invariant. The associated *chirality logic* is the map

$$\chi : \mathcal{C} \longrightarrow \{-1, 0, +1\}, \qquad \chi(x) := \operatorname{sgn}(\kappa(x)).$$

Thus:

- $\chi(x) = +1$ denotes a right-handed configuration,
- $\chi(x) = -1$ denotes a left-handed configuration,
- $\chi(x) = 0$ denotes the non-chiral or degenerate locus where the chiral invariant vanishes.

When the value 0 is admitted as a stable readable state, χ–logic is ternary. When the degenerate locus is excluded from the admissible state space, χ–logic reduces to a binary left/right channel.

The role of chirality. Unlike orientation, chirality is not naturally a cyclic n–ary phase coordinate. Geometrically, changing handedness is not a small rotation in a compact group; it amounts to changing the sign of a chiral invariant, such as torsion for a framed centerline or the sign of a local twist density. Consequently, any continuous interpolation between left- and right-handed configurations must pass through the non-chiral locus on which the relevant invariant vanishes.

At the level of coarse readout, this produces a natural three-way distinction: left-handed, right-handed, and degenerate/non-chiral. In the present paper, however, the twin helix register itself is taken to operate only on the nondegenerate binary sector

$$\{-1, +1\} \subset \{-1, 0, +1\},$$

so that the value 0 functions only as a boundary or transition state rather than as a stable logic value. This does not preclude other classes of GCMs from using the full ternary chirality space as a genuine computational channel; indeed, later gauge-theoretic examples (see appendix A.1) may best be understood in precisely that broader sense. This is the sense in which chirality forms a qualitatively different channel from orientation: ω–logic supports a rich cyclic phase structure, whereas χ–logic is organized by sign and degeneracy.

2.7 Syntax, Semantics, and Computation

We now synthesize three convergent lines of thought, alongside our geometric model.

First, following Felleisen's account of expressiveness [4], we distinguish *micro-* from *macro-expressiveness*: adding new local states or notations versus adding genuinely new global, compositional power.

Second, in the compositional semantics of processes of Categorical Quantum Mechanics: Abramsky–Coecke, with graphical calculi surveyed by Selinger, the meaning of a composite is determined by the wiring of its parts; contextuality and entanglement arise as *compositional* phenomena rather than add-ons [5, 6]. This gives us a precise, syntax-agnostic way to talk about how meanings compose.

Third, our geometric/groupoid models instantiate these ideas concretely: GCMs and decorated paths make composition, context, and macrostate emergence *geometric*. Macrostates are equivalence classes of geometric/computational microstates; macro-expression corresponds to traversing or constructing new geometric cobordisms.

Finally, in the spirit of Baez–Stay's "Rosetta Stone" perspective on physics, topology, logic, and computation [7], we regard computation itself as a process theory: objects are interfaces, morphisms are computations, and composition is gluing. In this language, the familiar tradeoffs become geometric/categorical phenomena rather than informal slogans. We do not merely interpret computation using geometry; we *construct the geometry of computation itself.* The following definitions guide our landscape.

Definition 2.9 (Micro-Expression)**.** A language extension is *micro-expressive* w.r.t. a base language L if every program P in the extended language L' can be mechanically translated (compiled/macro-expanded) to a program P_0 in L without changing its semantic structure. Formally: for all $P \in L'$, there exists a derivable $P_0 \in L$ with equivalent semantics; the extension is syntactic sugar, not added power.

Definition 2.10 (Macro-Expression)**.** A language extension is *macro-expressive* if it enables programs or control/data abstractions that cannot be obtained from the base language by any mechanical or structure-preserving translation. Formally: there exists $P \in L'$ such that no equivalent $P_0 \in L$ can be constructed by a uniform, semantics-preserving translation.

Felleisen's framework makes this distinction precise and operational [4], and we adopt it as the semantic backbone for what follows. They define precise criteria for when a language extension is merely syntactic (micro-expressive, or "eliminable by mechanical translation") versus when it adds genuine new expressive power (macro-expressive, or "semantically irreducible"). Just as in programming, where some features create behaviors not reducible to simpler forms—gluing submanifolds can create new topological invariants only if the gluing is smooth (singularity-free).[2] As systems grow more complex, obstructions to smooth composition proliferate, paralleling the limits of macro-expression in computation.

Definition 2.11 (Verifiability)**.** Let $\mathcal{S}$ be a computational/semantic system with predicates $\mathcal{P}$ over its states/behaviors. The *verifiability* $\mathcal{V}(\mathcal{S})$ is a normalized measure of how many *nontrivial* properties in $\mathcal{P}$ admit effective, pre-execution decision procedures. Equivalently: $\mathcal{V}(\mathcal{S})$ measures the size/structure of the subset of $\mathcal{P}$ consisting of properties decidable by algorithmic, structural, or analytic means (e.g., checkable invariants or geometric signatures), without brute-force evaluation.

Verifiability captures the capacity to certify properties *by construction.* In algebraic regimes, it correlates with rigid "ledgers" (e.g., line bundles/cohomology classes that distinguish states). Still, it is not identical to micro-expression: the former is semantic/algorithmic, whereas the latter is syntactic/generative. High verifiability corresponds to systems where many global or local properties can be decided by construction; low verifiability signals the presence of undecidable, chaotic, or non-algorithmic behavior.

In the process semantics of Abramsky–Coecke, contextual/entangled behavior is a *compositional* effect of how processes wire together. Our geometric program inverts the usual narrative: start from deterministic geometric computation, then see probabilistic/quantum-like structure *emerge* as the statistical/compositional shadow of complex gluing and flow.

Definition 2.12 (Tractability)**.** Let $\mathcal{S}$ be a formal system (e.g., a programming language, a computational model, or a geometric computation system). Tractability is the property that there exists a procedure (algorithmic or semantic) to:

- For machines: Execute or evaluate a statement/program/configuration in $\mathcal{S}$ with feasible computational resources (e.g., polynomial time or some agreed practical bound).
- For humans: Understand, reason about, or verify the intent and behavior of statements in $\mathcal{S}$ without overwhelming cognitive complexity.

More precisely: A property or decision problem in $\mathcal{S}$ is tractable if it can be decided or realized with computational resources (time, space, etc.) bounded by a polynomial (or similarly 'practical") function of the input size. A system is tractable if most of its intended computations and the process of programming/understanding them fall within humanly or machine-manageable complexity.

Tractability, in our framework, refers to the ease with which both a machine can execute and a human can understand or reason about a computational statement in a formal system. For a programming language or system of symbols, tractability is both a computational and a cognitive phenomenon. The degree of tractability depends not only on algorithmic complexity but also on the alignment between the system's formal structure and the way meaning is constructed and communicated among its users.

With these parts in place, we can now *macro-express* a new theorem.

[2]The geometric mismatches noted earlier explain why, in the language of Feynman diagrams, certain gluings can fail: the logic of the bordisms must agree.

Conjecture 2.6 (**CORE THESIS: Expressiveness/Verifiability/Tractability ("EVT") Hypothesis**). *The conjecture formally comprises two parts:*

(A) ***Unverifiability of Macro-expressive Systems.*** *Any sufficiently macro-expressive system of physical computation generically loses global algorithmic verifiability unless additional structural restrictions are imposed.*

(B) ***Expressiveness/Verifiability/Tractability Trilemma.*** *Within such a system, every deterministic computational model is fundamentally constrained by a three-way tradeoff among* expressiveness *(geometric or semantic generality),* verifiability *(algebraic checkability or formal runtime assurance), and* tractability *(algorithmic or computational efficiency). These attributes satisfy the following trilemma:*

> **No system can simultaneously maximize all three properties.** For any given model, at most two of the three can be optimized beyond a critical threshold, while the third must necessarily be sacrificed.
>
> *Formally, the achievable combinations of these properties are bounded by a 2-dimensional simplex in the space of attributes. The precise tradeoff is determined by the system's local symmetries and structural constraints.*

This is the central motivating result of this work. We take the phenomenon isolated in lemma 2.4 and generalize it from our geometric setting to physical deterministic computation more broadly. The claim is not that *all* computation is uniformly irreducible, nor that undecidability appears by mere complexity or by the use of a particular alphabet. Rather, we claim that there is a *structural threshold* in the design space of deterministic computational substrates.

The first half of the EVT Hypothesis should be read as a selection principle on substrate classes. It identifies which structural features drive a computational system toward global semantic opacity, and which restrictions recover verifiability. Later, in definition 5.2, we give a more precise geometric formulation of *macro–expressiveness*; for now, the term should be understood informally in the sense suggested by Felleisen: a substrate is macro-expressive when it supports genuinely new global compositional behavior rather than merely new local notation.

The key diagnostic is Rice's theorem [8]. In ordinary computability theory, Rice tells us that every non-trivial extensional semantic property of partial computable functions is undecidable. In the present framework, this is not treated as an isolated pathology of Turing-complete languages, but as the generic outcome of crossing the abelian threshold of lemma 2.4. Once a deterministic substrate supports self-interpretation and free composition, semantic predicates are forced into the same diagonal trap as halting.

To state the thesis informally:

> *Whenever a deterministic computational substrate encodes programs and data within a common semantic universe, supports self-reference through an evaluator-like mechanism, and interprets properties extensionally on computed behavior, Rice-style undecidability becomes generic.*

This statement does *not* replace Rice's theorem. Rather, it explains how to recognize in advance when Rice-type behavior should be expected to apply to systems *outside of typical discrete logic.* Computer science already possesses many local explanations of undecidability—via reductions, recursion-theoretic constructions, and semantic impossibility results. What the EVT contributes is a *global packaging* of these phenomena as a tradeoff in substrate design. It says, in effect, that once enough expressive freedom is admitted, global verifiability is no longer stable.

Why macro-expression? Felleisen's distinction between *micro-* and *macro-expressiveness* is essential for our purposes because the present framework is *not* restricted from the outset to digital syntax, natural number objects, or Turing-style encodings. If one begins directly with Kleene-style recursion-theoretic machinery, one has already assumed a symbolic regime rich enough to support self-interpretation. Our problem is prior to that: we need a language for describing when a *physical* or *geometric* substrate acquires genuinely new global computational power, independently of whether that power is presented discretely, continuously, or in mixed form.

Felleisen provides exactly this level of abstraction. His notion of *macro-expressiveness* isolates those extensions that do not merely add more local states or notation, but enlarge the global compositional possibilities of the system itself. This is the right conceptual tool for our purposes, because the onset of undecidability in our setting is not driven by the size of the local alphabet, nor by the mere richness of local geometry, but by the appearance of new *global* forms of composition. In our framework, this occurs precisely at the transition into non-abelian regimes, where order-sensitive transport, path dependence, and conjugation-sensitive observables become available.

Microstates, macrostates, micro-expression, and macro-expression. It is crucial to distinguish the state-based and expressive notions. A *microstate* is a local configuration of the substrate: a specific geometric, algebraic, or dynamical arrangement. A *macrostate*, by contrast, is an equivalence class of such configurations under the observational or symmetry relations relevant to the model. Thus, the micro/macrostate distinction is *ontic* or *configurational*: it concerns what states the system can occupy and how those states are grouped.

The distinction between *micro-expression* and *macro-expression* is different. It is *operational* and *compositional*. Micro-expression concerns the creation of new local states, labels, or local update rules without changing the global expressive capacity of the substrate. Macro-expression concerns the introduction of genuinely new global compositional freedoms: new ways of combining processes, transporting information, or building observables that cannot be reduced to the previous regime by a uniform local translation. In this sense, microstates/macrostates describe *what exists*, whereas micro-/macro-expression describes *what can be done*.

This distinction is exactly what the first half of the EVT must track. The relevant threshold is not the proliferation of microstates; a system may admit a very large or even continuous local state space while remaining compositionally tame. What matters is the onset of macro-expression: the point at which the substrate acquires new global compositional freedoms that support self-reference, order-sensitive control, or nontrivial semantic transport. In symbolic computation, this threshold is realized by evaluator-like mechanisms and unrestricted self-application. In geometric computation, it is realized by the transition from abelian to non-abelian structure.

Accordingly, the role of Felleisen in this paper is not to supply a recursion-theoretic proof of Rice's theorem. Rather, it is to identify the correct *threshold variable*. Rice-style undecidability does not appear simply because a substrate is rich, continuous, or high-dimensional. It appears when the substrate becomes macro-expressive in Felleisen's sense. Our contribution is to geometrize this observation: in our setting, the jump to macro-expression coincides with the onset of non-abelian compositional structure, and it is there that verifiability begins to fail in the precise sense articulated by the EVT.

Why alphabet size is not the issue. Recall the spectrum of n-logics generated from join symmetry in table 7. It is tempting to attribute the ubiquity of undecidability to binary logic alone: a two-symbol alphabet offers high compositional freedom while imposing very little local algebraic discipline. But the arity of the local alphabet is not the source of the Rice-style barrier; this is driven by *macro*-expressiveness, not by the number of local symbols.

Any finite n-ary digital substrate recovers the same Rice-style barriers as soon as it supports conditionals, memory, self-description, and unbounded composition. Ternary or quaternary hardware may enlarge the local state space, but they do not, by themselves, neutralize diagonalization. Put differently: richer alphabets change *micro*-expressiveness, while Rice is controlled by *macro*-expressiveness.

In principle, one could seek to purchase more verifiability in hardware by enforcing a richer local symbolic structure. In practice, device physics, noise margins, and error-correction heavily favor binary encodings. More importantly, even when local alphabets are enriched, universality and semantic opacity reappear as soon as the substrate admits unrestricted global composition. To mitigate Rice in a principled way, one must instead withdraw some macro-expressive feature: forbid general recursion, stratify effects, impose linear/affine discipline, or adopt total calculi. This is exactly how systems inspired by Coquand's Calculus of Constructions [9] recover semantic control: they do not eliminate local richness, but they constrain the global forms of self-application and composition that would otherwise make undecidability ambient.[3]

Mathematics and computation are not symmetric. There is a genuine parallel here with Gödel's incompleteness theorem, though the two results live on different sides of the logic/computation divide. Gödel says that sufficiently strong recursively axiomatized theories cannot prove all true arithmetical statements; Rice says that sufficiently expressive computational substrates cannot decide all non-trivial semantic properties of their own behaviors. Both are powered by self-reference. The point of the EVT is not to collapse these theorems into one another, but to place Rice in the broader context of substrate design: undecidability becomes generic not simply because "Turing machines are strange," but because certain structural freedoms inevitably reactivate diagonalization.

Rice's theorem feels harsher than Gödel's in practice. In mathematics, undecidability is often buffered by deliberate axiomatic discipline: many working theories are complete or decidable, and even incomplete ones frequently permit large islands of stable reasoning. In computation, by contrast, the default architecture of universal systems collapses program and data into a single semantic domain and allows unbounded self-application. That design choice places computation much closer to the edge of diagonalization than ordinary mathematical practice.

This is the real "impedance mismatch" between the two domains. It is not that computation is somehow metaphysically more dangerous than mathematics, but that mainstream computational models were deliberately engineered to maximize expressive freedom. Once one notices this, Rice no longer looks like an arbitrary theorem about machine languages; it looks like a warning label attached to a certain region of the design space.

Self-reference and calibration as substrate features. The deeper lesson is that self-reference is not merely an artifact of syntax. Its precise categorical expression may vary, but on the computational side, the operational content is clear: once a substrate can encode an evaluator acting on representations of its own processes, self-reference becomes an ambient feature of the system rather than an exceptional construction. Kleene's Recursion Theorem and the s-m-n theorem make this explicit for classical computation, while Turing's halting argument provides the canonical example. The first half of the EVT transports this observation from a theorem about one symbolic model to a structural criterion on computational substrates more generally.

Accordingly, the central question is not whether a system is merely "complex," but whether it has crossed the point at which its own semantic machinery can be re-fed into itself. Beyond that threshold, diagonalization is no longer a special trick; it becomes part of the ambient computational environment.

Rice's theorem is often stated more broadly than its hypotheses warrant. Its full force depends on extensionality together with a sufficiently rich self-referential structure. When those assumptions are withdrawn—for example, in total, strongly normalizing, staged, or heavily stratified systems—large classes of semantic questions become decidable, even though some equivalence problems may remain hard. For this reason, the first half of the EVT should be read as a *calibration principle*: increasing macro-expressiveness pushes a substrate toward semantic opacity, while recovering verifiability requires withdrawing the structural conditions that regenerate diagonalization.

In this sense, the claim is not that all interesting systems are irreducible. Rather, it is that certain architectural choices place a system in an irreducibility-like regime, while others—typing disciplines, totality restrictions, staged semantics, and strong normalization—restore islands of semantic control by limiting the forms of self-application and composition available to the substrate.

The second half of the EVT Hypothesis sheds light on the odd "dual nature" observed in functional programming languages. On the one hand, languages with strong static typing offer stronger compile-time reasoning principles (e.g. type safety and, in suitable fragments, parametricity), at the cost of meta-linguistic restrictions during construction (Haskell). On the other hand, dynamically typed languages maximize meta-linguistic expressiveness at construction time (macros, reflection), while giving up static guarantees at runtime and shifting assurance to testing and contracts (Scheme).

Bridging attempts proceed along multiple axes: *gradual typing* (Typed Racket) inserts types with higher-order contracts at typed/untyped boundaries, and *staging* (MetaOCaml) separates compile-

[3] One may heuristically regard an n-ary geometric computer as a kind of "hardware proof assistant": the local state space may be rich, but semantic control comes only from global discipline, not from alphabet size alone.

time code generation from runtime execution. From the proof-theoretic direction (Coquand), total dependent type theories recover strong guarantees by forbidding general recursion. From the pragmatic language research side, (Felleisen) shows how large untyped codebases can be migrated toward sound types via occurrence typing and higher-order contracts—at the cost of explicit blame and potential runtime overhead—an archetypal EVT tradeoff between meta-linguistic freedom and verifiability. Here, the goal is not to re-formalize the trilemma, but to show how the second half of the EVT manifests concretely in real language design. When expressive features are pushed far enough, the same structural tension repeatedly reappears in practice through a small number of recognizable engineering symptoms:

The limits of expressive semantics. These bridges are inherently delicate. When many powerful features coexist, three recurring pressure points tend to appear:

(a) **Static complexity.** Extremely rich type features (GADTs, type families, dependent types) that can render inference undecidable, forcing annotations; combining dependency with effects or staging threatens phase separation unless carefully constrained.

(b) **Compile-time liveness.** Macros or multi-stage generation to diverge at compile time; expressive compile-time evaluators reintroduce the very fixed-point phenomena the EVT warns about.

(c) **Runtime boundaries.** Gradual typing preserves type safety via contracts, but introduces blame and potentially unbounded overhead; interoperation across typed/untyped modules trades static guarantees for dynamic checks.

These are not accidents of particular implementations; they are manifestations of the EVT tradeoff: increased *macro-expressiveness* (reflection, self-interpretation, unconstrained composition) strains *global verifiability*, unless restrictions restore structure. Hence, an explicitly stated *conservation law for physical computational behavior* explains these difficulties.

The reason this theorem has long gone unformulated is that its essential tradeoff has long been hidden in plain sight—embedded in the very architecture of classical computation we all use in our day-to-day lives. The Von Neumann architecture, by treating program and data uniformly and allowing maximal flexibility (any memory can store code or data, and programs can modify themselves), enables *extraordinary* expressiveness. However, this very flexibility comes at the cost of global verifiability: reasoning about whole-system properties becomes fragile, partial, or expensive unless additional structure is imposed. When the hardware is instead fixed algebraically—restricting the set of possible states, operations, or transitions—the system gains predictability and becomes globally checkable, but loses expressive power.

The tradeoff does not assert a single conservation equality. There exist regimes in both classical (language-based) and physical (geometric) computation with both low expressiveness *and* low verifiability—the infamous "Turing tarpit," (e.g., bash, Malbolge), where everything is theoretically possible but practically inexpressible. The second half of the EVT is best read as a *no-free-lunch inequality* with phases: beyond a threshold expressiveness, Rice-type barriers are pervasive; and to regain verifiability, one must withdraw some of that structure. Between these extremes lie transitional corridors in which small feature additions can cause qualitative shifts in what can be verified.

Because these tradeoffs depend on ambient structure, the EVT treats *local* symmetries of a model or architecture rather than postulating a single global invariant across all substrates. In a Von Neumann setting, some of the most important local regularities are cognitive and conventional: programming idioms, abstractions, and composition patterns induce de facto invariants that temporarily "abelianize" parts of the system, while reflective or unrestricted features reintroduce non-commuting structure. Pushing too far past the barriers above yields systems that remain formally expressive but become progressively less usable in physical and human practice.[4]

Removing barriers to viewing computation. Regarding tractability, the present paper intentionally abstracts away many ordinary engineering bottlenecks. In an idealized geometric regime, one may treat the substrate as reconfigurable enough that the sharp distinction between hardware and program becomes less natural than it is in fixed Von Neumann architectures. This does *not* remove

[4]A fuller account of how the EVT Trilemma manifests in macrosopic programming practice is given in [10]. This outlines the stated connections to resource theory and can be read as a companion piece.

physical resource limits; rather, it allows the expressiveness/verifiability relation to be studied more directly, without immediately collapsing into familiar questions about instruction sets, compilers, and implementation overhead. For this reason, although the full trilemma must be stated as such, we do not pursue tractability in equal depth here: in the present paper, it functions chiefly as a reminder that human-built machines impose additional constraints beyond those intrinsic to the underlying geometric substrate.

So in our GCM-based setting, the expressiveness/verifiability relationship can be studied more transparently because there is *no fixed separation between logic and memory* of the usual Von Neumann sort. A GL is an idealized *non*-Von Neumann substrate defined directly in geometric and algebraic terms. By fusing logic and language into a single analog-geometric object, one can *see* how the operational semantics deforms the computational substrate, and model how the algebraic closure group constrains the geometry with every increasing bit!

This tension between expressive freedom and structural discipline is not unique to programming languages; it is a recurring theme whenever algebraic power begins to dominate geometric intuition. As Baez writes in *Getting to the Bottom of Noether's Theorem:* [11]

> "Atiyah [12] warned us of the dangers of algebra: *" ...algebra is to the geometer what you might call the Faustian offer. As you know, Faust in Goethe's story was offered whatever he wanted (in his case the love of a beautiful woman), by the devil, in return for selling his soul. Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvellous machine.*"

Baez notes that "While this is sometimes true, algebra is more than a computational tool: it lets us *express* concepts in a very clear and distilled way." We agree that the finest mathematics arises when algebra and geometry are brought into balance. But our contention here is that computation is one of the settings in which this balance becomes unusually unforgiving. Within an ordinary formal background such as ZFC, geometry and algebra can coexist without tension; but once a geometric system is required to *compute*, the act of doing so (in our case—the "gluing") *enforces constraints* on *how* the flow can be structured. The bargain, in this case, is not rhetorical but structural, and the devil shall not be cheated.

But why physics? It is reasonable for readers to ask why a theorem about computation is applied in a mathematical physics paper. Why do this? Two pragmatic reasons guide this choice. First, differential geometry and geometric analysis provide a language in which *flows, invariants, and gluing* are native: variational principles, conservation laws, and well-posed evolution already live there. Second, many of the sharpest constraints on computation are *physical*—finite precision, energy/entropy budgets, and measurement back-action—so a formulation that is testable against laboratory and numerical practice is preferable to one that remains purely syntactic (e.g. in terms of abstract syntax trees and Galois connections).

This does not diminish the interest of the EVT Hypothesis for computer science, education, or the philosophy of natural laws. Versions of the underlying tension have appeared in many domains, though rarely isolated as a single, explicit selection principle. A few recurring patterns (illustrative, not exhaustive) are:

- **Logic and programming.** As previously stated, total/strongly normalizing calculi gain verification power at the price of general recursion; reflective/metaprogramming gains flexibility at the cost of global guarantees.
- **Numerics and PDEs.** Stability constraints (e.g. CFL-type conditions) and structure-preserving schemes (e.g. symplectic integrators) enforce invariants but restrict admissible updates; aggressive damping improves robustness while destroying time-reversibility and fine invariants.
- **Many-body/field theory.** Integrable corners admit exact charges and factorization but are measure-zero in parameter space; generic regimes are expressive yet analytically resistant. Undecidability phenomena (e.g. spectral-gap questions) underscore that some semantic properties of dynamics cannot be decided uniformly.

But the most pressing applications of a *computational* selection principle are presently in *quantum gravity* and *lattice gauge theory*, where one must extract dynamics from large, ongoing physical computations (experiments or simulations) under resource and precision constraints. The EVT Trilemma is intended to bound *how* such processes can behave and what can be recovered from them, independently of any particular quantization scheme.

Mathematics has Gödel–Tarski style incompleteness, and computer science has the halting problem and Rice-type no-go results. Physics, despite its rich family of conservation laws, lacks a comparably explicit bound on the *simultaneous* attainability of global verifiability and unrestricted expressive power in *physical* information-processing. Various desiderata for such a bound have been voiced (e.g. calls for a "Gödelization" of fundamental theory in public remarks), and adjacent programs (such as Constructor Theory [13]) have sought axioms phrased in terms of counterfactual capabilities; our starting point diverges in ontology but shares the motivation.

The EVT Hypothesis is proposed as a *selection principle* for the behavior and modeling of computational processes: physical computation—discrete or analog—is not substrate-independent magic; it is a *gauge-constrained flow* carried by a geometric substrate, with invariant quantities and resource monotones that cannot be simultaneously optimized without limit. From this viewpoint, the structures that make computation possible (*how* we compute) are as consequential as the extensional outcomes (*what* we compute). Any physical theory of computation should, at a minimum, respect this geometric constraint.

3 Kleene Computation and Geometry

Stephen Kleene's work in the 1930s–40s [14] established the modern theory of computability via the partial recursive functions, providing a machine-independent foundation for classical computation. In this section, we show that the discrete sector of GCT recovers this classical framework. The purpose is not to reduce geometric computation to symbolic recursion, but to verify that GCT contains ordinary computability as a special case before passing to the richer geometric regime.

Our strategy is constructive. We first interpret the primitive recursive operations inside our geometric category, and then extend this interpretation to partial maps to capture Kleene's μ-operator. This yields a precise symbolic boundary for the theory: GCT agrees with classical computability in its discrete, NNO-based regime, while the latter geometric and groupoid-based sections develop the broader structure that goes beyond it. It is not intended as the final formal basis for later regimes, whose design is instead total and effect-controlled (cf. section 5.5).

3.1 Fibered Kets and Product States

To track both observable and internal degrees of freedom of a geobit, we introduce a notation reminiscent of definition 2.4, but adapted to the present geometric setting. This notation is purely organizational: it packages a pair of states, one observable and one internal. Throughout this section, the symbol $\otimes$ is mnemonic and denotes the cartesian product of state spaces.

Definition 3.1 (Fibered Ket for Geobits)**.** Let A be the finite observable state space encoding chirality and orientation, and let B be the internal state space of fluidic or other variables. The associated *state object* is

$$S := A \times B.$$

A *fibered ket* is the notational pairing

$$|\alpha\rangle \otimes |\beta\rangle \;\widehat{=}\; (\alpha, \beta) \in S,$$

where:

(a) $|\alpha\rangle \in A$ encodes the observable component;

(b) $|\beta\rangle \in B$ encodes the internal component;

(c) S is an object of the geometric category $\mathcal{G}$, equipped with a groupoid of admissible local symmetries (see section 4.5).

An elementary update, or local move, is a morphism $T : S \to S$, written

$$(|\alpha\rangle \otimes |\beta\rangle) \longmapsto |\alpha'\rangle \otimes |\beta'\rangle\,.$$

We write $G \triangleright T$ for the right action of an update T on a state $G \in S$, so that $G \triangleright T := T(G)$. When T factors by components, we write $T = T^\alpha \otimes T^\beta$ with $T^\alpha : A \to A$ and $T^\beta : B \to B$; more generally, T may couple the two factors.

Remark 3.1. In this section, the symbol $\otimes$ in the fibered-ket notation denotes the cartesian product in $\mathcal{G}$, not a tensor product in a linear or Hilbert-space sense. No inner-product structure is assumed here.

Operationally, we treat a GCM as an object subjected to computational flows, namely endomorphisms of S generated by discrete moves or finite vector-field updates. The recursor introduced below iterates such updates in a manner matching Kleene's schemes of composition and primitive recursion. In the categorical summary at the end of the section, this will appear as preservation of products and of the natural-numbers-object recursor in $\mathcal{G}$, with partiality handled in the associated partial-maps setting.

3.2 Partial Recursive Functions—Primitives and Composition

Classically, the partial recursive functions over $\mathbb{N}$ are generated from three primitives:

Zero $Z(\vec{x}) = 0$.

Successor $S(x) = x + 1$.

Projections $P^n_k(x_1, \dots, x_n) = x_k$.

and three closures:

Composition Given $f : \mathbb{N}^k \to \mathbb{N}$ and $g_i : \mathbb{N}^n \to \mathbb{N}$, define

$$h(\vec{x}) = f\big(g_1(\vec{x}), \dots, g_k(\vec{x})\big).$$

Primitive Recursion Given $f : \mathbb{N}^m \to \mathbb{N}$ and $g : \mathbb{N}^{m+2} \to \mathbb{N}$, define $h : \mathbb{N}^{m+1} \to \mathbb{N}$ by

$$h(0, \vec{x}) = f(\vec{x}), \qquad h(n{+}1, \vec{x}) = g\big(n,\, h(n, \vec{x}),\, \vec{x}\big).$$

Unbounded μ–Recursion Given $g : \mathbb{N}^{n+1} \to \mathbb{N}$, define the partial map $f : \mathbb{N}^n \rightharpoonup \mathbb{N}$ by

$$f(\vec{x}) = \mu\, y\ \big[\, g(\vec{x}, y) = 0 \,\big],$$

i.e. the least y (if any) with $g(\vec{x}, y) = 0$.

We begin with the three primitive constructors. Let $(N_G, 0_G, S_G)$ be an NNO in $\mathcal{G}$, and write N_G^n for finite products.

Zero. For each arity n, the Zero map is the constant arrow

$$Z_n := !\,;\, 0_G \ :\ N_G^n \longrightarrow 1 \longrightarrow N_G.$$

Example 3.1 (Zero as a Reset). Fix a readout $\mathrm{tw} : B \to N_G$ extracting a discrete orientation count (or any integer-coded internal variable), together with a setter $\mathrm{set0} : B \to B$ that overwrites this count with 0. Define the local move $T_0^\beta : B \to B$ by

$$T_0^\beta(\beta) = \mathrm{set0}(\beta),$$

and let it act trivially on A. Then

$$(|\alpha\rangle \otimes |\beta\rangle)\ \triangleright\ (\mathrm{id}_A \otimes T_0^\beta)\ =\ |\alpha\rangle \otimes |\,\mathrm{set0}(\beta)\,\rangle\,.$$

At the controller level, the associated numeric output is $Z_n(\vec{x}) = 0_G$, independent of the input. In this sense, the Zero function is realized geometrically as a reset of the relevant internal register.

Successor. The Successor map is simply

$$S_G \;:\; N_G \longrightarrow N_G.$$

Example 3.2 (Successor as a Single Rotation). Let $\mathrm{tw} : B \to N_G$ be as above, and let $\mathrm{inc} : B \to B$ implement a unit increase in the relevant discrete variable:

$$\mathrm{tw}(\mathrm{inc}(\beta)) = \mathrm{tw}(\beta) + 1.$$

Define $T^{\beta}_{+1} : B \to B$ by $T^{\beta}_{+1} = \mathrm{inc}$, again acting trivially on A. Then

$$(|\alpha\rangle \otimes |\beta\rangle) \,\triangleright\, (\mathrm{id}_A \otimes T^{\beta}_{+1}) \;=\; |\alpha\rangle \otimes |\, \mathrm{inc}(\beta)\,\rangle\,.$$

On the numeric side, this is governed by $S_G : N_G \to N_G$, i.e. the Successor on the chosen orientation or twist counter.

Projections. For $1 \le k \le n$, the Projection maps are the Cartesian projections

$$\pi^n_k \;:\; N^n_G \longrightarrow N_G, \qquad \pi^n_k(x_1, \ldots, x_n) = x_k.$$

Example 3.3 (Projection as Readout/Selection). Given an input tuple $(x_1, \ldots, x_n) \in N^n_G$ serving as control parameters—for instance, planned numbers of twists, swaps, or steps—the projection π^n_k selects the kth control:

$$\pi^n_k(x_1, \ldots, x_n) = x_k.$$

Operationally, one can route $\pi^n_k(\vec{x})$ to an update family $\{T_y\}_{y \in N_G}$ to choose which local move is applied next. Projection itself does not alter the current geobit state; it simply selects the relevant control datum.

Primitive Composition. Composition is interpreted as the sequential assembly of local controllers and updates. Write $\pi_A : S \to A$ and $\pi_B : S \to B$ for the projections from the state object $S = A \times B$.

Fix a total map

$$v : B \to B,$$

thought of as a one-step update, and define

$$\beta' := v(\beta).$$

From this update, define a discrete controller

$$\sigma : B \to \{+1, -1\}$$

by

$$\mathrm{sgn}(\beta') = \begin{cases} +1, & \text{if the chosen readout of } \beta' \text{ is } > 0, \\ -1, & \text{otherwise.} \end{cases}$$

Any two-valued partition would suffice; the sign function is simply the most convenient example.

Now define the observable update

$$f : A \times \{+1, -1\} \to A$$

by

$$f(\alpha, +1) = \alpha, \qquad f(\alpha, -1) = -\alpha.$$

Then the observable component updates as

$$\alpha' := f\big(\pi_A(\alpha, \beta),\ \mathrm{sgn}(v(\pi_B(\alpha, \beta)))\big).$$

The full composite step is therefore

$$T := \big\langle\, f \circ \langle \pi_A,\ \mathrm{sgn} \circ v \circ \pi_B \rangle, \quad v \circ \pi_B \,\big\rangle : S \to S,$$

which is exactly of the classical composition form

$$f \circ \langle g_1, \ldots, g_k \rangle.$$

Example 3.4 (Flow-Controlled Chirality Flip). Take the initial state

$$|\alpha\rangle \otimes |\beta\rangle = |{+1}\rangle \otimes |0.7\rangle .$$

Let

$$v(\beta) = \beta - 1,$$

so that it undergoes a one-step slowdown. Then

$$\beta' = v(0.7) = -0.3, \qquad \operatorname{sgn}(\beta') = -1.$$

It follows that

$$\alpha' = f(+1, -1) = -1.$$

Thus, the updated state is

$$|{-1}\rangle \otimes |{-0.3}\rangle .$$

Basic linear encoding. For readers who prefer an explicit finite coding, choose a two-point basis for A:

$$|{+1}\rangle = (1, 0), \qquad |{-1}\rangle = (0, 1),$$

and a finite code for flow levels in B:

$$|\text{low}\rangle = (1, 0, 0), \qquad |\text{med}\rangle = (0, 1, 0), \qquad |\text{high}\rangle = (0, 0, 1).$$

Together with a coarse-graining

$$q : B \to \{\text{low}, \text{med}, \text{high}\},$$

for example

$$q(0.7) = \text{med}, \qquad q(-0.3) = \text{low},$$

the pair (α, β) is represented by the Kronecker product

$$|{+1}\rangle \otimes |\text{med}\rangle = (1, 0) \otimes (0, 1, 0) = (0, 1, 0,\ 0, 0, 0).$$

In this finite encoding, the updates are represented by block-permutation matrices implementing f on A and v (followed by q) on B.

Primitive Recursion as Recursive Composition. Fix an initializer

$$i : X \to S,$$

thought of as the geometric constructor determined by inputs $\vec{x} \in X$, and a family of local moves

$$T_n : S \to S$$

indexed by $n \in N_G$. Each T_n is an endomorphism of S and, in reversible regimes, may be an automorphism. We continue to write the right action as

$$G \triangleright T := T(G).$$

The resulting state sequence is defined by

$$G_0(\vec{x}) = i(\vec{x}), \qquad G_{n+1}(\vec{x}) = G_n(\vec{x}) \triangleright T_n.$$

If T_n factorizes by components, we may write

$$T_n = T_n^{\alpha} \otimes T_n^{\beta},$$

although more general coupled updates on $A \times B$ are also allowed.

In fibered-ket notation,

$$|G_n\rangle = |\alpha_n\rangle \otimes |\beta_n\rangle , \qquad |G_{n+1}\rangle = (T_n^{\alpha} |\alpha_n\rangle) \otimes (T_n^{\beta} |\beta_n\rangle),$$

with the obvious modification in the coupled case. After k steps,

$$|G_k\rangle = \big(\cdots((|G_0\rangle \triangleright T_0) \triangleright T_1) \cdots\big) \triangleright T_{k-1}.$$

Equivalence with the PRF Recursor. Let

$$u : S \times N_G \to S$$

be the step map defined by

$$u(s, n) = T_n(s).$$

Then the NNO recursor yields a unique map

$$h = \mathrm{rec}(i, u) : X \times N_G \to S$$

such that

$$h(\vec{x}, 0) = i(\vec{x}), \qquad h(\vec{x}, n{+}1) = u(h(\vec{x}, n), n).$$

Hence for all k,

$$G_k(\vec{x}) = h(\vec{x}, k).$$

This is exactly the geometric realization of Primitive Recursion.

Example 3.5 (Commuting Flips on Separate Factors). Let

$$|\Psi_0\rangle = |\mathrm{left}\rangle \otimes |\mathrm{up}\rangle .$$

Let

$$C : A \to A$$

swap left↔right and

$$O : B \to B$$

flip up↔down. Define

$$T_0 = C \otimes I, \qquad T_1 = I \otimes O.$$

Then

$$|\Psi_1\rangle = (C \otimes I) |\Psi_0\rangle , \qquad |\Psi_2\rangle = (I \otimes O) |\Psi_1\rangle .$$

Thus

$$|\Psi_2\rangle = (I \otimes O)(C \otimes I) |\Psi_0\rangle = (C \otimes O) |\Psi_0\rangle ,$$

since C and O act on different factors and therefore commute.

3.3 Unbounded Recursion as Geometric Search

Unbounded μ–Recursion is interpreted as geometric search together with genuine partiality. We work in the partial-maps completion $\mathsf{Par}(\mathcal{G})$. Fix a total map

$$g : N_G^{n+1} \to N_G$$

in $\mathcal{G}$, viewed as a decidable numeric predicate with 0_G playing the role of "true". The associated partial map

$$\mu(g) : N_G^n \rightharpoonup N_G$$

is defined on $\vec{x}$ precisely when there exists y such that $g(\vec{x}, y) = 0_G$, in which case

$$\mu(g)(\vec{x}) = \min\{\, y \in N_G \mid g(\vec{x}, y) = 0_G \,\}.$$

Constructively, $\mu(g)$ is realized as the increasing union of finite-search partial maps

$$w_k : N_G^n \rightharpoonup N_G$$

defined by Primitive Recursion:

$$w_0(\vec{x}) = \begin{cases} 0, & \text{if } g(\vec{x}, 0) = 0_G, \\ \text{undefined}, & \text{otherwise,} \end{cases}$$

$$w_{k+1}(\vec{x}) = \begin{cases} w_k(\vec{x}), & \text{if } w_k(\vec{x}) \text{ is already defined,} \\ k{+}1, & \text{if } g(\vec{x}, k{+}1) = 0_G, \\ \text{undefined}, & \text{otherwise.} \end{cases}$$

Then $\mu(g)$ is the partial map whose graph is

$$\bigcup_k \mathrm{graph}(w_k).$$

Geometric reading. For inputs $\vec{x} \in N_G^n$, let an initializer

$$i : X \to S$$

choose a starting state, and let a family of local moves

$$U_y : S \to S$$

generate candidate configurations

$$|G_y\rangle = U_y\big(i(\vec{x})\big).$$

A geometric halting predicate

$$H : S \to N_G$$

such as "symmetry satisfied" or "entered a distinguished region" then defines

$$g(\vec{x}, y) := H(|G_y\rangle).$$

Accordingly, $\mu(g)(\vec{x})$ is the least y for which

$$H(|G_y\rangle) = 0_G,$$

if such a y exists. Otherwise the search is undefined at $\vec{x}$, i.e. it diverges.

Example 3.6 (Divergence via Mutual Triggering in a Minimal GL). Consider two neighboring GCMs A and B with state factors

$$|\alpha_\bullet\rangle \in A \qquad \text{(orientation: up/down)}$$

and

$$|\beta_\bullet\rangle \in B \qquad \text{(continuous flow)}.$$

The joint state is

$$|\Psi\rangle = (|\alpha_A\rangle \otimes |\beta_A\rangle) \otimes (|\alpha_B\rangle \otimes |\beta_B\rangle) \in (A \times B) \times (A \times B).$$

Let

$$T : (A \times B) \times (A \times B) \to (A \times B) \times (A \times B)$$

be the local transition that, at each step,

- increases β_A; if it crosses the threshold 1, flips α_A and triggers an increase of β_B;
- symmetrically increases β_B; if it crosses the threshold 1, flips α_B and triggers an increase of β_A.

Set

$$|\Psi_{n+1}\rangle := T\,|\Psi_n\rangle\,,$$

and let the halting predicate be given by

$$H(|\Psi\rangle) = 0_G$$

iff a designated stop condition holds, for example both flows vanish or a marked symmetry cell is reached.

If T perpetually re-triggers the threshold crossings, then for all n one has

$$H(|\Psi_n\rangle) \neq 0_G,$$

and the search never halts:

$$|\Psi_0\rangle \xrightarrow{T} |\Psi_1\rangle \xrightarrow{T} |\Psi_2\rangle \xrightarrow{T} \cdots$$

Hence, for the associated predicate

$$g(\vec{x}, y) := H(|\Psi_y\rangle),$$

there does not exist any y with $g(\vec{x}, y) = 0_G$. Therefore

$$\mu(g)(\vec{x})$$

is undefined as a partial arrow in $\mathsf{Par}(\mathcal{G})$. In this way, unbounded μ–Recursion is realized geometrically as an endless sequence of fibered-ket transitions with no halting configuration.

3.4 Kleene Computability and Computational Groupoids

We now summarize the preceding constructions in a compact categorical form. The point of this subsection is twofold: first, to package the PRF correspondence internally to the geometric category $\mathcal{G}$; second, to clarify the distinction between reversible local geometry, which is naturally groupoidal, and genuine computational partiality, which belongs to the partial-maps setting.

Definition 3.2 (Lawvere Theory for Primitive Recursion)**.** Let LPR be the small cartesian category freely generated by an object N with arrows $0 : 1 \to N$, $S : N \to N$, finite products, and the primitive recursion operator characterized by its usual universal property.

Definition 3.3 (Geometric NNO)**.** In our semantic category $\mathcal{G}$ of geometric computations, a natural numbers object is an object N_G equipped with maps $0_G : 1 \to N_G$ and $S_G : N_G \to N_G$ such that for every object X, every $f : X \to N_G$, and every $g : X \times N_G \to N_G$, there exists a unique $\mathrm{rec}(f, g) : X \times N_G \to N_G$ satisfying the usual recursion equations.

Theorem 3.1 (Representation of PRF)**.** *There exists a strict cartesian functor*

$$F : \mathsf{LPR} \to \mathcal{G}$$

with

$$F(N) = N_G, \qquad F(0) = 0_G, \qquad F(S) = S_G,$$

preserving finite products and primitive recursion. Consequently, every primitive recursive function $f : \mathbb{N}^k \to \mathbb{N}$ *is realized by a unique morphism*

$$F(\ulcorner f \urcorner) : N_G^k \to N_G.$$

Corollary 3.2. *Zero, Successor, Projections, Composition, and Primitive Recursion are all realized in* $\mathcal{G}$*.*

To pass from primitive recursion to partial recursion, we move from $\mathcal{G}$ to the partial-maps category $\mathsf{Par}(\mathcal{G})$. This is the correct setting for unbounded search and possible non-termination.

Theorem 3.3 (Kleene Normal Form in $\mathsf{Par}(\mathcal{G})$)**.** *Let* $\mathsf{Par}(\mathcal{G})$ *denote the partial-maps category over* $\mathcal{G}$*. It is cartesian, inherits* N_G*, and admits a partial unbounded search operator* μ *for decidable primitive recursive predicates. Hence there exist primitive recursive maps* T, U *in* $\mathcal{G}$ *such that every partial recursive function* ϕ_e *is represented as*

$$\phi_e(\vec{x}) = U(\mu\, y.\, T(e, \vec{x}, y) = 0)$$

in $\mathsf{Par}(\mathcal{G})$*.*

Thus the classical Kleene normal form is recovered internally to the geometric setting. At this point, the distinction between reversible geometry and computational partiality becomes important.

Reversible, state-dependent micro-moves are modeled by a groupoid

$$\mathcal{H} \rightrightarrows S$$

acting on the geobit state object $S = A \times B$: an arrow exists only when the move is admissible for that source state, and every such arrow is invertible. In this way, groupoids encode local symmetries, chart changes, and reversible twists.

By contrast, irreversible evolution, failed transitions, and unbounded search are not groupoid phenomena. These belong to the partial-maps completion $\mathsf{Par}(\mathcal{G})$, whose arrows

$$X \rightharpoonup Y$$

are partial morphisms, represented for example by spans or by the partiality monad $1+-$. In particular, the μ-operator lives in $\mathsf{Par}(\mathcal{G})$, not in the groupoid itself.

Remark 3.2. Groupoids capture *where* reversible moves are allowed: no arrow means that the move is geometrically forbidden. The category $\mathsf{Par}(\mathcal{G})$, on the other hand, captures *when* a computation may legitimately be undefined, whether through divergence, failure, or the absence of a solution.

We can now also isolate the correct self-referential content of the Kleene layer. The point is not that every geometric state admits a symmetry-fixed representative, but rather that once coding and evaluation are available internally, the usual recursion-theoretic fixed-point mechanism reappears in geometric form.

Fix a state object S in $\mathcal{G}$. Let C be an object of codes for realizable partial endomaps of S, equipped with an evaluator

$$\mathsf{eval} : C \times S \rightharpoonup S \qquad \text{in } \mathsf{Par}(\mathcal{G}).$$

Assume there is a surjective coding

$$\langle - \rangle : N_G \twoheadrightarrow C$$

and a universal partial evaluator

$$U : N_G \times S \rightharpoonup S \qquad \text{with} \qquad \mathsf{eval}(\langle e \rangle, s) = U(e, s).$$

Finally, assume that total computable transformers on codes are represented by total arrows

$$\psi_C : C \to C$$

induced from total maps

$$\psi_N : N_G \to N_G$$

through $\langle - \rangle$. These are the standard hypotheses of realizability semantics: the NNO provides internal arithmetic and coding, the μ-operator supplies unbounded search, and U plays the role of a universal machine.

Theorem 3.4 (Internal Kleene Recursion)**.** *For every total computable transformer $\psi_C : C \to C$, there exists a code $c^* \in C$ such that*

$$\mathsf{eval}(c^*, -) = \mathsf{eval}\big(\psi_C(c^*), -\big) \qquad \textit{as partial endomaps } S \rightharpoonup S.$$

Proof sketch. Work internally over N_G. By the usual s–m–n construction for partial recursive functions, there exists a total map

$$\mathbf{s} : N_G \to N_G$$

such that

$$U(\mathbf{s}(e), s) = U(\psi_N(e), s)$$

for all e and s. Let $e^* = \mathbf{s}(e^*)$ be the diagonal fixed point supplied by Kleene's classical argument, internalized through the NNO. Setting $c^* = \langle e^* \rangle$ and transporting along $\langle - \rangle$ yields the stated equality for eval. ☐

Corollary 3.5 (Geometric Quine)**.** *Let*

$$W : S \rightharpoonup S$$

be the partial update that writes the presented code into a designated register of S and then proceeds. Define ψ_C to prepend W to the behavior of any code. By theorem 3.4, there exists c^* such that

$$\mathsf{eval}(c^*, -) = \mathsf{eval}(\psi_C(c^*), -).$$

Running c^* therefore causes the system to write an encoding of c^* into its own state before continuing. In this sense, c^* is a geometric quine.

The next section passes from this discrete computability-theoretic layer to the analytic layer: groupoid flows and ODE-driven updates acting as total or partial arrows, with existence and uniqueness assumptions made explicit. The PRF/μ semantics developed here remain the controlling discrete interface, while the continuum dynamics provide their geometric realization in time.

4 Abelian Gauge Dynamics and Phase Geometry

Recent work on the computability and complexity of fluid flows (e.g. Cardona et al. [15]), together with Tao's conjectural program on universality phenomena for the Euler equations [16], highlights a recurring theme: *continuous media can support computation*, but often only through dynamics whose analytic behavior is difficult to control.

In this section, we introduce the *quasi-fluidic band* as an *abelian gauge* mode of computation. Informally: a GL is a lattice of geometric cells whose internal degrees of freedom carry abelian phase data (typically $U(1) \simeq S^1$, or a finite cyclic approximation C_n), together with discrete labels and quantized counters.

As established earlier, a geometric computer is neither strictly a Turing machine nor a quantum computer. Quantum computation is organized around a global state vector and coherence constraints; locality and scaling are mediated by entanglement. By contrast, a GL scales by composition: cells are wired together by interfaces and boundary conditions, and the operative degrees of freedom are abelian phases and discrete readouts that compose functorially across the lattice. There are no complex amplitudes and no Born-rule collapse; instead, one reads gauge-invariant features of geometry (e.g. phase differences and winding).

As a motivating example, consider the classic experiment in which a hose filled with running water is placed across a vibrating speaker. The resulting jet forms a visible oscillating pattern: the driving frequency controls coil tightness, amplitude controls the spiral radius, and phase controls the initial offset. This is an instructive caricature of the quasi-fluidic regime: periodic forcing induces an effective $U(1)$-phase along a geometric substrate, and the observable shape provides a coarse readout of that phase. In the language adopted later (4.1), the phase offset belongs to the T-layer (S^1 data), while winding/twist counts correspond to quantized Z-invariants; discrete readouts arise from coarse partitions of these quantities. Helical and twisted structures appear here not as an aesthetic choice, but as energetically natural carriers of phase and twist under periodic driving. A visual demonstration by brusspup [17] makes this phenomenon concrete.[5]

Computational geometric bordisms. The vibrating-hose experiment may be viewed as an analog of a *computational geometric bordism*: a smooth evolution mediating between boundary conditions (fixed endpoints, imposed phase/frequency, initial and final configurations). Here, the bordism is "internal" to the geometry of the evolving flow, and the computable content is not the PDE solution itself but the stable, gauge-invariant features one can read from it. See appendix A.4 for cobordism formalisms.

Proposition 4.1. *The geometric topology of a GCM constrains its information-bearing degrees of freedom, but the* effective *information content is relative to a chosen readout (coarse-graining). In particular, geometric computation is deterministic at the microstate level, while information is defined relative to an encoding.*

Unlike in classical computation—where the bit is treated as an absolute, context-free primitive—in geometric computation, the notion of a "geobit" is always specified with respect to a readout map (or partition) on the underlying state space. Concretely, one fixes an observable or coarse-graining

$$\rho : \Sigma \longrightarrow B,$$

where Σ is the (possibly structured) microstate space of the GCM and B is a finite label set used for symbolic computation. Once ρ is fixed, information becomes an objective property of the pair (Σ, ρ): the dynamics may be fully deterministic on Σ, yet the induced evolution on B depends on which features are chosen as readable and which are treated as hidden or irrelevant.

This is directly analogous to physics, where measurements and reference choices enter through specified observables rather than through subjectivity. The boundary between symbolic and quasi-fluidic computation is therefore not a matter of taste. Still, of modeling choice: symbolic regimes arise when the operative degrees of freedom are purely discrete (a finite B with static invariants), while the quasi-fluidic regime appears when one enlarges the local fiber to include compact abelian phases and quantized topological counters (the $B/T/Z$ structure of definition 4.1). Geometric deformation

[5] Regarding an unconventional reference; the point is purely pedagogical: it is a compact way to visualize phase, winding, and coarse readout in a physical system.

then permits interpolation between discrete and continuous logic *without* introducing probabilistic amplitudes.

This does not mean the quasi-fluidic regime is vague. In many ways, it has already appeared implicitly: even in earlier symbolic constructions, evolution was driven by an explicit action stream (global pulses and local update schedules) acting on finite geometric labels. The quasi-fluidic mode makes this action stream geometric by allowing abelian phase transport and thresholded topological events, while retaining the core principle of GCT: computability is realized by local, deterministic transition structure rather than by solving global differential equations.

4.1 A New Kind of Computation

A central motivation of GCT is to avoid the classical reliance on global PDE evolution as the *definition* of computation over continuous media (fluids, fields, elastic bodies). As Tao has repeatedly emphasized, even well-posed PDE models may develop loss of regularity, extreme sensitivity to initial conditions, or finite-time singular behavior; these analytic pathologies are not merely technicalities, but can become *computationally relevant* in unfavorable ways.

The paradox of analog logic flow. Historically, analog and fluidic devices exploited continuous physics to compute. Mechanical differential analyzers and wartime fire-control machines, as well as later fluidic and hydraulic integrators, demonstrate that a physical medium can implement nontrivial transforms. Yet this tradition also exposes a basic paradox of analog logic: the computational substrate is not insulated from the mathematics it is tasked with realizing. If the governing model admits blow-up, stiffness, or chaotic amplification, then the physical device inherits the same fragility. In contrast to digital machines—which often fail noisily via discrete faults—analog computation can fail by gradual loss of identifiability: the state remains *well-defined physically* while becoming *unreadable* with respect to the intended encoding.

GCT responds by shifting the locus of computability from *solving equations* to *enforcing structure*. A GCM is designed so that the fundamental computational step is a local, deterministic update at an interface: an equivariant transformation of a structured state fiber (in the quasi-fluidic band, the $B/T/Z$ decomposition of definition 4.1), driven by controlled inputs (pulses, boundary perturbations, or local couplings). The ambient dynamics may be smooth, but the computation is carried by the admissible morphisms of the system—what changes, what cannot change, and what invariants must persist. In this sense, GCMs are naturally piecewise smooth: not because we approximate a PDE, but because computation is organized around local transitions and thresholded events rather than global analytic integration. In a GL, the computer is the problem, and the problem is the solution.

Accordingly, we do not attempt to compute Euler solutions by "bit-flips" internal to the manifold. Instead, we encode logic into the topology, phase structure, and allowable deformations of embedded geometry, and we read outputs through fixed observables (coarse-grainings) of that geometry. In the coming sections, we will model how complex behavior can emerge from simple shapes.

This structural viewpoint is not without precedent. Goriely and collaborators, whose earlier work on nonlinear geometry and helices we will cite later, have recently advocated a related shift in applied modeling: replacing monolithic PDE descriptions by networked, local interaction rules in neuroscience and biophysics. Their work illustrates how complex global behavior can arise from local coupling laws on a discrete architecture, providing a concrete example of how one may trade PDE-centric descriptions for robust, compositional dynamics (see, e.g., Thompson et al. [18]).

4.2 Boundaries in Quasi-Fluidic GCMs

At scale, computation in a GL emerges from the coordinated action of many such units coupled by local interfaces. In this setting, μ-recursion should be understood as a *structured search* over geometric configurations and their induced readouts, driven by local update laws and global pulses, and terminating when a designated halting condition is detected at the readout layer. We now make a precise phase boundary.

Definition 4.1 (Quasi–Fluidic Computation and Regime Phases)**.** Given a GCM, we say it operates in the *quasi–fluidic regime* if each region R_i carries a finite–dimensional state fiber:

$$\mathcal{M} \;\cong\; B_i \times T_i \times Z_i,$$

as in definition 2.1 with the following structure:

(QF1) **Discrete core (symbolic).** B_i is finite, typically $B_i \cong (C_2)^{k_i}$; in the twin helix unit one has $k_i = 4$ for chirality/orientation bits.

(QF2) **Phase layer.** T_i is a continuous but compact abelian Lie group (a torus), $T_i \cong (S^1)^{m_i} \simeq U(1)^{m_i}$, encoding axial phase(s), orientation angles, or other bounded "analog" degrees of freedom.

(QF3) **Quantized invariants.** Z_i is a finitely generated abelian group (e.g. $\mathbb{Z}^{p_i}$), recording twist/winding/linking numbers or other topological charges.

(QF4) **Abelian symmetry and equivariance.** The local symmetry acting on Σ_i is

$$K_i \;\cong\; (C_2)^{k_i} \times (S^1)^{m_i},$$

and every interface map $I_{ij} \in \mathcal{I}$ and the induced flow on Σ_i are K–equivariant, that is, they depend on phase differences and commute with flips/rotations.

(QF5) **Flow coupling.** The ambient flow $\{\mathsf{Ev}_t\}$ transports the T–coordinates continuously (phase evolution) and can modify Z only by isolated, energy–threshold events (phase slips) prescribed by the interface rules; B updates by the same local rules. No non-abelian monodromy is present.

(QF6) **Non–singularity and bounded energy.** Trajectories have no finite–time blow-ups; curvature/torsion remain bounded; state updates at interfaces are well defined for all t.

A system is symbolic when $m_i = 0$ and Z_i is static; it is purely fluidic when B_i is trivial and non-abelian holonomy appears. The quasi–fluidic band sits between these: it combines a finite discrete core with $U(1)$–type phases and integer topological counters, all governed by abelian symmetry.

Composition of GCMs. In the symbolic model, a GL is composed of *discrete interconnect*: neighboring GCMs exchange a finite readout and update by a local rule, precisely as in a cellular automaton with richer internal state. In the quasi-fluidic regime, this picture remains valid. Still, each cell now carries an additional compact phase and a quantized invariant structure (the $B/T/Z$ fiber of definition 4.1), so the internal dynamics of a single unit are no longer adequately described by a purely discrete transition table.

There is also a second, more geometric notion of composition which becomes natural in this band: *flow-circuit composition*. Rather than identifying adjacency inside a single ambient lattice, one regards a GCM as a module with an *input interface* and an *output interface*, together with a local update law. The module, therefore, induces a (generally state-dependent) transfer map on the relevant readouts and phase variables, and connecting two modules corresponds to wiring output interfaces into input interfaces. At the level of these transfer maps, composition is ordinary function composition.

Operationally, "wiring" is encoded by the choice of interfaces $\mathcal{I}$ and their coupling strength: one may couple two neighboring regions by a nontrivial interface map I_{ij}, or effectively *disconnect* them by selecting a trivial (e.g. identity/no-op) interface, or by imposing a boundary condition that forces the coupling term to vanish. In this way, reconfiguration is expressed within the same functorial framework: one modifies the *interface structure* (the allowed local morphisms). This viewpoint aligns with the guiding principle of the quasi-fluidic regime: computation is carried by the admissible local moves and their compositions, while the ambient drive (pulses, boundary perturbations) supplies controlled input to those moves.

4.3 Abelian Lattice Gauge Dynamics in the Quasi–Fluidic Mode

In the QF band, the appropriate "computational" physics we should work with is not a discrete logic model as the symbolic band but an *abelian lattice gauge dynamics*: compact phases transported and coupled locally, with evolution governed by a bounded local energy and (optionally) a driven-damped relaxation law. This section is *interpretive*; it provides a stable dynamical model for analyzing QF behavior, without elevating differential equations to the definition of geometric computability.

Lattice variables (QF state). Let $\Gamma = (V, E)$ be the interaction graph of a GL (cells/modules are vertices, interfaces are edges). For each cell $i \in V$, we take an internal phase (orientation layer)

$$\theta_i \in U(1) \cong \mathbb{R}/2\pi\mathbb{Z},$$

and for each interface $(i, j) \in E$ we take a *link phase* (a discrete connection 1–form)

$$A_{ij} \in \mathbb{R}/2\pi\mathbb{Z}, \qquad A_{ji} = -A_{ij},$$

equivalently $U_{ij} := \exp(iA_{ij}) \in U(1)$. (If one wishes to remain in the finite cyclic subband, replace $U(1)$ by C_N and interpret all equations below as a continuous approximation to a stroboscopically driven C_N update.)

Gauge redundancy. A $U(1)$ gauge transformation is a choice of angles $\alpha_i \in \mathbb{R}/2\pi\mathbb{Z}$ at each cell,

$$\theta_i \mapsto \theta_i + \alpha_i, \qquad A_{ij} \mapsto A_{ij} + \alpha_i - \alpha_j,$$

under which the *gauge–invariant phase difference*

$$\Delta_{ij} \;:=\; \theta_i - \theta_j - A_{ij} \pmod{2\pi}$$

is unchanged. In this sense, "independence of local modules" is a choice of gauge/frame, while couplings are encoded invariantly by Δ_{ij}.

Local energy (interface functional). A minimal gauge–invariant energy on Γ is

$$\mathcal{E}(\theta, A) \;=\; \sum_{(i,j)\in E} J_{ij}\left(1 - \cos(\Delta_{ij})\right) \;+\; \sum_{p\in \mathsf{P}} \kappa_p \left(1 - \cos((dA)_p)\right), \tag{1}$$

where $J_{ij} \geq 0$ are interface couplings and P is a chosen set of plaquettes (2–cells) with discrete curvature

$$(dA)_p \;:=\; \sum_{(i,j)\in \partial p} s_{p,ij}\, A_{ij} \pmod{2\pi}, \qquad s_{p,ij} \in \{\pm 1\}$$

the oriented sum around ∂p. The first term penalizes a mismatch between neighboring phases modulo the link variable; the second penalizes nonzero curvature ("flux").

Gradient–flow dynamics. A simplest well-behaved evolution law is driven by gradient flow on the compact state space:

$$\dot{\theta}_i \;=\; -\gamma_\theta \frac{\partial \mathcal{E}}{\partial \theta_i} \;+\; \eta_i(t), \qquad \dot{A}_{ij} \;=\; -\gamma_A \frac{\partial \mathcal{E}}{\partial A_{ij}} \;+\; \eta_{ij}(t), \tag{2}$$

with mobilities $\gamma_\theta, \gamma_A > 0$ and external drives $\eta_i(t), \eta_{ij}(t)$ (global pulses, boundary forcing, or localized injections). Concretely, the Euler–Lagrange components are

$$\frac{\partial \mathcal{E}}{\partial \theta_i} = \sum_{j\sim i} J_{ij}\, \sin(\Delta_{ij}), \tag{3}$$

$$\frac{\partial \mathcal{E}}{\partial A_{ij}} = -J_{ij}\, \sin(\Delta_{ij}) \;+\; \sum_{p\ni(i,j)} \kappa_p\, s_{p,ij}\, \sin((dA)_p), \tag{4}$$

so (2) relaxes toward gauge–invariant consistency ($\Delta_{ij} \to 0$) and low curvature ($(dA)_p \to 0$) unless forced otherwise. Because the variables are compact and $\mathcal{E}$ is bounded below, this regime is dynamically stable and aligns with the QF "no blow–ups" philosophy.

Remark 4.1 (Scope and Interpretation). Equations (1)–(2) are not proposed as fundamental laws of GCT; they are a *normal form* for the abelian corridor: local couplings, compact phases, gauge redundancy, and dissipative relaxation. In particular, continuous flows of this type do *not* implement jump–style updates by themselves; any discrete "computational" layer is induced later by readout/-partitioning (return maps).

Auxiliary geometric ODE variables. To retain an explicit connection to helix geometry, one may attach slowly varying, bounded geometric parameters to each cell,

$$r_i(t) \in \mathbb{R}_{>0} \qquad \text{(effective radius/scale),} \tag{5}$$
$$\tau_i(t) \in \mathbb{R} \qquad \text{(effective twist rate),} \tag{6}$$
$$\mathcal{L}_i(t) \in \mathbb{R}_{>0} \qquad \text{(effective length)} \tag{7}$$

and couple them to *local energy density* and/or *local phase current*. A simple choice is:

$$\dot{r}_i = -\lambda_r\,(r_i - r_0) \;+\; \alpha_r \sum_{j\sim i} \big(1 - \cos(\Delta_{ij})\big), \tag{8}$$
$$\dot{\tau}_i = -\lambda_\tau\,(\tau_i - \tau_0) \;+\; \alpha_\tau \sum_{j\sim i} \sin(\Delta_{ij}), \tag{9}$$
$$\dot{\mathcal{L}}_i = -\lambda_L\,(\mathcal{L}_i - \mathcal{L}_0) \;+\; \alpha_L\, r_i\, |\tau_i|, \tag{10}$$

with $\lambda_\bullet > 0$ ensuring relaxation to baseline values $(r_0, \tau_0, \mathcal{L}_0)$ and $\alpha_\bullet$ setting coupling strength. These equations encode the informal idea that mismatch/coupling stress can thicken the effective geometry (8), phase transport can bias twist (9), and geometry co-scales (10). One may also let the mobilities in (2) depend on τ_i (twist as a local "clock" factor), e.g. $\gamma_\theta = \gamma_\theta(\tau_i)$, without changing the gauge structure.

Synchronization limit: Kuramoto reduction. A useful special case of the QF phase dynamics is obtained by restricting to nearest-neighbor coupling on the $U(1)$ phase layer and passing to a co-rotating frame. In that case, the local phase evolution reduces to a Kuramoto-type system [19, 20]

$$\frac{d\theta_i}{dt} = \Omega_i + K \sum_{j\sim i} \sin(\theta_j - \theta_i),$$

where $\theta_i \in S^1$ is the phase of cell i, Ω_i is its intrinsic frequency, and K is the coupling strength.

This should be understood as a *synchronization limit* of the abelian gauge dynamics, not as the full semantics of quasi–fluidic computation. Physically, it describes the tendency of neighboring helices to phase-lock under local coupling; computationally, it captures the emergence of a shared clock or coherent phase carrier across a region of the lattice. In particular, when the $U(1)$ corridor is used only as a transport layer, Kuramoto synchronization provides a natural mechanism by which continuous phase variables may stabilize before a stroboscopic readout is applied.

Equivalently, if the local energy is taken to be of XY type,

$$E(\theta) = \sum_{(i,j)\in E(\Lambda)} \kappa_{ij} \big(1 - \cos(\theta_i - \theta_j)\big),$$

then the dissipative phase flow

$$\dot{\theta}_i = -\frac{\partial E}{\partial \theta_i}$$

recovers the Kuramoto form above up to the choice of sign convention and the addition of the drift terms Ω_i. Thus, the Kuramoto system is a familiar, physically interpretable subcase of the general QF lattice dynamics.

Remark 4.2 (Event Exclusions in the Pure QF band). In the strict QF corridor, one typically *excludes* threshold events (phase slips/defects) so that integer topological counters remain frozen and the dynamics stay smooth and energy–bounded. Allowing such events corresponds to leaving the pure abelian corridor toward more expressive regimes. The return–map/readout layer (introduced next) is the mechanism by which discrete computation is recovered without invoking singular PDE behavior.

4.4 Vector Cellular Automata and Stroboscopic Readout

The QF dynamics of section 4.3 are continuous (compact phases with local energy and gradient flow). As such, they do *not* natively implement jump–style updates. The bridge back to classical symbolic computation is therefore *stroboscopic*: we sample the system once per pulse and apply a *discretized readout* (a quantizer) to obtain an induced update on a finite alphabet. In this sense, discrete computation in the QF band is a *return–map phenomenon*, not an ODE phenomenon.

Definition 4.2 (Pulse Times and Stroboscopic Map)**.** Fix a pulse period $\Delta t > 0$ and define $t_n := n\Delta t$. Let $P : L \to L$ denote the time–Δt map on the QF state space L induced by one pulse:

$$P \ := \ \mathsf{Ev}_{\Delta t}.$$

We write $x^n \in L$ for the post–pulse state at time t_n, so $x^{n+1} = P(x^n)$.

Because the abelian corridor carries gauge redundancy, any robust readout should be phrased in terms of *gauge–invariant local observables.* In the lattice–gauge normal form, the basic invariants are:

$$\Delta_{ij} = \theta_i - \theta_j - A_{ij} \pmod{2\pi} \qquad \text{and} \qquad (dA)_p = \sum_{(i,j)\in\partial p} s_{p,ij} A_{ij} \pmod{2\pi}.$$

We therefore define a cellwise observable $y_i = y_i(x)$ as a finite list of such invariants in a bounded neighborhood (e.g. the Δ_{ij} on links incident to i, and optionally nearby plaquette fluxes $(dA)_p$).

Definition 4.3 (Discretized Readout/Quantizer)**.** Fix a finite alphabet B and a (piecewise constant) quantizer

$$Q : \ \mathcal{Y} \longrightarrow B,$$

where $\mathcal{Y}$ is the range of local invariants y_i. The induced discrete configuration for a state $x \in L$ is

$$b = \mathrm{Read}(x) \in B^V, \qquad b_i := Q\big(y_i(x)\big).$$

For example, one may partition S^1 into arcs and threshold a chosen invariant angle, or partition the flux $(dA)_p$ into bins. The specific partition is a modeling choice.

Definition 4.4 (Vector Cellular Automata on an Induced Symbolic Return Map)**.** Write x_i^n for the local continuous state in cell i at pulse n (phases θ, links A, and any auxiliary variables such as $r, \tau, \mathcal{L}$). Because the QF energy and interfaces are finite-range, there exists a finite neighborhood $\mathcal{N}(i)$ such that the stroboscopic update is local:

$$x_i^{n+1} \ = \ \mathsf{U}\Big(\{x_j^n\}_{j\in\mathcal{N}(i)}\Big), \qquad x_i^n \in \mathsf{X},$$

for some compact state manifold X (a product of circles and bounded geometric variables). This is precisely a *vector cellular automaton* (continuous alphabet, local rule, synchronous clock).

Composing with the readout gives an induced discrete return map on B^V:

$$\mathcal{F}_B \ := \ \mathrm{Read} \circ P \circ \mathrm{Lift} \ : \ B^V \longrightarrow B^V,$$

where Lift selects a representative microstate $x \in L$ consistent with a discrete labeling (e.g. by choosing canonical phases inside each bin).

Remark 4.3 (What is "Computed"?). In the QF corridor, the *continuous* dynamics are the physical substrate, and the *symbolic* computation appears only after (a) stroboscopic sampling and (b) a discretized readout. This is the same conceptual move used in electronics: continuous circuit laws are real, while digital logic is the stable quotient obtained by thresholding and clocking.

The $U(1)$ Corridor and an E/V Paradox. It is at this point necessary to issue some disclaimers regarding the use of the full S^1 state spaces, that is, the $U(1)$ corridor. As established, within the quasi–fluidic band, the ω-logic layer may be taken either as a finite cyclic alphabet C_N or as a continuous phase $U(1)$. Passing to the full S^1 limit is mathematically tidy (abelian, compact), but it also oddly backfires, creating a regime with atypical E/V trade-off behavior. The following remarks record the current hypothesis of this paper:

- **Low geometric expressiveness.** Pure $U(1)$ phase composition is commutative. As a result, sequencing power collapses: one cannot exploit non-commuting compositions to build richer path-dependent behavior, and the phase layer alone tends to support only limited families of smooth, energetically bounded dynamics. In this corridor, most "computational complexity" must therefore be carried either by the discrete core B (via readout and interface thresholds) or by leaving the abelian setting (e.g. via threshold events/topological defects or non-abelian holonomy).

- **Low algebraic verifiability.** Any *finite* readout must discretize phase (explicitly or implicitly) by thresholds. In the $U(1)$ corridor, near-resonant phase evolution makes this discretization brittle: small phase perturbations can move trajectories across readout boundaries, producing significant changes in the inferred discrete state. Moreover, phase-locking reduces redundancy (many degrees of freedom share a common "clock"), weakening independent cross-checks.

Transition, not destination. For these reasons, the $U(1)$ phase limit is best regarded as a *critical corridor*: visually orderly and analytically tractable, yet brittle under discretized readout and shallow in purely geometric expressiveness unless supplemented by discrete structure. It serves as a staging layer for robust computation on B and as a boundary marker before the onset of genuinely non-abelian (fluidic/gauge) behavior.

4.5 Application of Lie Groupoids in Geometric Computation

As the finite cyclic phase spaces used in large abelian geobit groups are refined, they naturally approach a continuous $U(1)$ phase description. At the same time, Kleene's theory of partial recursion suggests that partial transformations, local domains of definition, and composability should be built into the formalism from the outset. These requirements are brought together naturally by the notion of a *Lie groupoid*—introduced by Charles Ehresmann in the 1950s—which combines smooth structure with local, partially defined symmetry.

Informally, a Lie groupoid is a groupoid in which the space of objects and the space of morphisms are smooth manifolds, and all structural maps (source, target, identity, inversion, and composition) are smooth. In contrast to a single global group action, a groupoid allows invertible morphisms only where they are geometrically meaningful. This flexibility is well-suited to GCMs, whose computation is local, interface-driven, and often only partially defined.

We begin with a general decorated formalism, and then specialize it to the quasi-fluidic abelian regime.

Definition 4.5 (S–decorated Groupoid)**.** Let $(S, *)$ be a monoid with unit e, and let $\mathbf{B}S$ denote the one-object category whose endomorphism monoid is S. An *S–decorated groupoid* is a pair $(\mathcal{G}, d)$ consisting of a small groupoid $\mathcal{G}$ together with a functor

$$d : \mathcal{G} \longrightarrow \mathbf{B}S.$$

Equivalently, each morphism $f \in \mathrm{Mor}(\mathcal{G})$ is assigned a label $d(f) \in S$ such that

$$d(g \circ f) = d(g) * d(f), \qquad d(\mathrm{id}_x) = e$$

for all composable f, g and all objects x. Since functors preserve isomorphisms, every decoration of an arrow lies in the unit group $S^\times \subseteq S$.

Definition 4.6 (Morphisms of S–decorated Groupoids)**.** A morphism

$$(\mathcal{G}, d) \longrightarrow (\mathcal{H}, \delta)$$

of S–decorated groupoids is a functor $F : \mathcal{G} \to \mathcal{H}$ such that

$$\delta \circ F = d.$$

This defines the category $\mathrm{DecGrpd}_S$ of S–decorated groupoids.

Remark 4.4. The use of the term "decorated" follows modern geometric and categorical usage (see, e.g., Penner [21] and Fong [22]). The point is to enrich a base compositional structure with auxiliary data while preserving functorial composition. In the present context, the decorations record geometric or computational quantities associated with paths, interfaces, or local transformations.

General decorated path data. Let M be a smooth manifold (or a stratified manifold when needed), and let $\mathcal{S}$ denote a specified collection of geometric or computational structures on M—for example, twist counts, phase accumulation, chirality labels, or local field values. The classical fundamental groupoid remembers only path homotopy classes. In geometric computation, however, one often needs a refinement that also records such auxiliary data.

Definition 4.7 (Decorated Fundamental Groupoid)**.** Let M be a smooth manifold and let S be a monoid. An *S–decorated fundamental groupoid* of M is an S–decorated groupoid

$$(\Pi_1^*(M), d)$$

whose underlying groupoid $\Pi_1^*(M)$ is chosen from a path groupoid naturally associated to M (for example, the ordinary fundamental groupoid, or the thin fundamental groupoid when differential invariants are to be preserved), and whose decoration functor

$$d : \Pi_1^*(M) \to \mathbf{B}S$$

records the specified geometric data along paths. Composition is induced by path concatenation together with the monoid law in S.

Remark 4.5. When the decoration depends only on ordinary homotopy classes, one may work with the usual fundamental groupoid $\Pi_1(M)$. When the decoration depends on differential data such as connection holonomy or thin geometric transport, one should instead use the thin fundamental groupoid $\Pi_1^{\mathrm{thin}}(M)$, since such invariants are constant under thin homotopy but need not factor through ordinary homotopy classes.

A geometric model for the twin helix. For the purposes of geometric computation, a finite twin helix module is best modeled as a regular neighborhood of an embedded arc. Topologically, this gives

$$THM \;\cong\; D^2 \times I \;\cong\; B^3.$$

Thus, the underlying space is contractible: its ordinary homotopy type is trivial, and the undecorated fundamental groupoid carries no nontrivial loop information.

It means that the computationally relevant structure of a twin helix module is not encoded by the bare topology of the carrier space, but by the additional geometric data placed on it: phase fields, twist laws, chirality assignments, framings, and interface rules. In the symbolic and general decorated setting, one may attach labels such as chirality or twist to paths in this contractible space. In the quasi-fluidic abelian reduction developed later in this chapter, chirality is no longer treated as an actively updated path label, but instead as a choice of orientation convention for link variables.

Example 4.1 (Decorated Loops in a Finite Module)*.* Let

$$THM \;\cong\; D^2_{(r,\varphi)} \times I_z,$$

and choose a smooth phase 1–form on the cross-sectional angular direction,

$$A \;=\; \Omega(r,\varphi,z)\, d\varphi.$$

For fixed (r,z), let $\gamma_{r,z}$ denote the loop obtained by traversing once around the circle of radius r in the cross-section $D^2 \times \{z\}$. As a loop in the contractible manifold THM, $\gamma_{r,z}$ is null-homotopic in the ordinary sense. Nevertheless, it may carry nontrivial geometric data.

For example, taking

$$S = U(1) \times C_2,$$

one may define a decoration on an appropriate refined path groupoid (for example, the thin fundamental groupoid) by

$$d([\gamma_{r,z}]) \;:=\; \left(\exp\!\Big(i \int_{\gamma_{r,z}} A\Big),\; \chi(\gamma_{r,z}) \right),$$

where the first component records the accumulated phase and the second records a chirality label determined by the chosen geometric structure. The resulting decorated class

$$([\gamma_{r,z}],\, d([\gamma_{r,z}]))$$

is computationally nontrivial even though the underlying ordinary homotopy class of $\gamma_{r,z}$ is trivial.

Thus, in a finite twin–helix module, the computational content lives in the decoration rather than in the bare fundamental groupoid.

Remark 4.6. This is precisely why the general decorated picture and the later quasi-fluidic abelian reduction must be kept distinct. In the symbolic/general setting, chirality may legitimately appear as part of the decoration target S, and one works with a refined path groupoid because ordinary homotopy forgets too much. In the specialized $U(1)$ phase regime, by contrast, chirality is frozen into an orientation convention and the active structure reduces to abelian phase transport.

Why Lie groupoids appear in the QF regime. The preceding decorated picture is intentionally broad. In the quasi-fluidic regime, however, the relevant structure simplifies dramatically. The active degree of freedom is no longer a freely manipulated chirality/twist label, but an abelian phase field. The correct object is therefore not a general decorated path groupoid, but a Lie groupoid encoding *gauge redundancy* of local $U(1)$ phase assignments on a lattice.

To provide a rigorous setting for this maximal abelian regime, we explicitly construct the corresponding Lie groupoid.

Definition 4.8 (S–decorated Lie Groupoid)**.** If $\mathcal{G} \rightrightarrows M$ is a Lie groupoid and S is a Lie group, an *S–decorated Lie groupoid* is a smooth functor

$$d : \mathcal{G} \to \mathbf{B}S,$$

where $\mathbf{B}S$ is the one–object Lie groupoid with endomorphism group S.

Definition 4.9 (Underlying Lattice/Graph Datum)**.** Fix a finite connected graph

$$\Gamma = (V, E),$$

for example a finite patch of a GL, where V is the set of sites and E the set of undirected adjacency links. Choose an orientation on each undirected edge, written as a subset

$$E^+ \subseteq \vec{E},$$

so that every edge $\{i, j\} \in E$ appears exactly once as either $e : i \to j$ or $\bar{e} : j \to i$ in E^+. Write $s(e), t(e) \in V$ for the source and target of an oriented edge e.

Definition 4.10 (Core Configuration Space in the QF Regime)**.** The quasi-fluidic core state of the lattice is a *link phase assignment*

$$M_{\mathrm{QF}} \;:=\; U(1)^{E^+} \;\cong\; (S^1)^{|E^+|}.$$

An element $U \in M_{\mathrm{QF}}$ is a family $U = (U_e)_{e \in E^+}$ with $U_e \in U(1)$. The value on the oppositely oriented edge is determined by inversion:

$$U_{\bar{e}} \;:=\; U_e^{-1}.$$

Remark 4.7 (Chirality as Orientation Convention). In this quasi-fluidic interpretation, chirality is *not* a separate state coordinate. Instead, the handedness of the underlying helix selects (or suggests) a preferred orientation for reading the link phase; reversing that convention replaces U_e by U_e^{-1}. Thus the earlier symbolic chirality label is absorbed into the orientation convention on each link and disappears from gauge–invariant observables.

Definition 4.11 (Gauge Group and Action on Link Phases)**.** Let the site gauge group be

$$G_{\mathrm{QF}} \;:=\; U(1)^V.$$

For $g = (g_i)_{i \in V} \in G_{\mathrm{QF}}$ and $U = (U_e)_{e \in E^+} \in M_{\mathrm{QF}}$, define the action

$$g \cdot U \in M_{\mathrm{QF}}$$

by the standard link transformation law

$$(g \cdot U)_e \;:=\; g_{s(e)}\, U_e\, g_{t(e)}^{-1}, \qquad e \in E^+.$$

Definition 4.12 (QF Core Lie Groupoid: the Gauge Action Groupoid)**.** The quasi-fluidic core Lie groupoid is the action groupoid

$$\mathcal{G}_{\mathrm{QF}} \;:=\; G_{\mathrm{QF}} \ltimes M_{\mathrm{QF}} \rightrightarrows M_{\mathrm{QF}}.$$

An arrow is a pair (g, U) with $g \in G_{\mathrm{QF}}$ and $U \in M_{\mathrm{QF}}$. The source and target maps are

$$s(g, U) = U, \qquad t(g, U) = g \cdot U,$$

and composition is

$$(h,\, g \cdot U) \circ (g, U) = (hg,\, U), \qquad h, g \in G_{\mathrm{QF}}.$$

For finite Γ, this is a finite-dimensional Lie groupoid.

Remark 4.8 (Gauge Invariants Live on the Quotient Stack)*.* The physically meaningful quantities in the quasi-fluidic abelian regime are gauge–invariant functions on M_{QF} (equivalently, functions on the quotient stack

$$[M_{\mathrm{QF}}/G_{\mathrm{QF}}]).$$

The groupoid $\mathcal{G}_{\mathrm{QF}}$ records *gauge redundancy*, that is, equivalences of description rather than physical time evolution. Dynamics—for example, the gradient flow of a local gauge–invariant energy built from plaquette terms—is additional structure: a G_{QF}–equivariant flow on M_{QF} that descends to the quotient. We treat this dynamical layer separately from the gauge groupoid itself; see also [23] as reference.

The viewpoint adopted here is therefore two-tiered. At the general level, decorated groupoids package the local geometric and combinatorial data relevant to symbolic or hybrid computation. At the quasi-fluidic abelian level, this broad picture reduces to a concrete lattice $U(1)$ gauge groupoid whose arrows encode only local phase-frame redundancy. This distinction is intentional: transcendental or geometric data may decorate computation in general, whereas the $U(1)$ corridor isolates the simplest smooth abelian core on which the later return-map and phase-ledger constructions are built.

Conjecture 4.2 (Extended Grothendieck Homotopy Hypothesis)**.** *Let $\mathcal{C}$ be a category of smooth or stratified manifolds equipped with additional geometric structure (for example, decorations by phase, twist, chirality, flow, or other local computational data), and let*

$$(M, \mathrm{str}) \longmapsto \Pi_1^S(M, \mathrm{str})$$

denote the corresponding decorated path groupoid (or, in higher form, decorated ∞–groupoid) obtained by adjoining this structure to the ordinary homotopy data.

Then the following principle is expected to hold.

(a) ***Classical recovery.*** *When the decoration is trivial, or factors entirely through the ordinary homotopy type, the decorated object reduces to the usual homotopy-theoretic one. In this sense, the classical Grothendieck Homotopy Hypothesis is recovered as the undecorated case.*

(b) ***Enriched classification.*** *When the decoration records genuinely geometric or computational data not visible to the bare homotopy type, the resulting decorated groupoid refines the ordinary homotopy invariant. Thus two spaces may be weakly equivalent as topological spaces while remaining inequivalent as decorated computational spaces.*

(c) ***Failure of undecorated classification.*** *In general, the passage*

$$(M, \mathrm{str}) \longmapsto \Pi_1(M)$$

forgets computationally relevant structure. A faithful classifier for such spaces therefore requires an enriched object, such as a decorated groupoid, Lie groupoid, or higher groupoid carrying the additional geometric data explicitly.

(d) ***Computational interpretation.*** *In favorable cases, decorated paths may be read simultaneously as homotopies and as computational processes: composition corresponds both to path concatenation and to execution of successive geometric updates, while invariance under decorated homotopy expresses computational equivalence.*

It would be amiss not to mention the Grothendieck Homotopy Hypothesis. However, while decorated groupoids should be viewed not as replacements for ordinary homotopy types, a natural extension of them is relevant when geometry carries computational meaning. This conjectural principle is the homotopy-theoretic background for the later in appendix A.7 when the decoration factors through ordinary homotopy, one expects to recover the classical cobordism-type picture, whereas non-factorizing decorations lead to the enriched computational case. This suggests a possible enrichment of the homotopy hypothesis, in which ∞-groupoids may encode intrinsic computational or dynamical structure—a new bridge between geometry, logic, and computation foreshadowed by Grothendieck's original vision.

Definition 4.13 (Homotopy Computation)**.** Fix a GCM lattice in the quasi-fluidic regime with configuration space M_{QF}, pulse map

$$P : M_{\mathrm{QF}} \to M_{\mathrm{QF}},$$

and readout

$$\mathrm{Read} : M_{\mathrm{QF}} \to B^V.$$

A *homotopy computation* is a finite or continuous state evolution

$$\gamma : [0,1] \longrightarrow M_{\mathrm{QF}}$$

such that γ is generated by admissible local dynamics of the model (for example, gradient flow segments, pulse updates, or their hybrid composition), together with its induced readout trajectory

$$\mathrm{Read} \circ \gamma : [0,1] \to B^V.$$

Two homotopy computations $\gamma_0, \gamma_1 : [0,1] \to M_{\mathrm{QF}}$ with the same endpoints are said to be *computationally equivalent* if there exists a homotopy

$$H : [0,1] \times [0,1] \longrightarrow M_{\mathrm{QF}}$$

such that:

(a) for each fixed $u \in [0,1]$, the path $t \mapsto H(t,u)$ is again an admissible computation,

(b) $H(t,0) = \gamma_0(t)$ and $H(t,1) = \gamma_1(t)$,

(c) the relevant observable data are preserved along the deformation (for example, the same decorated invariants, or the same induced symbolic readout up to the chosen notion of equivalence).

Interpretation. This definition is intentionally regime-dependent. In the symbolic regime, a homotopy computation reduces to a path through a finite transition graph. In the quasi-fluidic regime, it is a path in the continuous phase configuration space M_{QF}, typically observed only through the stroboscopic return map and the readout Read. Thus, the same underlying idea—computation as an admissible path modulo invariant-preserving deformation—specializes to both discrete and continuous settings.

Remark 4.9. The point of the terminology is not that every computation is literally a topological homotopy, but that admissible evolutions in configuration space form path-like objects whose equivalence is controlled by the invariants one chooses to preserve. In the abelian quasi-fluidic corridor, these are typically gauge-invariant phase observables and their induced symbolic readouts.

Lie algebroid chart and Atiyah's linearization of flow. Readers from differential geometry may reasonably ask why we do not take Lie algebras and Lie brackets as the primary algebraic language throughout. The short answer is that our computational semantics is organized by *global, composable transport data* (paths, holonomy, winding/phase, and gluing across interfaces), and these are most naturally encoded at the groupoid level rather than by infinitesimal generators. Nothing here rejects the Lie-algebraic viewpoint; rather, it appears as a *linearization chart.* Given a Lie groupoid $\mathcal{G} \rightrightarrows M$, its infinitesimal object is the Lie algebroid

$$A(\mathcal{G}) \;:=\; \ker(ds)\big|_M \;\longrightarrow\; M, \qquad \rho := dt\big|_{A(\mathcal{G})} : A(\mathcal{G}) \to TM,$$

whose bracket on sections encodes first-order commutator information of local bisections. This is the many-object analogue of “Lie algebra = infinitesimal Lie group” (see, e.g., Weinstein’s expository discussion [24]). Thus, whenever one restricts to a sufficiently smooth regime where a genuine local symmetry pseudogroup governs evolution, one can pass from the groupoid semantics to an algebroid bracket calculus.

Atiyah’s gauge-theoretic instance. In particular, for a principal G–bundle $P \to M$, the gauge groupoid $(P \times P)/G \rightrightarrows M$ has associated Lie algebroid

$$A \;\cong\; TP/G \;\to\; M,$$

fitting into the short exact “Atiyah sequence”

$$0 \;\to\; \mathrm{ad}(P) \;\to\; TP/G \;\to\; TM \;\to\; 0,$$

introduced by Atiyah in his study of connections. This is precisely the setting in which the Lie bracket formalism is maximally effective: it governs infinitesimal deformations of connections and local curvature evolution [25].

Why we do not build the theory around brackets. The point of the “algebra of flow” emphasized here (cf. 5.2) is that many of the invariants relevant to computation are inherently global and persist under operations for which infinitesimal linearization is either insufficient or not stable. Accordingly, we treat Lie algebroids (and brackets) as a *derived* local chart available in the upcoming fluidic/gauge regimes, while keeping the groupoid/decorated-path level as the primitive semantic layer. Unlike Lie algebras, Lie algebroids need not integrate to global Lie groupoids without additional conditions; integrability is controlled by explicit obstructions (Crainic–Fernandes [26]). This reinforces the methodological choice: our semantics is formulated on the global/compositional side, with infinitesimal calculus used when (and only when) the regime admits it.

This work presents one possible homotopy-theoretic reading of geometric computation, offered in the hope that it may be refined, challenged, or extended by the mathematical, computer science, and physics communities.

4.6 Connections and Phase Ledgers on a Trivial Helical Bundle

The finite twin–helix module used in the basic construction is locally modeled by a regular neighborhood of an embedded arc, and is therefore topologically

$$D^2 \times I \;\cong\; B^3.$$

Thus, as a D^2–bundle over the interval I, the local helical model is topologically trivial. In particular, there is no intrinsic bundle monodromy in the D^2 fibers. This does *not* mean that the geometry is computationally trivial. Rather, it means that the information relevant to computation is not carried by the bundle class itself, but by additional geometric structure placed on this trivial scaffold: chosen phase fields, connections, deformations, and local interface rules.

This is the correct abelian analogue of the Aharonov–Bohm (AB) viewpoint [27]. In the AB effect, the underlying space may be locally simple while the physically meaningful information is carried by phase accumulation of a $U(1)$ connection around loops. The same lesson applies here: even on a topologically trivial helical module, one may impose a phase field on the transverse circular directions in each cross-section, and the resulting loop integral defines a nontrivial $U(1)$ response. Thus, the relevant invariant in the present setting is not topological twisting of the bundle, but phase accumulation determined by the chosen geometric control data.

We therefore use the above product structure to isolate the abelian structure that will persist in later sections: an S^1–valued phase accumulated around transverse circles in a fixed cross-section. In modern language, this is most naturally described by a $U(1)$ connection on a trivial circle bundle over such loops (equivalently, a Hermitian line bundle with unitary connection), although in the present discussion only the elementary notions of phase, curvature, and holonomy are required. As an organizational comparison, the product structure also means that local differential data may be written globally in coordinates; in this limited sense, one avoids the extra cocycle bookkeeping that would arise on a genuinely nontrivial fibration.

Fix a longitudinal parameter value $z_0 \in I$, and work in the transverse disk

$$D^2 \times \{z_0\}.$$

For each radius r for which the circle

$$\gamma_r(\theta) \;:=\; (r,\theta,z_0), \qquad \theta \in [0,2\pi],$$

lies in the interior of the disk, let

$$A \;:=\; \Omega(r,\theta)\,d\theta$$

be a $U(1)$ connection one–form on that transverse slice. Define the associated phase ledger by

$$U(r) \;:=\; \exp\Big(i\int_{\gamma_r} A\Big) \;=\; \exp\Big(i\int_0^{2\pi} \Omega(r,\theta)\,d\theta\Big) \;\in\; U(1).$$

Proposition 4.3 (Radial Shear Identity)**.** *Let*

$$A := \Omega(r,\theta)\,d\theta \qquad \textit{and} \qquad F := dA = \partial_r\Omega(r,\theta)\,dr \wedge d\theta$$

on a fixed transverse slice $D^2 \times \{z_0\}$. For $r_1 < r_2$, let γ_{r_1} and γ_{r_2} denote the circles of radii r_1 and r_2, and let

$$\mathcal{A}_{r_1,r_2} := [r_1,r_2] \times S^1$$

denote the annulus between them. Then

$$U(r_2)\,U(r_1)^{-1} \;=\; \exp\Big(i\int_{\mathcal{A}_{r_1,r_2}} F\Big).$$

Equivalently, the relative phase between the two radii is determined by the curvature flux through the annulus. In particular, if $\partial_r\Omega \equiv 0$ (no radial shear), then $F = 0$ on the annulus and $U(r)$ is independent of r.

Proof. By definition,

$$U(r_k) = \exp\Big(i\int_{\gamma_{r_k}} A\Big), \qquad k = 1,2.$$

Hence

$$U(r_2)\,U(r_1)^{-1} = \exp\Big(i\int_{\gamma_{r_2}} A - i\int_{\gamma_{r_1}} A\Big) = \exp\Big(i\int_{\partial\mathcal{A}_{r_1,r_2}} A\Big).$$

Applying Stokes' theorem on the annulus gives

$$\int_{\partial\mathcal{A}_{r_1,r_2}} A = \int_{\mathcal{A}_{r_1,r_2}} dA = \int_{\mathcal{A}_{r_1,r_2}} F,$$

which proves the identity. If $\partial_r\Omega \equiv 0$, then $F = 0$ and therefore the phase ledger $U(r)$ is constant in r. □

Example 4.2 (Gluing and Factorization of Phase Ledgers)*.* Let A_1 and A_2 be two annular regions in transverse slices, each equipped with a $U(1)$ connection one–form

$$A^{(1)} = \Omega_1(r,\theta)\,d\theta, \qquad A^{(2)} = \Omega_2(r,\theta)\,d\theta,$$

defined in compatible boundary trivializations. Suppose the outer boundary circle of A_1 is identified with the inner boundary circle of A_2 so that the connection data glues smoothly across the common interface. Then the glued annulus

$$A_{\mathrm{gl}} = A_1 \cup A_2$$

inherits a connection A_{gl} obtained by patching $A^{(1)}$ and $A^{(2)}$.

Let γ_1 and γ_2 denote the fundamental loops in A_1 and A_2, and let γ_{gl} denote the corresponding loop in the glued annulus. The associated holonomies satisfy the factorization law

$$\mathrm{Hol}_{A_{\mathrm{gl}}}(\gamma_{\mathrm{gl}}) = \mathrm{Hol}_{A^{(2)}}(\gamma_2)\,\mathrm{Hol}_{A^{(1)}}(\gamma_1).$$

Equivalently, writing

$$U_k := \exp\Big(i \int_{\gamma_k} A^{(k)}\Big), \qquad k = 1, 2,$$

one has

$$U_{\text{gl}} = U_2\, U_1.$$

Likewise, if

$$F^{(k)} = dA^{(k)}, \qquad k = 1, 2,$$

are the curvatures on A_k, then the curvature on the glued annulus is

$$F_{\text{gl}} = dA_{\text{gl}},$$

and Stokes' theorem gives additivity of flux:

$$\int_{A_{\text{gl}}} F_{\text{gl}} = \int_{A_1} F^{(1)} + \int_{A_2} F^{(2)}.$$

Thus, in the abelian $U(1)$ corridor, phase accumulation behaves exactly as a ledger: holonomies multiply under composition, while curvature contributions add under gluing.

4.7 Lancret's Ghost?

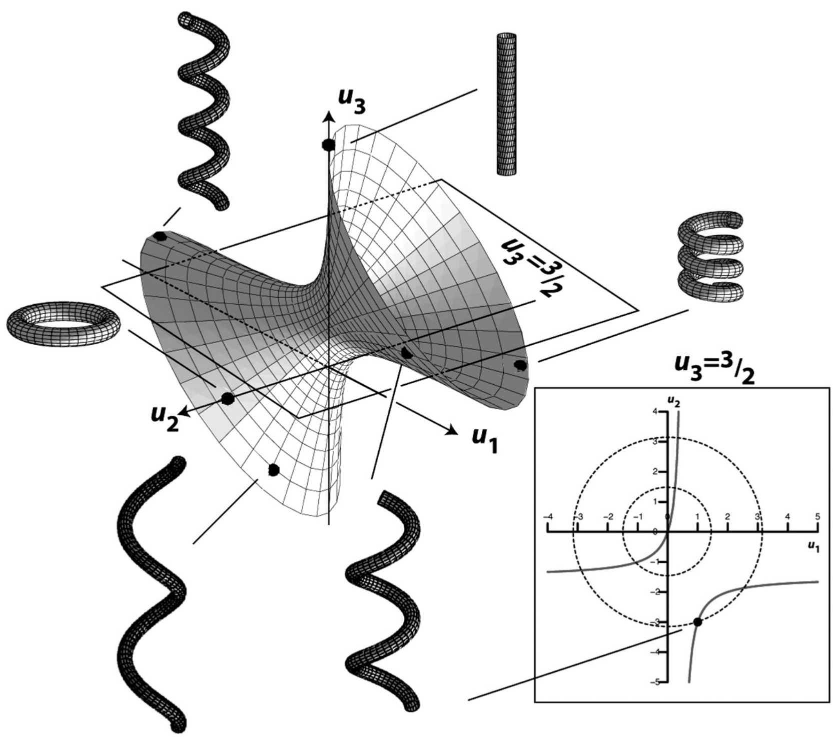


Figure 4: Homology, Helicity, and Hyperbolicity: a geometric "Rosetta Stone." *(Image Courtesy of Alain Goriely, "Helices" 2006)*

Helices are mathematical objects rich in dynamical variables—as explored in this section—whose properties interact synergistically but without the constraints of maximal symmetry. This flexibility has arguably obscured their structural significance. A helix is not merely a curve in space, but a framed, oriented 1–manifold carrying both axial and cyclic data. As earlier sections suggest, maximal symmetry tends to suppress geometric expressiveness, whereas physically useful computation requires enough asymmetry to support nontrivial state encoding and controlled transitions. Helices can dance with ones and zeros, but a sphere will never lie.

The helix presents a unique challenge in mathematical physics: despite its ubiquity in nature—from DNA, to protein folding, to solitons—it resists easy classification in modern foundational frameworks.

The standard helix relies on *transcendental* functions (e.g., $\gamma(t) = (a\cos t, a\sin t, bt)$), and as a result, it falls outside the traditional scope of algebraic geometry, which deals natively with polynomial varieties. As a result, the helix constitutes something of a disciplinary orphan: too smooth for pure combinatorics, too transcendental for algebraic geometry, and too continuous for discrete computation. Yet, for a theory of physical computation—especially one grounded in constructible geometric objects—this specific structure must take center stage.

Emergence of helices in nature. Helices occur constantly in physics, chemistry, and biology precisely because they integrate two forms of structure at once: cyclic phase and longitudinal transport. This tension may even be computationally significant. Recent work on the internal representations of large language models [28] suggests that gradient-based optimization can give rise to helical or helix-like internal organization for arithmetic tasks. Though such findings do not admit a universal interpretation, they at least reinforce the plausibility that helices are not merely common physical shapes, but natural carriers of computation whenever cyclic and linear degrees of freedom must be coordinated.

And while helices offer remarkable efficiency for encoding computational structure, it is essential to note that not all helices in nature are used for computation, nor do they dominate large-scale architectures. Helical forms excel at packing information, facilitating transmission, and exploiting symmetry at small scales (e.g., DNA and chiral molecules). However, at larger scales, nature often favors other symmetries—branching, fractal, planar, or modular structures—driven by stability, energetic, and functional constraints. Thus, the computational potential of helices arises not from their ubiquity but from their unique ability to localize and propagate information under appropriate conditions.

The helical moduli problem. Lancret's theorem (1802) already hints that "being helical" is not a local accident: a space curve $\gamma \subset \mathbb{R}^3$ is a circular helix iff the Frenet invariants satisfy $\tau/\kappa \equiv C$. Equivalently, helices are integral curves of a geometric constraint in the 2-jet bundle of immersed curves [29, 30]. In our setting, however, the computational objects are *thickened* and *decorated*: a helical *spine* carries a framed tubular neighborhood (cross-section data), marked boundary ports for gluing, and (in higher regimes) connection/holonomy data. The natural "moduli space" is therefore not a single finite-dimensional manifold, but a quotient of an infinite-dimensional configuration space by reparametrization and symmetry. Once one allows variation in framing, cross-section, pitch, chirality, and coupling data, the resulting parameter spaces become extremely large.

Definition 4.14 (Helical Moduli Stack, Working Model)**.** Fix a smooth compact *template* helical body H (e.g. a capped tube with marked base disk and chosen framing class) and a regularity class C^k (or Sobolev H^s) appropriate to the regime. Let $\mathrm{Emb}^k_{\mathrm{fr}}(H, \mathbb{R}^m)$ be the space of framed C^k-embeddings, and let $\mathrm{Diff}^k_{\mathrm{fr}}(H)$ act by reparametrization (preserving the marked boundary structure), together with the ambient Euclidean group $E(m)$ acting by rigid motion. We define the moduli *stack*

$$\mathfrak{M}^k_H \;:=\; \Big[\mathrm{Emb}^k_{\mathrm{fr}}(H, \mathbb{R}^m) / \big(E(m) \times \mathrm{Diff}^k_{\mathrm{fr}}(H)\big)\Big],$$

and its coarse moduli space M^k_H as the corresponding orbit space when it is well-behaved. Allowing variable cross-sections amounts to fibering this construction over a shape space of embedded disks (or contours), which increases the functional degrees of freedom.

This formulation is concrete enough to state what must be constructed: local models (Banach/Fréchet charts depending on k or s), a stratification by singular phenomena (self-contact, self-intersection, knotting/linking type, degeneration to the torus-like and straight-rod limits), and gluing maps induced by the marked boundary ports. In particular, the difficulty is not naming the quotient; it is extracting computable invariants from it in a way that is stable under deformation and compatible with composition [31].

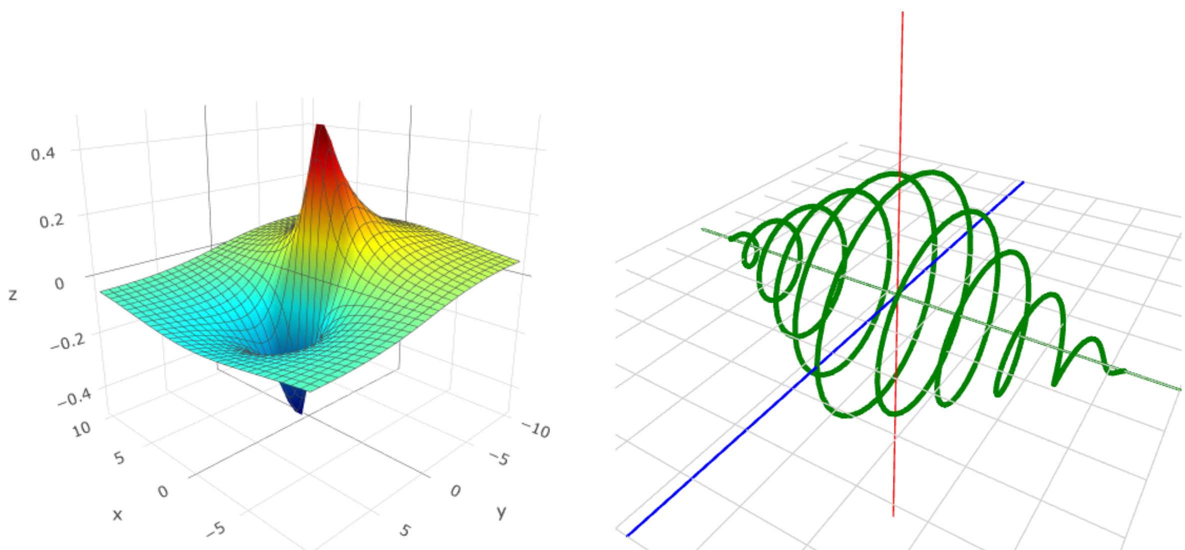
Once one treats computation or physics as *local-to-global* with observables defined locally on helical pieces and composed by gluing, one is forced toward an integration/pushforward formalism on a stratified quotient object. In other words, the analogue of a path integral or partition function in geometric computation is a functorial assignment built from (a) passage to moduli (quotienting by symmetry), and (b) a gluing law that behaves like a Fubini theorem for composition. Developing such an "integration on moduli of helical histories" is exactly the point at which one should expect derived/stacky and constructible techniques to enter, much as they do in Kontsevich-style theories of integration over highly singular, infinite-dimensional parameter spaces [32].

Overlooked but important reading on helicity. For a continuum–mechanics foundation on why helices are generic energy minimizers (and how curvature/torsion couple), see Goriely [33]; for a crisp, canonical derivation of when minimizers are helical with pitch/radius set by constraints, see Chouaieb et al. [34]. Developable–ribbon energetics and torsion–curvature locking—which motivate "channel" formation in cross–sections—are captured by Starostin–van der Heijden [35]. On the geometric side that is often missed, Adachi [36] provides a leading example of how (helical) Killing fields and foliations structure flows, aligning with our base–join transport picture. Ferrández Izquierdo's recent monograph [37] will also provide a focused, modern treatment of helical geometries to unify these threads for newcomers.

For all we know? Perhaps the key to modern quantum theory lurks quietly in the diagrams of a century-old German treatise (Loria [38]) on transcendental curves—waiting for someone to notice that helices and hyperboloids speak a universal geometric tongue.

5 Geometric Computational Field Theory

With the geometric and computational preliminaries now in place, we turn to the intersection of modern quantum physics. The aim of the present section is not to reopen the full problem of quantization, but to show that the structures developed above—holonomy, bordism, curvature, chirality, and coarse-grained state spaces—naturally dovetail with questions that have also appeared in the work of Kontsevich [39], Connes [40], and Witten [41], among others. Our claim is not that these programs are identical, but that they touch a common boundary at which geometry, topology, and computation can no longer be cleanly separated. It is from that boundary that the final section proceeds.



(a) BPST instanton: a finite-action, localized topological solution to the Yang–Mills equations, exhibiting geometric twist and winding.

(b) Complex quantum wave: a superposition of oscillations, illustrating the intertwined, helical modes of computation.

Figure 5: Familiar examples of geometric configurations corresponding to Type 3 and Type 1 THM.

5.1 Curvature as Computation

After assembling THM in the abelian, quasi–fluidic band—where computation is well captured by finite local operations on geometric primitives—we now enter the *fully fluidic* regime. Regard the helix lattice as a bundle with a connection. Let

$$\pi : P \to B$$

be a principal G–bundle over a computably presented base B (the working lattice region), with associated state bundle $E := P \times_G \mathsf{F}$ whose fiber F encodes the finite–dimensional internal variables of a twin helix cell. A connection $A \in \Omega^1(B; \mathfrak{g})$ prescribes parallel transport, and its curvature

$$F \ = \ dA + A \wedge A \ \in \ \Omega^2(B; \mathfrak{g})$$

measures path–dependence. In this language, a *join* is no longer purely combinatorial: it is the composition of parallel transports along adjacent interfaces. When G is abelian (e.g. $U(1) \simeq SO(2)$), joins commute; once G is non–abelian, joins become order–sensitive and composition acquires memory.

Example 5.1 (Holonomy as Computational Memory). For a loop γ based at $x \in B$, the parallel transport

$$U_\gamma \ = \ \mathcal{P} \exp\Big(\int_\gamma A \Big) \in G$$

is the *holonomy*. The subgroup $\mathrm{Hol}_x(A) \subseteq G$ generated by all such U_γ is the effective symmetry seen at x. In the abelian limit $U_{\gamma_1} U_{\gamma_2} = U_{\gamma_2} U_{\gamma_1}$ and only additive flux matters; in the non–abelian case, path ordering is essential and different execution paths may yield nonconjugate elements, even for homotopic loops when $F \neq 0$. The gauge–invariant readouts are the *conjugacy classes* $[U_\gamma]$ (or class functions such as $\mathrm{tr}_\rho(U_\gamma)$). The following now hold:

(a) **Registers as loops.** Minimal cycles in the lattice carry holonomies $\{U_{\gamma_k}\}$ serving as state registers; the state is the family $\{[U_{\gamma_k}]\}$.

(b) **Updates as transport.** Driving fields update A and hence $\{U_{\gamma_k}\}$; gluing cells changes the loop basis.

(c) **Curvature as storage.** F encodes what cannot be gauged away—persistent, path–dependent context (geometric memory).

This is the geometric analogue of Wilson loops: the observable is the holonomy, *not* the potential itself. In the quasi–fluidic abelian band, holonomy reduces to a scalar phase (orientation logic, integer twist counters). Entering the fluidic/gauge band promotes these to matrix phases (non–abelian Berry/Wilczek–Zee transport) [42], and join symmetry breaks via $[U_{\gamma_1}, U_{\gamma_2}] \neq e$.

From abelian phases to non–abelian transport. Each twin helix cell has a finite internal state space; as parameters vary over B, these states form a vector bundle $E \to B$ (the *band bundle*). Adiabatic transport induces a connection on E. If $\mathrm{rk}\, E = 1$ over a region, the induced connection is abelian (Berry phase), hence commuting joins. When degeneracy persists ($\mathrm{rk}\, E \geq 2$ on an open set), the connection becomes non–abelian (Wilczek–Zee): parallel transports no longer commute and path ordering becomes a resource. Base anisotropy and helix–helix coupling are precisely the mechanisms that create protected degeneracy and thus promote $U(1)$ to $SU(2)$–type dynamics.

Definition 5.1 (Join Commutator in the Interface Groupoid)**.** Let $\mathcal{J}$ be the interface groupoid whose objects are boundary faces and whose morphisms are admissible joins (parallel transports along interfaces). For $J_1, J_2 \in \mathrm{Hom}_{\mathcal{J}}(X, Y)$ with common source/target, set

$$[J_1, J_2] \ := \ J_1 J_2 J_1^{-1} J_2^{-1}.$$

Joins *commute* if $[J_1, J_2] = \mathrm{id}$ in $\mathrm{Hom}_{\mathcal{J}}(X, X)$.

Computation now proceeds by progressively internalizing curvature; symbolic joins become ordered transports, and the state is read off from conjugacy classes of holonomies. Expressivity exceeds purely symbolic models because time–ordering, loop creation/annihilation, and curvature redistribution generate behaviours not captured by classical finite–step semantics.[6] Physically, the twin helix fiber is the matter degree of freedom, and A is the control field; noncommutativity (path dependence) turns the join algebra into a genuine gauge computation.

[6] We refrain from machine–model claims. Under continuum idealizations, non-abelian gauge dynamics are known to encode phenomena beyond classical Turing models; our interest here is programmability via $(A, F, \mathrm{Hol}(A))$.

Spin textures as fluidic defects. A THM defines an internal orthonormal 2–frame; choosing a unit normal gives a map $n : B \to S^2$. Its defects organize memory channels:

- **Vortex/phase defects** (*abelian*). Windings in $U(1)$ along loops (π_1),
- **Skyrmion–type textures** (*non–abelian*). Degree of n on embedded 2–spheres ($\pi_2(S^2) \cong \mathbb{Z}$),
- **Linked textures** (*Hopf–type*). Linking of preimages $n^{-1}(p)$ and $n^{-1}(q)$ for $p \neq q$ ($\pi_3(S^2) \cong \mathbb{Z}$ in 3D thickenings).

These classes appear in the holonomy spectrum: abelian vortices shift scalar phases; non–abelian textures produce non-conjugate holonomies around homologous loops, yielding distinct class–function readouts. The central split in summary:

(a) Symbolic/Quasi–fluidic. G is effectively $U(1)$, joins commute, memory is a scalar phase or integer flux.

(b) Fluidic/Gauge. G is non–abelian (from helix symmetry and interface groupoid), joins do not commute, memory lives in holonomy conjugacy classes; curvature stores context.

The base topology controls available memory channels: on contractible B with $F = 0$ holonomy is trivial up to conjugacy; nontrivial F, defects, or handles produce noncommuting generators. In the terms of lemma 2.4 and table 7, flipping across from the quasi–fluidic abelian band to the fluidic/-gauge band trades raw orientation microstates for noncommuting holonomy, where the true expressive capacity of geometric computability is realized. Fully fluidic GCMs begin to resemble endless streams of information as increasingly chaotic computational phenomena appear.

Example 5.2 (Spectral Flow as a Computable Readout). Driving the system around a control cycle S^1 promotes B to $B \times S^1$. The net change in holonomy class across a period is governed by the integral of curvature over spanned 2–cycles (a Thouless–type pump):

$$\Delta\big(\text{class function on } U_\gamma\big) \;=\; \int_{\Sigma \subset B \times S^1} \operatorname{tr}(F) \quad \text{(abelian)}, \qquad \Delta\big(\operatorname{spec}(U_\gamma)\big) \text{ by } \int \operatorname{tr}(F \wedge F) \text{ (non–abelian)}.$$

Thus, periodic control implements a bordism in parameter space, and evaluation of class functions on Wilson loops gives an integer (or discrete spectrum) that is stable and algorithmically retrievable in our symbolic presentation.

Symplectic vs. contact presentation. Although we have not formulated the above regime in the standard symplectic/contact dialect, much of the structure we use admits a direct translation. A fully fluidic "state space" is naturally modeled as a symplectic manifold (X, ω) (phase space), with evolution given by Hamiltonian dynamics or a controlled perturbation thereof; see [43]. When one restricts to an energy hypersurface $\Sigma \subset X$ of *contact type* (equivalently: there exists a primitive λ with $d\lambda = \omega$ near Σ such that $\alpha := \lambda|_\Sigma$ is a contact form), the induced characteristic dynamics on Σ is canonically described by the *Reeb vector field* R_α determined by

$$\alpha(R_\alpha) = 1, \qquad (d\alpha)(R_\alpha, \cdot) = 0,$$

(cf. Geiges [44]). In this setting, Hamiltonian flow restricted to Σ is a reparametrization of Reeb flow: for suitable defining Hamiltonians, one has

$$X_H|_\Sigma = f \cdot R_\alpha \quad \text{for some positive function } f,$$

so closed Hamiltonian characteristics on Σ are the same as periodic Reeb orbits.

Periodic cycles as discrete readouts. From this perspective, the "periodic control cycles" and robust closed trajectories used as readouts in our fluidic discussion can be viewed (after choosing a contact-type boundary chart) as periodic Reeb orbits and their associated action/index data. This is precisely the regime in which symplectic/contact methods (Floer-theoretic and SFT-type constructions) extract deformation-invariant algebraic information from Reeb dynamics [45]. We do *not* invoke the full machinery of symplectic field theory here; the point is only that the fluidic limit admits a recognized contact-geometric presentation, and that the periodic-orbit language provides a standard interface for readers coming from Hofer-style dynamics.

Interpretation. In our terminology, passing to a contact/Reeb chart is best viewed as a *change of presentation* (a local coordinate system on the same process space) rather than an additional ontological assumption. It supplies a conventional vocabulary for "time-like" evolution on contact-type boundaries and for the discrete invariants carried by periodic cycles.

5.2 The Algebra of Flow

Starting in the quasi–fluidic regime, interactions are not purely symbolic: they are geometric and dynamical—the information is encoded in the differential structures of both flow and form. In the fully fluidic regime, the cross–sectional geometry, as discussed in section 2.6, reappears. Why the focus?

Two kinds of analog computation. Computation as we know it splits as such:

- *(Algebraic analog flow)* Rank–1 bundles with abelian holonomy ($U(1)$): states are continuous but updates commute; residual symmetry imposes selection rules; behavior is predictable.
- *(Geometric analog chaos)* Rank–≥ 2 degeneracy with non–abelian holonomy ($SU(2)$–type): path ordering matters; joins do not commute; internal mixing encodes control flow; behavior is history–sensitive. In Felleisen's sense [4], these differ at the level of *core constructs*, not mere "data types."

Symmetry selection rules (abelian kinematics). When a local annulus $A \simeq S^1 \times I$ around a helix cell carries a residual C_n symmetry, transport admits a harmonic decomposition $e^{\mathrm{i}m\theta}$ in which the connection couples only modes $m \equiv m' \pmod{n}$. Transport block–diagonalizes by angular–momentum class modulo n, and the induced holonomies on each block commute. In this quasi–fluidic band, joins commute and memory reduces to scalar phases or integer twist counters. Breaking C_n (or leaving a round annulus) removes these selection rules; see appendix A.8 for the calculation.

Promoting $U(1) \to SU(2)$ by degeneracy. Helix–helix coupling plus base anisotropy can maintain a rank–≥ 2 degenerate internal subspace across a region of B. Adiabatic transport on this subbundle is then *non–abelian* (Wilczek–Zee): path–ordered products of transports do not commute, and the state carried by a loop is a conjugacy class $[U_\gamma]$ rather than a single phase. Practically, anisotropy supplies off–diagonal couplings while coupling between the twin strands stabilizes the degeneracy that upgrades $U(1)$ to $SU(2)$–type dynamics.

Definition 5.2 (Macro–expressiveness)**.** For two geometric computation models A, B, write $A \succeq_{\mathrm{macro}} B$ if there is a finite library of local transports/joins/defect insertions in A whose *gauge–invariant readouts* (class functions on holonomy) can *macro–express* every primitive of B with bounded overhead, preserving observational equivalence at the boundary. Abelian $U(1)$ lacks path ordering and conjugation gates; the non–abelian regime supplies them, hence the jump in macro–expressiveness.

Base topology and gauge–invariant readouts. On contractible patches with $F = 0$, holonomy is trivial up to conjugacy, so no persistent state is stored. Nontrivial memory channels arise from curvature, defects, or handles. Gauge transformations act by conjugation on holonomies; the physically meaningful readouts are *class functions* of U_γ (e.g. traces in fixed representations).

Computational tradeoffs: algebraic vs. geometric analog.

- **Memory carrier.** *Algebraic analog ($U(1)$)*: a scalar phase per loop. *Geometric analog ($SU(2)$)*: a conjugacy class per loop (equivalently, spectra or traces $\mathrm{tr}(U_\gamma), \mathrm{tr}(U_\gamma^2), \dots$).
- **Registers and gates.** Registers are chosen loops $\{\gamma_k\}$. In $U(1)$ gates add phases (commuting joins); in $SU(2)$ gates are ordered transports with nontrivial commutators $[U_{\gamma_i}, U_{\gamma_j}] \neq \mathrm{id}$.
- **Control knobs.** (a) introduce base anisotropy beyond the first harmonic; (b) maintain internal degeneracy (rank ≥ 2 band bundle); (c) add defects/handles so $\pi_1(B)$ generates noncommuting holonomies; (d) redistribute curvature to reprogram stored context.

- **Verifiability.** $U(1)$: readout is a phase/integer. $SU(2)$: readout uses class functions (e.g. traces of Wilson loops) that remain algorithmically retrievable in our symbolic presentation.

Since the base geometry controls capacity, on simply–connected regions with $F = 0$, holonomy classes are trivial; introducing curvature, handles, or protected degeneracy creates noncommuting generators and promotes the system from the quasi–fluidic abelian band to the fluidic/gauge band. This is the bridge where join algebra becomes order–sensitive and holonomy conjugacy classes serve as programmable memory.

Now the internal geometry of the cross–section becomes a new axis of calculation. Beyond circles, one can admit distinct flow channels, branch lines, and defect cores within the same cross–section; these change the *isotypic decomposition* of transport and create/maintain degeneracies. Higher symmetry (round annulus) yields fewer angular sectors and abelian blocks; broken symmetry and multi–channel shapes create protected mixing and non–commuting gates. Thus cross–section complexity controls the number and kind of "channels" available for computation: not just *which* states, but *how* transport moves *within* a state.

Remark 5.1. Only the geometric analog can outstrip algebraic field–operation models under standard readouts: chaotic/PDE dynamics can exceed polynomial/field–arithmetic semantics as shown by Pour-El and Richards study of the wave equation [46]. In our constructive calculus (section 5.5), such steps appear as *effects* with witnesses (no hidden oracles), preserving normalization and observational equivalence on gauge–invariant readouts.

5.3 Helicity and the Origin of Quantum Spin Networks

Although our program does not aim to adjudicate unification debates, aspects of the geometric computation model do resonate with the combinatorial, categorical, and background-independent *perspectives* emphasized in Loop Quantum Gravity.

As developed throughout this work, helices exhibit exceptionally rich geometric behaviour. The pre-quantum computational geometry of twin helices—where information resides in flows, holonomies, and deformations—can supply a substrate from which discrete boundary data (as in spin-foam state spaces) may emerge after coarse-graining. Rather than postulating quantum discreteness as primitive, it is reasonable to view quantum structure as an invariant/stable phase of the underlying geometric dynamics, much as solitons or topological defects arise from continuous systems. In this picture, coarse-graining the micro-geometry of helices can yield spin-network–like graphs (cf. section 2.5); the effective pure-state manifold of a qubit is $\mathbb{CP}^1 \cong S^2$ (the Bloch sphere), while discreteness enters at the level of projective readout rather than the underlying state space.

Our viewpoint also aligns with strands of work beyond LQG: non-Markovian/correspondence ideas in quantum foundations (e.g. Barandes [47]), relativistic quantum information (Fuentes-Schuller [48]), the positive geometry program in scattering amplitudes (Arkani-Hamed et al. [49]), holonomy-first approaches (Grimstrup-Aastrup [50]), and deterministic explorations ('t Hooft [51]).[7] In our reading, discrete quantum geometry then appears as a phase transition within a more general continuous geometric logic—without discarding the manifold substrate.

Example 5.3 (The Quantum Tetrahedron). Fix a working region B and a principal G–bundle $P \to B$ with connection A, as in the fluidic discussion. Choose a cellular decomposition dual to the helix lattice (plaquettes/handles adapted to the THM_4 pattern), and let Γ be the resulting 4–valent graph:

- **Edges** $e \in E(\Gamma)$. Homology classes of minimal loops threading twin helix channels. The nonabelian holonomy U_e around e has conjugacy class $[U_e]$; its spectrum (eigenphases) is the gauge–invariant "register value."

- **Edge labels.** Are extracted by coarse–graining class functions of U_e. For $G = SU(2)$, the spectrum organizes into unitary irreps; we record the isotypic content by an $SU(2)$ spin $j_e \in \frac{1}{2}\mathbb{N}$. Operationally, j_e is the minimal label whose character χ_{j_e} matches the measured class functions $\{\mathrm{tr}(U_e^k)\}$ on e.

- **Vertices** $v \in V(\Gamma)$. These arise where four channels meet (the V_4 cell). Gauge covariance imposes the Gauss law $\prod_{e \ni v} U_e \simeq \mathrm{id}$ (up to conjugation); in representation terms this singles out an $SU(2)$ intertwiner $\iota_v \in \mathrm{hom}\big(\bigotimes_{e\to v} V_{j_e}, \bigotimes_{e \leftarrow v} V_{j_e}\big)$.

[7] Regarding superdeterminism: GCT rejects this. Measurement settings are treated as exogenous boundary data; at the ontic level, this induces measurement dependence via a global, time-symmetric constraint, while linear/gauge symmetries preserve operational free choice and no-signalling.

Thus each coarse cell yields a labeled pair $(\Gamma, \{j_e\}, \{\iota_v\})$, i.e. a spin network state in the standard sense. No discreteness hypothesis was used: it is the *stability* of the holonomy conjugacy classes under small deformations (and the V_4 join rule) that makes the labels robust.

Choice of group: Z_4 vs V_4? The two independent binary symmetries of a twin helix cell (left/right, up/down) generate $V_4 \simeq \mathbb{Z}_2 \times \mathbb{Z}_2$. Classifying surface modes or boundary excitations by $\mathbb{Z}_4$ collapses these axes; phrasing them as V_4 cleanly factors the parity data that seed the four–valent node. In practice, this matches the base 2–geobit state space and yields a sharper route from micro–geometry to the 4–valent "quantum tetrahedron" intertwiners.

Spin foams as computational bordisms. Helix updates (gluing, defect motion, curvature redistribution) generate a 2–complex $\mathcal{F}$ whose faces track the evolution of edge holonomies and whose vertices encode local join moves. Coarse–graining sends such an update to a labeled 2–complex (a spin foam) whose faces carry j–labels and whose edges carry intertwiners, precisely the data of LQG transition amplitudes. In our setting, amplitudes arise from products of local class functions evaluated on Wilson loops, making the construction functorial with respect to computable bordisms. The "foam" is therefore an emergent bookkeeping of gauge computation, not a fundamental discretization of spacetime.

Spin networks correctly capture the *gauge–invariant* content (conjugacy classes and intertwiners) of nonabelian holonomy on coarse loops and junctions; spin foams capture its elementary updates. In the helix picture, these are *phase data of transport* through THM_4 cells, stabilized by join rules and curvature, rather than postulated "atoms of space." Background independence is recovered at the coarse level because all readouts are class functions (insensitive to coordinate choices), even though the micro–dynamics live on a smooth 4–manifold.

Proposition 5.1 (Coarse–graining to Spin Networks)**.** *There is a canonical coarse–graining functor*

$$\mathcal{C}: \frac{\textit{helix configs on } (B, A)}{\textit{computable gauge bordism}} \longrightarrow \frac{\textit{spin networks } (\Gamma, \{j_e\}, \{\iota_v\})}{\textit{graph isomorphism}}$$

sending minimal loop holonomies to edge labels and join constraints to intertwiners. Moreover, $\mathcal{C}$ is compatible with the composition of updates: a computable bordism of helix configs maps to a spin foam between the corresponding spin networks.

Proof sketch. Gauge–invariant readouts are class functions of holonomy; spectral data of U_e define j_e via the $SU(2)$ character table. Local gauge constraints at V_4 junctions select the invariant subspace, i.e. an intertwiner. Elementary updates act by conjugation and fusion/splitting, which descend to foam faces/edges. Functoriality follows from the locality of joins and the multiplicativity of class functions along concatenated loops. □

Baez's and Barrett's Quantum Tetrahedra inside THM_4 At a single V_4 cell, four edge labels $\{j_1, \ldots, j_4\}$ and an intertwiner ι define a quantum tetrahedron. In the helix model, these labels come from the spectra of the four minimal loop holonomies, while the join law enforces ι. The usual area/angle relations are thus coarse summaries of underlying nonabelian transport around the cell.

Physical evidence of note for GCT/LQG. Recent work by Bliokh et al. [52] has demonstrated that the surface Maxwell modes at media interfaces are governed by a topological invariant valued in $\mathbb{Z}_4$. Their approach, rooted in phase winding, represents a significant advance in our understanding of topological invariants in photonic systems. However, in my view, this classification does not fully account for the independent binary symmetries present in the physical configuration—specifically, the left/right and up/down combinations encoded in a THM_4 construction. While the choice of $\mathbb{Z}_4$ as an invariant is logical within their framework, a reformulation in terms of the Klein four-group V_4 more accurately reflects the underlying symmetry structure. The viewpoint directly parallels the base 2-geobit state space and LQG tetrahedra, providing both a more granular symmetry classification and a richer foundation for both physical insight and computational applications.

Summary. Spin networks/foams are not competitors to geometric computation; they *are* its coarse invariants. The helix micro–geometry supplies the nonabelian holonomy that the LQG combinatorics were designed to encode. In this sense, LQG's discrete data are recovered as a stable phase of a continuous, programmable gauge dynamics, and need not be elevated to fundamental kinematics.

5.4 On the Expressive Power of Physical Computation

Before stating our final claim, we collect a few canonical gauge–theoretic solutions whose geometry already exhibits the helical/twisting features emphasized in this work. The point is not to assert equivalence, but to note recurring *mechanisms*—periodicity, compactification twists, unstable saddles, and chiral transport—by which the connection A accumulates phase or matrix holonomy along preferred directions.

- **Calorons.** Periodic instantons on $\mathbb{R}^3 \times S^1$, whose Euclidean time compactification enforces a helical/phase structure along the circle [53].
- **Twisted solutions.** Gauge fields on compactified or cylindrical backgrounds with nontrivial boundary conditions; the twist enforces a globally enforced phase (or matrix) winding [54].
- **Sphalerons.** Unstable saddle points in electroweak theory; they sit atop energy barriers and encode large, topological reconfigurations (holonomy changes) along interpolation paths [55].
- **Witten's twisted constructions.** Lower–dimensional/topologically twisted reductions where the effective degrees of freedom are organized by holonomy classes and cohomological data [56].
- **Non-abelian plane waves.** Chiral, order-sensitive transport in non-abelian gauge theory; parallel transport along wavefronts produces noncommuting holonomies [57].

Taken together, these exemplars highlight three recurring themes: (a) *periodic/compactified directions* that act as helical axes (calorons, twisted sectors), (b) *unstable saddles* that mediate large holonomy moves (sphalerons), and (c) *intrinsically chiral, non-abelian transport* (plane waves). In what follows, we leverage exactly these mechanisms—now encoded by holonomy class functions on chosen loops—to motivate the usage of geometric computation.

While this discussion does not claim to exhaust the intricate connections between geometric computation and modern field theory, it is instructive (cf. fig. 5a) that gluing two helical manifolds along a common boundary produces patching data directly analogous to the clutching that underlies BPST instantons. We now enter the final of the four computational regimes, *emergent gauge*.

Each unit helix manifold H_A, H_B is modeled by open sets $U_A, U_B \subset M$ carrying local trivializations of a principal G–bundle $\pi : P \to M$. Gluing the helices along their common base is encoded by a transition map

$$g_{AB} : U_A \cap U_B \longrightarrow G,$$

which acts on the fibers; on the overlap, the base identification is the identity. In gauge theory, g_{AB} specifies how local data patches are related; geometrically, it records how internal frames are rotated when crossing the interface. Deformations of the gluing (and of g_{AB}) change the induced holonomies on linking loops and thus realize discrete computational updates. In this sense, a unit helix behaves solitonically, and computation in this regime naturally traffics in the same patching/moduli structures that organize the instanton sector of Yang–Mills theory. This positions geometric computability as a concrete starting point for a *computational Yang–Mills* framework: a GCM presented as a bundle with connection whose updates are local in the bundle charts and functorial under gluing.

$$\begin{array}{ccc} \text{Local helix chart } (U_A) & \xrightarrow{\quad g_{AB} \quad} & \text{Local helix chart } (U_B) \\ \downarrow & & \downarrow \\ \text{Trivialization } P|_{U_A} & & \text{Trivialization } P|_{U_B} \end{array}$$

The diagram summarizes the configuration space of gluings: once a connection is chosen and equations (e.g., ASD) are imposed, one arrives at a moduli space modulo gauge—the familiar organizing object for instantons.

Example 5.4 (Minimal Bundle–Connection Data). Let $P \to M$ be a principal G–bundle with local 1–forms $A_A \in \Omega^1(U_A, \mathfrak{g})$ and $A_B \in \Omega^1(U_B, \mathfrak{g})$ related on overlaps by

$$A_B \;=\; g_{AB}^{-1} A_A \, g_{AB} \;+\; g_{AB}^{-1} d g_{AB}, \qquad F_B \;=\; g_{AB}^{-1} F_A \, g_{AB},$$

with $F_A = dA_A + A_A \wedge A_A$. Viewed as a time-bordism on $M \times I$ (temporal gauge), the BPST sector corresponds to $G = SU(2)$ and anti-self-dual curvature,

$$F + {*F} \;=\; 0.$$

On a compact 4-manifold $\widehat{M}$ (a compactification of $M \times I$), the instanton number is

$$k \;=\; \frac{1}{8\pi^2} \int_{\widehat{M}} \mathrm{tr}(F \wedge F) \in \mathbb{Z},$$

and the boundary Chern–Simons functional changes by $\Delta \mathrm{CS} \;=\; \int_{\widehat{M}} \mathrm{tr}(F \wedge F) \;=\; 8\pi^2 k$ under standard normalizations.

In the abelian/quasi–fluidic limit $G = U(1)$, a loop $\gamma \subset M$ carries the computational phase

$$d_\tau([\gamma]) \;=\; \mathrm{Hol}_\gamma(A) \;=\; \exp\Big(\mathrm{i} \int_\gamma A\Big), \qquad d_\chi \in C_2 \text{ (chirality)},$$

and these labels transform covariantly under a change of patching g_{AB}. Gluing two helices along $U_A \cap U_B$ modifies the transition map ($g_{AB} \mapsto h\, g_{AB}$) and hence the induced holonomies on linking loops—implementing a discrete update at the interface. In the $SU(2)$ sector, a computational step can be idealized as traversing an ASD configuration on $M \times I$; the induced change in the boundary Chern–Simons functional is governed by k, providing a topological gate in the emergent-gauge regime.

At the level of *computational role*, effectively abelian regimes correspond to the stable carriage of information: state is propagated in a periodic, order-insensitive, and comparatively reversible manner, so that the dominant behavior is storage, transmission, or recurrence rather than structural transformation. By contrast, non-abelian regimes are characterized by order-sensitive evolution, where the flow itself performs the computation: composition is no longer reducible to simple commutative propagation, and state is actively reshaped through mixing, routing, and nontrivial transition. In the highest regime—*emergent gauge*—computation becomes genuinely history-dependent and topological. There, the relevant logical content is carried not only by the states themselves but by the global manner in which one state is connected to another, so that transition processes—for example, sphaleron-type barrier crossings—become part of the computation itself (see Manton [58]). The current hypothesis in this paper, that the geometry of a Yang-Mills field is, in all likelihood, some form of analog computation comprised of helices, is outlined in table 8.[8]

Interpretation of the regime labels. The classifications in table 8 are *computational* rather than group-theoretic. They do not assert that all examples in a given column are abelian or non-abelian as gauge fields. Instead, they classify the *role* played by the configuration in the computational picture.

In particular, calorons are genuinely non-abelian Yang–Mills configurations, but they are placed in the quasi-fluidic regime because their operative structure is cyclic and holonomy-driven: state is transported around a compact S^1 direction, and memory is encoded by recurrence and phase accumulation. By contrast, instantons appear in the gauge regime when the emphasis is on isolated topological sector change.

Example 5.5 (The Computational Consequences of Non-Abelian Plane Waves). As a motivating example of one solution cited above, consider the appearance of non-abelian plane waves as described by Coleman:

$$L = -\frac{1}{4}\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) = \frac{1}{2}\sum_{a=1}^{n}\left[(E^a)^2 - (B^a)^2\right] =: \frac{1}{2}(E^2 - B^2).$$

This is discussed by Kastrup [60], where he notes (with some exasperation) that plane–wave solutions with equal electric and magnetic energy densities satisfy $E^2 = B^2$ and hence $L = 0$. In the non-abelian

[8] As to why the Universe chooses to *express itself* this way? The author kindly asks that questions be directed *elsewhere* (cf. Penrose [59]).

Table 8: Computational regimes in GCT and the geometric structures that realize them.

Regime	**Computational signature**	**Primary state carrier**	**Topological role**	**Representative physical exemplars**
Symbolic	Discrete, stable, addressable states; local read/write behavior; robust storage.	Local occupation values, boundary conditions, or effectively commuting observables.	Topology is present but inactive; it serves as a fixed background structure rather than as the active computational variable.	Classical bits, digital memory, effectively abelian electromagnetic storage modes.
Quasi-fluidic	Cyclic or clocked evolution; recirculating transport; finite memory carried through repeated traversal of a loop.	Holonomy, phase accumulation, or other history-bearing data associated with motion around a compact direction.	Topological content is accumulated and reassembled over a cycle rather than expressed as a single isolated transition.	Calorons, periodic instantons, finite-temperature Yang–Mills configurations with nontrivial holonomy.
Fluidic	Distributed continuous transformation; order-sensitive mixing; non-commutative transport across an extended medium.	Flow profile, relative ordering of non-commuting updates, chiral or braided propagation patterns.	Topology is secondary to transport dynamics; the main effect is transformation by extended flow rather than sector change.	Non-abelian plane waves, helical or twisted Yang–Mills solutions, propagating chiral field configurations.
Gauge	Global, history-dependent update between sectors; computation by topological event rather than by local rewriting.	Bundle class, winding data, Chern-type invariants, or other global sector labels.	Topology is primary and active: the computation is identified with tunneling, obstruction crossing, or sector change.	Instantons, sphalerons, vacuum tunneling configurations, topological transition events.

case, such waves carry internal structure: along the propagation direction, the gauge potential undergoes nontrivial parallel transport, yielding non-vanishing holonomy in the gauge group. Equivalently, the field configuration traces a nontrivial path in the group manifold even though the local scalar invariant $E^2 - B^2$ vanishes.

Within a geometric–computational viewpoint, this slice $L = 0$ may be regarded as a *switching hypersurface* in configuration space: crossing it changes the group–theoretic state realized by the parallel transport while preserving the light-like character of the wave. The family of such plane waves thus furnishes a controlled alphabet of *signal trajectories*: each solution implements a definite group action (holonomy) along its world-volume, so compositions of waves correspond to compositions of group elements, with path ordering in the non-abelian case. In this sense, classical Yang–Mills solutions encode a concrete logic of state transitions, with the group providing the algebra and the spacetime evolution providing the flow.

5.5 Constructive Computational Gauge Theory

We feel that it is a useful exercise to frame gauge theory (and, by extension, quantum gravity) as a problem of *expressiveness* versus *verifiability*. If we allow "all histories" (arbitrary geometries, topology change, gauge redundancy, unbounded recursion in the construction of spacetimes), amplitudes become ill-defined and intractable. If we clamp down too hard, we lose physically relevant states and dynamics. **Functional programming offers a blueprint for balancing these extremes**. Our proposal is for a constructive computational gauge theory that strikes a principled middle ground: a typed, linear, total, effect-controlled calculus of geometries. Concretely, boundary data (3-geometries with gauge labels) are the types; spacetime regions (4-dimensional histories/cobordisms) are the terms; and gluing is composition. This gives a compositional semantics familiar from TQFTs but designed to scale beyond purely verifiable/topological settings (e.g., Chern-Simons). Readers are asked to consider the correspondences in table 9. The primitives are as follows:

Table 9: Programming concepts and quantum gravity analogues

Programming Concept	**Quantum Gravity Analogue**
Types	Boundary states (3-geometries with gauge data)
Terms / Programs	4-geometries (cobordisms, histories)
Composition	Gluing of spacetime regions
Linear types	Conservation laws, unitarity (no-cloning of boundary data)
Totality	Termination of the "geometry evaluator" (finite amplitudes)
Effects & handlers	Coarse-graining and renormalization
Dependent types	Gauge and diffeomorphism constraints

Types. The proposed helical primitives provide the concrete generators of histories: smooth, orientable flows that carry discrete topological labels (orientation/chirality) alongside continuous geometry. This marries the "continuous versus discrete" tension: spectra and curvature are continuous objects; quantum numbers arise as stable, counted winding data. In this sense, a type is not merely a boundary condition in the ordinary functional-analytic sense, but a geometric interface condition: a declaration of what sort of flow, symmetry, and conserved structure may consistently enter or leave a region. Types therefore package both kinematic admissibility and computational resource discipline.

Terms. *Terms* are the actual geometric histories realizing computation. Concretely, a term is a decorated bordism—a gluable spacetime region assembled from helical primitives, equipped with whatever bundle, connection, flow, and readout data are required by the regime under study. In the symbolic and quasi-fluidic bands, such a term may be little more than a controlled phase-carrying cylinder or junction; in the fluidic and gauge regimes it may carry nontrivial holonomy, curvature, or defect data. The point is that terms are not symbolic expressions standing in for geometry: they *are* the geometry, regarded operationally as a computation from one typed boundary to another. See appendix A.4 regarding the usage of standard topological cobordisms in this regard.

Composition. *Composition* is gluing. Given two terms whose outgoing and incoming interfaces match at the level of type, one composes them by identifying the relevant boundary components and extending the declared geometric structure across the seam. This is the physical and categorical content of "running one computation after another": the output flow of the first region becomes the admissible input flow of the second. At the semantic level, composition must preserve the chosen observational equivalence on boundaries—for example, class functions of holonomy, flux, or winding data—so that gluing does not merely concatenate regions topologically, but yields a well-defined composite computation. Parallel composition is given by disjoint union, while sequential composition is given by boundary matching; together they form the minimal compositional skeleton needed for a constructive gauge calculus.

Three design choices enforce computability and physics: linearity, totality, and effects. Linearity tracks boundary degrees of freedom as conserved resources (no cloning/erasure). At the same time, totality means the "geometry evaluator" (our state-sum/variational executor) always normalizes. The phenomena that usually force uncontrolled manipulations, such as coarse-graining, stochastic mixing, and renormalization, are modeled explicitly as algebraic effects with handlers. Dependent types encode gauge and diffeomorphism constraints at the level of well-typedness, so invariances propagate mechanically through compositions. We elaborate each in turn:

Linearity. *Linearity* is the discipline that boundary degrees of freedom are treated as conserved resources under gluing: no implicit copying, no implicit erasure. This is the categorical analogue of "no cloning, " so in this way, conservation laws (charge, flux) are enforced by typing rather than imposed post hoc. In particular, it ensures that "assembling" a region from primitives is compatible with conservation constraints at interfaces: what enters a glued boundary must exit somewhere, unless an explicit effect declares otherwise.

Totality. By *totality* we mean: the *core* evaluator always normalizes. Concretely, the "geometry evaluator" (state-sum/variational executor) is required to terminate on the core fragment and return a well-defined, finite semantic value (e.g. a boundary object, a morphism, or a computable readout). This is not the claim that nature is finite; it is the claim that the *core language* is designed so that evaluation is mechanically checkable and compositional. The point which will be expounded upon later is guardrails: the core fragment is the part of the semantics on which we can prove locality and gluing theorems without importing the full analytic burden of continuum QFT.

Effects and handlers. The phenomena that typically force uncontrolled manipulations—coarse-graining, stochastic mixing, renormalization, metric dependence, and (in ambitious extensions) topology change—are introduced as *algebraic effects* with *handlers*. In this way, "renormalization" is not treated as an ad hoc subtraction, but as an explicit *program transformation*: a handler takes a well-typed geometric term and returns a new well-typed term, together with a stated observational equivalence on the boundary.

Example 5.6 (Coarse-Graining as a Handled Effect.). For example, let a geometric term $W : \Sigma_{\mathrm{in}} \to \Sigma_{\mathrm{out}}$ contain a family of microscopic helical loops $\gamma_1, \ldots, \gamma_k$ in some interior region. A coarse-graining handler replaces this subterm by an effective loop system W_{eff} together with a certificate that the chosen boundary observables are preserved. In the abelian regime, this means preserving total phase/flux; in the non-abelian regime, it means preserving specified class functions of holonomy (for instance, Wilson-loop traces) on the boundary. The transformation is therefore not an arbitrary simplification of geometry, but a semantics-preserving rewrite of the program represented by W.

Soundness obligations for handlers. To prevent the effect layer from collapsing back into folklore, we impose the following contract on any declared handler (RG, coarse-graining, metric insertion, noise):

(a) **Functoriality.** Handling commutes with gluing/composition: handling a composite computation agrees (up to stated equivalence) with composing the handled pieces.

(b) **Monoidality.** Handling respects parallel composition (disjoint union): independent subsystems remain independent.

(c) **Gauge/diffeomorphism compatibility.** Handling preserves invariance of boundary readouts: observable semantics are phrased in gauge-invariant class functions of holonomy (e.g. Wilson-loop traces), and handlers must preserve these up to the declared equivalence relation.

(d) **Resource discipline.** Handling does not implicitly violate linearity; any creation/erasure of degrees of freedom must be explicit (an effect, not a side-effect).

Dependent types. Gauge and diffeomorphism constraints are carried as dependent typing data on boundaries and interfaces: a term is admissible only if it respects the declared constraint structure. The payoff is the mechanical propagation of invariances through composition: once a boundary constraint is well-typed, gluing does not "forget" it.

Practically, the workflow is: specify typed boundary data; assemble regions from helical primitives; compose by gluing; evaluate the *core* semantics (the total fragment); and only then, where required, apply declared effect handlers with an explicit statement of what equivalence of boundary readouts they preserve. In this sense, renormalization and coarse-graining are treated as structured code transforms on geometric terms, rather than informal manipulations of integrals.

Our aim is a language that is *expressive* enough to describe nontrivial gauge dynamics and background independence, yet *restricted* enough to prove normalization, locality/compositionality, and anomaly-freeness in the core. Extensions (matter content, topology change, nonperturbative sectors) are added modularly as new effects or controlled type extensions, preserving verification theorems as we widen scope. In this way, constructive computational gauge theory provides a semantics where we can calculate, compose, and certify. For the foundational work on constructive quantum field theory, see Baez, Segal, and Zhou [61]. Our approach here is in this spirit, but computational. Likewise, see Adlam [62] for recent philosophical work on constructive spacetime axiomatics.

On QFT and foundational description. We do not dispute the empirical success of quantum field theory, nor its central role as a unifying descriptive framework in modern physics. Per the EVT 2.6, our concern is narrower. When QFT is taken as a single, continuum-based probabilistic formalism in four dimensions, it appears to exceed the class of systems that can be globally verified without encountering tractability barriers (exponential numbers of Feynman diagrams, for instance). Infinite-dimensional functional measures inherit substantial analytic and measure-theoretic difficulties, while renormalization already shows that the formalism is intrinsically scale-dependent: the effective description is repeatedly rewritten as one changes resolution. In this sense, QFT behaves less like a fixed axiomatic object than like a computational process whose semantics are only partially stable across scales.

From this perspective, the issue is not that QFT is incorrect, but that it is unlikely to serve as the final foundational layer for a theory of physical computation. The deeper problem lies in the substrate it attempts to describe. Herein lies the same issue as pointed toward in section 4.1. If the underlying geometric dynamics are *themselves* computationally highly expressive, then any continuum formalism built to capture them in full generality will tend to inherit the same verification barriers. Our proposal is therefore not a rejection of QFT, but a change in starting point: rather than taking continuum probability and field measures as primitive, we begin from a constructive local-to-global calculus in which admissible operations are restricted from the outset, and in which probabilistic and field-theoretic behavior arise only after coarse-graining. In this way, QFT is retained as an emergent descriptive interface, while the foundational burden is shifted to a more controlled computational substrate that alleviates the cognitive and time burdens on human physicists.

From external lattices to an intrinsic, lattice gauge calculus. Lattice gauge theory discretizes a continuum on an external grid: links carry group elements, plaquettes carry curvature, and one hopes to recover physics in a subtle continuum limit. But computability is not baked in; it is delegated to numerics.

By contrast, GCM lattices are of a different kind. The "lattice" is the 1–skeleton generated by minimal holonomy loops of the helix geometry itself; nodes are join interfaces (Gauss constraints), edges are Wilson–loop registers, faces carry curvature flux. No continuum limit is imposed from outside: the cell structure is induced by the manifold's own group–theoretic and topological constraints. Spectral gaps and forbidden transitions then arise from intrinsic geometry rather than from a choice of mesh.

Operationally, this translates to:

- **State space.** A presheaf of local moduli (connections/holonomy classes on cells) over a helix–adapted cover; *sheafification* enforces gluing laws (Gauss, Bianchi) and yields global states.
- **Evaluation.** Amplitudes are functorial evaluations of class functions on Wilson loops (edge registers) subject to node intertwiners; normalization is a theorem of the typed core.
- **Scale change.** Refinement/coarse–graining is an explicit algebraic effect that changes the cover and updates moduli by certified handlers; proofs of invariance live in the type/effect system.

In this setting, the "sum over histories" is replaced by *sheafification + evaluation*: one computes global sections (or their invariants) of the sheafified moduli rather than integrating over an externally supplied configuration measure.[9]

Example 5.7 (Sheafification/Evaluation on an Intrinsic Helical Chain). Let $X = U_1 \cup U_2 \cup U_3$ be a helical working region whose open sets are centered on three consecutive intrinsic holonomy loops $\gamma_1, \gamma_2, \gamma_3$, with overlaps $U_{12} = U_1 \cap U_2$ and $U_{23} = U_2 \cap U_3$, and no triple overlap. Assume these opens as coming from the geometry of the GCM itself, rather than from an externally imposed cubic grid.

Let $\mathcal{M}(U)$ be the presheaf assigning to each open set $U \subseteq X$ the groupoid of $U(1)$-connections on the trivial line bundle over U, modulo gauge, together with the induced Wilson-loop values on the intrinsic loops contained in U. A local section over U_i is represented by a connection A_i and its holonomy

$$u_i := \mathrm{Hol}_{\gamma_i}(A_i) \in U(1).$$

[9] "It's nice to have a framework where you can only have one Lagrangian, but it would be even nicer to have some reason that this particular algebra would inevitably appear as an answer to some question." —E. Witten [63]

The local data glue precisely when the restrictions align to gauge overlaps:

$$A_2|_{U_{12}} = A_1|_{U_{12}} + g_{12}^{-1} dg_{12}, \qquad A_3|_{U_{23}} = A_2|_{U_{23}} + g_{23}^{-1} dg_{23}.$$

After sheafification, the admissible global states are therefore not arbitrary triples (u_1, u_2, u_3), but compatible glued sections

$$A \in \Gamma(X, \mathcal{M}^{\#}).$$

Now choose the outgoing loop γ_{out} obtained by concatenating the three intrinsic channels. In the abelian corridor, functoriality of holonomy gives

$$\text{Hol}_{\gamma_{\text{out}}}(A) = u_3 u_2 u_1.$$

Thus, the intrinsic lattice calculus does not "sum over histories" on an external mesh; it first constructs the global section by gluing local moduli, and only then evaluates the observable.

The constructive evaluator is the gauge-invariant map

$$\text{ev}_X : \Gamma(X, \mathcal{M}^{\#}) \longrightarrow U(1), \qquad \text{ev}_X(A) := \chi(\text{Hol}_{\gamma_{\text{out}}}(A)),$$

where in the simplest abelian case one may take $\chi(z) = z$, or more generally a thresholded readout to a finite alphabet.

This also exhibits the effect-handler viewpoint. A coarse-graining handler may replace the pair (U_1, U_2) by a single effective region U_{12} with effective holonomy

$$u_{12} := u_2 u_1.$$

The rewritten term has local data (u_{12}, u_3), but the boundary observable is unchanged:

$$\chi(u_3 u_{12}) = \chi(u_3 u_2 u_1).$$

Hence, coarse-graining is not an informal simplification; it is a semantics-preserving rewrite certified by preservation of the boundary class function.

In the non-abelian lift, the same example survives with ordered transport:

$$U_{\text{out}} = U_3 U_2 U_1, \qquad \text{ev}_X(A) = \text{tr}_\rho(U_3 U_2 U_1),$$

so the only change is that composition becomes path-ordered, and the evaluator passes from scalar phase to a class function on holonomy.

Remark 5.2 (Why Not Classical $\mathbb{R}$?)*.* In the fluidic/gauge band, one is tempted to model directly over the classical continuum. However, unrestricted real computation (cf. Blum–Shub–Smale A.5) admits hyper–Turing power and brittle pathologies (exact tests, branch–on–sign over nonrobust inputs, oracle–like primitives). This undermines the two pillars we require: *executability* (every step is effectful but total) and *verifiability* (small perturbations do not change the decision class). Our constructive stance preserves smooth geometry while preventing hidden oracles.

Geometric reals? We separate *semantics* (smooth reasoning) from *execution* (effective approximation):

(a) **Computable reals** $\mathbb{R}_{\text{comp}}$. externally, numerical data (couplings, boundary values, readouts) live as computable reals given by fast Cauchy names (or, when possible, algebraic numbers with exact arithmetic). Equality is replaced by interval refinement; separation proofs guard branching. This is the target of all executable algorithms.

(b) **Synthetic/smooth reals.** $\mathbb{R}_{\text{syn}}$ internally, differential reasoning uses the smooth reals of a well–adapted SDG model (Kock–Lawvere axiom, nilpotent $d\varepsilon = 0$), so vector calculus and curvature remain first–class without appeal to pointwise limits.[10]

[10] Any well–adapted model suffices; no commitment to a specific topos is needed, but an implementation is provided in appendix A.7 for convenience.

(c) **A bridge.** A semantic–to–execution functor sends internal smooth expressions (curvatures, class functions on holonomy) to external names in $\mathbb{R}_{\mathrm{comp}}$, with *proof obligations* discharged by our effect handlers (for limits/coarse–graining).

The aim is not to deny the continuum but to choose a base field that respects computability. Parameters are measured to finite precision; algorithms require effective representations. Thus we use rationals/algebraics/$\mathbb{R}_{\mathrm{comp}}$ for execution, while $\mathbb{R}_{\mathrm{syn}}$ supplies calculus internally. This pragmatic split aligns with the observation (cf. Szudzik [64]) that physical models do not require arbitrary reals to be *expressed* or *simulated*, even if smooth structure is invaluable for *reasoning*.

Design axioms: equality, limits, and effects. We enforce three constraints in the core calculus:

- **No exact branching on undecidable predicates.** Tests are certified by separation bounds (or deferred as obligations).
- **Limits are effects.** Completion, closure, and renormalization are algebraic effects with handlers that carry witnesses of convergence/stability.
- **Gauge invariants only.** Readouts are class functions of holonomy (e.g. traces), ensuring stability under small perturbations and gauge changes.

Remark 5.3 (Disclaimers on Classical QFT). Our setting concerns analog geometrodynamics on a noncompact configuration space. "Gaps" (mixing rates, spectral edges of transfer generators) are typically defined by nested sup / inf over unbounded domains. In general, these constants need not be uniformly computable from the defining data. We therefore (a) state results in terms of *semi–decidable* or *effectively approximable* quantities, or (b) confine genuinely noncomputable constants to effectful interfaces outside the core (they cannot influence totality/normalization proofs). As a result, this is a computability warning label for our own continuum models; we cannot use our model to solve the Clay Yang–Mills problem [65], which requires a fully rigorous quantum theory.

On finite information and intuitionist physics. We concur with the operational point that experimental data carry finite information, and that uncountable precision should not be a primitive of algorithms, as brought up by Prof. Gisin [66]. Where we differ is methodological: we retain a smooth internal semantics ($\mathbb{R}_{\mathrm{syn}}$) to keep geometry local and functorial, while ensuring that *every* externally visible quantity is realized in $\mathbb{R}_{\mathrm{comp}}$ with certified error control. Thus, the continuum remains a modeling tool; computation proceeds on effectively presentable numbers.

Choosing a constructive base allows us to retain the familiar analytic language where needed while ensuring that each step of the formalism remains executable. The framework developed here treats physical dynamics as smooth, often chiral/transcendental flow, but it also treats topology and quantization as mechanisms of coarse-graining that project this flow onto the discrete structures accessible to observation. In this sense, the continuum substrate and the discrete emergent layer are not competitors but complementary levels of description. What we call "physical computation" is the geometric process that occurs at their interface, with its behavior constrained by the EVT Trilemma. Nothing in the present approach depends on sub-Planckian assumptions, nor do we take uncountable structure as primitive in the dynamics. Our hope is that the fully constructive classification of topological field theories outlined here (section 5.6) provides a more precise setting in which longstanding questions, including those surrounding Yang–Mills, can be reformulated and studied. A high-level road map toward the Yang–Mills mass gap in computational gauge theory is outlined below:

(a) **Define computational semantics for gauge theory.** Define a typed, linear, total calculus whose types encode boundary data (3-geometries with gauge labels) and whose terms encode histories (4-dimensional "programs" composed by gluing).

(b) **Build computable geometric bundles.** Give the calculus a denotational semantics as a computable fiber-bundle (connection, parallel transport) formulated in a constructive/synthetic differential setting over a computable base field. This preserves the differential structure while ensuring that all primitives are evaluated effectively.

(c) **Assess the scaling limit to continuum Yang-Mills.** Prove that families of programs approximate continuum Yang–Mills in a controlled scaling limit. Concretely, a "differential computational jet calculus" provides error bounds showing convergence of correlation functions/Wilson loops to those of continuum Yang-Mills as the program lattice is refined (see appendix A.2.)

(d) **Prove the gap and its stability.** Establish a spectral gap (uniform lower bound on excitation energies) for the constructive model using the calculus' linear/compactness properties; then show this lower bound survives the scaling limit, yielding a continuum mass gap. Here, the constructive setting replaces global non-compact searches with local-to-global, type-theoretic invariants.

All physically relevant predictions are recoverable with parameters and observables taken from an effective number system. This keeps the continuum as a limit, not a prerequisite, and allows "proof by construction" to coexist with the analytic tools physicists use.

This approach reframes the Yang–Mills mass-gap problem as an information-theoretic problem. Can the dynamics of a gauge field theory, defined over the continuum and acted on by a compact simple Lie group, enforce a nontrivial discrete gap in its computational state space? This translation from an energy gap to an information gap offers a new perspective on both the foundations of gauge theory and the limits of computability in physical systems. **In essence, to attempt to solve the mass gap, we cannot just try to compute the solutions *to* the Yang Mills equations, but rather we may have to compute *with* the solutions to the equations themselves.**

Euclideanization and clocking. The physical Yang–Mills mass gap is posed on Minkowski spacetime $\mathbb{R}^{1,3}$, but our constructive model is formulated on a *Euclidean* 4-manifold. Concretely, we work after a Wick rotation $t \mapsto -i\tau$. Hence, the background metric is Riemannian, and the fourth coordinate can be used to encode an extended, chirality-resolving structure (e.g., Type 2 *THM*) without committing to a Lorentzian time foliation.

Operationally, temporal evolution is replaced by global pulses/flow propagation that provide an internal clocking of evaluations. Lorentzian observables are recovered by analytic continuation: correlation functions are computed in the Euclidean model and (where reflection positivity holds) reconstructed to Minkowski amplitudes via the usual Osterwalder–Schrader logic [67]. In our calculus, "Wick rotation" is a controlled effect: Euclidean evaluation (for existence/compactness) and Lorentzian reconstruction (for scattering/causality) are separate, typed phases of the same program. This retains the computational advantages of a 4D Euclidean setting while maintaining a principled path back to $\mathbb{R}^{1,3}$ for physical predictions.

The landscape of twistor theory. While our constructions emphasize categorical and geometric composition, it would be amiss not to note that *twistor theory*, pioneered by Penrose and collaborators [68], gives a striking *holomorphic* encoding of four-dimensional conformal geometry and massless field equations. Concretely, the Penrose transform and the Ward correspondence identify cohomological data on projective twistor space with solutions of linear massless equations and, in the non-abelian case, (anti-)self-dual Yang–Mills on spacetime.

Reader's compass. For our purposes, "twistor phase" means a regime where inherently noncommuting spacetime/gauge data admit a holomorphic reparametrization that turns analysis into cohomology. The symmetry group at play is the full conformal group $\mathrm{SU}(2,2) \cong \mathrm{Spin}(2,4)$; it is *not* an abelian corner. Thus, the twistor limit is best read as a complex-analytic chart in which verification improves because equations become integrable/holomorphic—even though the underlying gauge content remains non-abelian. Nothing in this viewpoint licenses literal real-valued "infinite computation": finite-information bounds (e.g. Bekenstein–Bousso [69]) still apply. The twistor phase should be read as a *cohomological* linearization window, not as physical hypercomputation.

Finally, our framework is naturally compatible with holomorphic tools that also appear in parts of the Geometric Langlands story (cf. Kapustin–Witten [70]) and with twistor approaches to scattering amplitudes; see, e.g., the mathematical discussion noted by Woit [71]. The correspondence is evidence that the gauge regime admits nontrivial equivalences between presentations of the same boundary observable algebra; we treat such dualities as chart changes. Our work does not require the full program, but it motivates the principle that physical/gauge computation should be invariant under these categorical re-factorizations. Hence, we do not claim a direct realization of Langlands duality;

rather, we point to a common structural mechanism—holomorphic/complex-analytic repackaging of non-abelian field data—that GCT will interface with in future work.

On Calabi–Yau manifolds and stable loci. Our perspective is not string-theoretic, but it sits squarely within the geometric-analysis paradigm inaugurated by Calabi: seek canonical metrics by analytic deformation and flow, then read structure from the resulting moduli. On a compact Kähler manifold with $c_1 = 0$, Yau's theorem [72] guarantees a unique Ricci-flat Kähler metric in each Kähler class. This furnishes a canonical, *integrable* background with powerful linear tools (Hodge theory, period maps, variation of Hodge structure) and controlled deformations.

Reader's compass. In our setting, Calabi–Yau (CY) [73] structures are *stable bookkeeping loci*: places in the global moduli where curvature decouples enough that the effective local algebra is as close to commutative as the dimension allows. The harmonic representative of each cohomology class provides a linear "ledger" for fluxes and charges, and the Hodge decomposition/intersection form supplies rigid accounting for admissible operations. For threefolds, the $(1,1)$ and $(2,1)$ sectors (Kähler vs. complex moduli) control, respectively, capacities and morphisms available to the substrate; at such loci, symbolic projections are maximally checkable because key questions reduce to cohomological linear algebra.

Technically, this is the state space where:

(a) the effective coefficient algebra is (up to mild twists) E_∞ in abelian and maximally-abelian quasi-fluidic regimes, so excision holds, and parallel composition is transparent;[11]

(b) twisted and equivariant K-theory encodes stable resource classes (channels, phases) and their integral constraints via periods, while variations of complex structure are recorded by period integrals and Picard–Fuchs equations;

(c) moving away from a CY locus (e.g. conifold degenerations or large-complex-structure limits) introduces monodromy/Picard–Lefschetz transformations, which reintroduce noncommuting transport: checks become globally harder even as the space of behaviors enlarges.

Hence, CY manifolds are not a generic case; they are *integrable gate points*—symmetry-enhanced regions that abelianize the accounting and make verification accessible. They play an essential role in the lower (abelian) and quasi-fluidic phases as calibration loci and phase boundaries: near a CY stratum, the dynamics are governed by linear/cohomological data; moving away activates non-abelian holonomy and the twistor calculus native to the more non-commutative regimes.[12]

Table 10: Classification of computational regimes by data type and algebraic structure

Symmetry Regime	**Algebraic Structures**	**Data Types**
Discrete / symbolic (finite cyclic C_n)	Finite rings/Boolean algebras	Bits, n-its
Analog abelian ($U(1)$)	Real/complex fields, Lie algebra $\mathfrak{u}(1)$	Circle phases, amplitudes
Analog non-abelian (($SU(2)$, $SU(3)$); gauge group actions)	Principle bundles, moduli spaces, ∞-groupoids; (co)homology operations	Global classes (characteristic classes/Chern, Floer/Donaldson-type)

Interactions of regimes within emergent gauge behavior. The presentations of the discrete and abelian regimes in sections 1 and 2 are intentionally given at the level of their *readouts*—coarse projections that isolate the symbolic (C_n) and commutative ($U(1)$) components of the substrate. In the full interaction picture, where non-abelian transport and defect/gluing data are active, the resulting dynamics are *not* adequately described by a standalone $C_n/U(1)$-valued cellular automaton. For expository purposes, we can collapse the final four-regime classification of section 5.6 into the three-band summary of table 10, and then indicate how the emergent gauge regime reorganizes the lower readouts.

[11] E_∞ here just means "homotopy-commutative"; readers may safely read it as "fully abelian up to coherent homotopy." See appendix A.5.

[12] For readers fluent in string theory: one may regard a geobit as a deterministic analogue of a worldsheet defect. This viewpoint suggests a geometric mechanism reminiscent of the weak gravity conjecture [74]; we do not attempt a derivation here.

Regimes as readouts of a single substrate. As emphasized in proposition 4.1, the notion of a geobit is readout-dependent, so table 10 should not be read as a list of unrelated theories, but as a hierarchy of *coarse-grained readouts* of one underlying geometric computing substrate. Concretely, let $\mathcal{S}$ denote the space of full substrate states (helical geometry, joins/defects, and transport data). The three rows of table 10 correspond to progressively richer "decoding maps"

$$\pi_{\mathrm{C_n}} : \mathcal{S} \to \mathcal{S}_{\mathrm{C_n}}, \qquad \pi_{U(1)} : \mathcal{S} \to \mathcal{S}_{U(1)}, \qquad \pi_{\mathrm{gauge}} : \mathcal{S} \to \mathcal{S}_{\mathrm{gauge}},$$

where $\pi_{\mathrm{C_n}}$ forgets geometric transport and retains only local symbolic degrees of freedom (a cellular automaton shadow), $\pi_{U(1)}$ retains a commutative phase/flux bookkeeping (an "abelianized" quasi-fluidic shadow), and π_{gauge} retains full non-abelian parallel transport and defect/gluing data (the fluidic/gauge regime). Thus, entering into π_{gauge} does not replace the lower regimes; it means we stop applying the early projections. The lower regimes remain present as *shadows* of the full dynamics, i.e. as the observables that survive after discarding non-commuting transport information.

Gauge back-reacts on lower readouts. Once non-abelian transport is active, the lower readouts inherit three systematic features:(a) *sectorization* by global topological labels (bundle/defect classes); (b) *twisting* of coefficients (fields become sections of associated bundles/local systems); (c) *center/trace observables* (the surviving abelian data is typically class-function data of holonomy). Concerning cohomology, near integrable/CY strata, cohomology acts mainly as a linear *ledger* within a sector; in the full gauge/twistor regime, it acts as *governance* (sectors, obstructions, monodromy/wall-crossing).

Example 5.8 (A Three-band Toy Model on T^3: Abelian Ledger Twisted Gauge Readout). This example makes explicit how a non-abelian gauge regime induces (a) sector labels, (b) twisted coefficients, and (c) trace/center observables in the lower abelian/quasi-fluidic readouts.

Base. Let $X = T^3 = \mathbb{R}^3/\mathbb{Z}^3$ with coordinates $x_1, x_2, x_3 \in \mathbb{R}/\mathbb{Z}$. Then

$$H^1(T^3;\mathbb{Z}) \cong \mathbb{Z}^3, \qquad H^2(T^3;\mathbb{Z}) \cong \mathbb{Z}^3, \qquad H^2(T^3;\mathbb{Z}_2) \cong (\mathbb{Z}_2)^3.$$

Band A: $U(1)$ ledger (commutative transport). A $U(1)$-connection A on a line bundle $L \to T^3$ splits into topological flux and a flat holonomy part:

(a) *Flux ledger (H^2).* The Chern class $c_1(L) \in H^2(T^3;\mathbb{Z}) \cong \mathbb{Z}^3$ may be represented by an integral curvature form

$$\frac{F}{2\pi} = n_1\, dx_2 \wedge dx_3 \;+\; n_2\, dx_3 \wedge dx_1 \;+\; n_3\, dx_1 \wedge dx_2, \qquad n_i \in \mathbb{Z},$$

so (n_1, n_2, n_3) is a literal cohomological "ledger" of $U(1)$ fluxes.

(b) *Holonomy ledger (H^1).* A flat connection may be written

$$A_0 = 2\pi(\alpha_1\, dx_1 + \alpha_2\, dx_2 + \alpha_3\, dx_3), \qquad \alpha_i \in \mathbb{R}/\mathbb{Z},$$

with holonomies around the coordinate 1-cycles γ_i given by

$$\mathrm{Hol}_{A_0}(\gamma_i) = e^{2\pi i \alpha_i} \in U(1).$$

Thus, the abelian/quasi-fluidic readout is linear: integers in H^2 plus phases in $H^1(-;\mathbb{R}/\mathbb{Z})$.

Band B: Flat $SU(2)$ holonomy (non-commutative transport). A flat $SU(2)$-connection on T^3 is equivalent to a representation

$$\rho : \pi_1(T^3) = \mathbb{Z}^3 \longrightarrow SU(2),$$

i.e. a commuting triple $(g_1, g_2, g_3) \in SU(2)^3$ modulo simultaneous conjugation. Since commuting elements in $SU(2)$ lie in a common maximal torus, we may conjugate to

$$g_i \sim \begin{pmatrix} e^{i\theta_i} & 0 \\ 0 & e^{-i\theta_i} \end{pmatrix}, \qquad \theta_i \in [0, \pi],$$

and conjugation identifies $\theta_i \sim -\theta_i$ (Weyl action). Hence

$$\mathcal{M}_{\mathrm{flat}}(T^3, SU(2)) \;\cong\; (S^1)^3/\{\theta \sim -\theta\}.$$

This immediately exhibits two of the three claimed effects:

(c) Trace/center observables. Gauge-invariant observables factor through conjugacy classes, e.g. via traces:

$$\mathrm{tr}(g_i) = 2\cos(\theta_i),$$

so the "abelian shadow" of non-abelian transport is class-function data (center/trace information).

(b) Twisted coefficients. Fixing ρ, infinitesimal deformations are governed by cohomology with *twisted* coefficients:

$$T_{[\rho]}\mathcal{M}_{\mathrm{flat}} \;\cong\; H^1\big(T^3; \mathrm{ad}_\rho\big),$$

where ad_ρ is the local system with fiber $\mathfrak{su}(2)$ and monodromy $\mathrm{Ad}\circ\rho$. For generic angles $\theta_i \notin \{0, \pi\}$, the adjoint local system splits as

$$\mathrm{ad}_\rho \;\cong\; \underline{\mathbb{R}}\cdot\sigma_3 \;\oplus\; \mathcal{L}_{2\theta},$$

with $\underline{\mathbb{R}}$ a trivial rank-1 system (Cartan direction) and $\mathcal{L}_{2\theta}$ a rotating rank-2 system. In particular, the rotating part contributes no global invariants in degree 1 for generic holonomy, so one recovers a *reduced* linear tangent space

$$H^1(T^3; \mathrm{ad}_\rho) \;\cong\; H^1(T^3; \mathbb{R}) \;\cong\; \mathbb{R}^3,$$

illustrating that "linear" behavior persists only after accounting for twisting by holonomy.

Band C: sector labels via $SO(3)$ and w_2. On T^3, every $SU(2)$-bundle is topologically trivial, so bundle-class sectorization is muted at the simply-connected level. Passing to the adjoint form $SO(3) \cong SU(2)/\{\pm I\}$ restores genuine topological sectors: principal $SO(3)$-bundles are classified by

$$w_2(P) \in H^2(T^3; \mathbb{Z}_2) \cong (\mathbb{Z}_2)^3.$$

Thus, the same H^2-type bookkeeping that served as an abelian "flux ledger" becomes, in the emergent gauge regime, a discrete *sector label* (a lifting obstruction for holonomy data), i.e., cohomology functions as governance rather than ledger. [75]

Interpretation. This toy model exhibits the general mechanism used throughout the paper: the abelian/quasi-fluidic regimes record linear ledger data (classes/periods), while turning on non-abelian transport forces (a) decomposition into global sectors, (b) twisted coefficient systems for linearization, and (c) passage to center/trace observables as the stable abelian shadows of gauge-invariant content.

Diffeomorphism invariance and the forced move to moduli invariants. In the highest (gauge) regime, the effective closure is not merely a compact structure group G acting on fibers, but the full redundancy of the geometric substrate: diffeomorphisms of the base (and their induced action on bundles and connections) as gestured towards in table 7. Concretely, the relevant symmetry is the gauge–diffeomorphism groupoid generated by $\mathrm{Diff}(M)$ together with bundle gauge transformations. Once computation must be invariant under this action, coordinate-level quantities cease to be meaningful: only invariants of the resulting quotient spaces (moduli of connections/flows up to gauge and diffeomorphism) remain as stable readouts. This is precisely the point at which Floer/Donaldson-type constructions become natural [76, 77], since they are designed to extract computable cohomological data from such moduli spaces.

On Seiberg-Witten and Atiyah-Singer. In the continuum, anomaly cancellation and spectral flow are governed by index theory: the analytic index of Dirac-type operators equals a topological index (Atiyah–Singer [78]). In our framework, the corresponding Ward/typing identities can be seen as constructive index constraints: zero-mode counts and anomaly coefficients are enforced at the level of types and persist under gluing. Thus, index-theoretic stability is reflected as compositional invariants of our evaluator rather than as a posteriori analytic facts.

Likewise, the Seiberg–Witten equations [79] exemplify how gauge-theoretic PDEs on 4-manifolds can yield compact moduli spaces with energy bounds and computable invariants. Our Euclidean computational formulation seeks the same structural features: finite-energy solution classes generated by helical programs, compactness under refinement, and invariants derived from holonomy/typing. We emphasize that we do not claim new SW invariants; instead, we adopt the SW template with compactness and abelianized control as the target shape for our constructive certificates (e.g., gap lower bounds and well-behaved moduli).

Disclaimer regarding EFT, entropic gravity, and the Lorentzian. Lest our proposals be misinterpreted, it should be specified that geometric computation works at the pre-metric substrate and derives causal/optical structure via an effective constitutive tensor χ. In near-equilibrium regimes χ becomes metric-induced, recovering Lorentzian geometry and Schuller's pre-metric electromagnetism [80] as an effective semantics. On larger scales, the thermodynamic equation of state reproduces Jacobson's derivation of Einstein's equations. So Schuller's and Jacobson's perspectives [81] appear as complementary limits of a single constructive framework: EM fixes the light cone via χ; coarse-graining of the underlying computation promotes χ to $\chi[g]$ and yields GR. No contradiction is implied—only a layering of descriptions.

But we do not take Markovian effective field theory to be foundational. Rather, it is treated as an emergent approximation that becomes valid only after a suitable coarse-graining of the underlying geometric bulk. For this reason, clean IR/UV decoupling is not assumed a priori, and as a result, it is likely that questions of strongly coupled gauge dynamics will be addressed before one can reliably speak about thermodynamic or black hole observables. Likewise, we do not quantize gravity by postulating a continuum path integral over Lorentzian metrics, nor do we treat the metric as a fundamental Yang–Mills field.

In this sense, gravity is not "quantized" at the fundamental level. What appears in perturbative regimes as a graviton should instead be understood as an effective spin-2 excitation of the emergent Lorentzian metric sector, valid in the same restricted sense that phonons are valid excitations of a condensed-matter medium. Such excitations are therefore not denied, but classified as an approximation. The foundational role is played not by a quantum field of gravity, but by the constraint architecture and of the underlying computation in the context of sound gluing rules; causal, metric, and field-theoretic structures arise only as effective descriptions of their large-scale organization.

What would count as success or failure? The framework proposed here should be judged neither as a metaphor nor as a purely speculative reinterpretation, but by whether its structural claims survive contact with mathematics and physics. It gains support if its phase distinctions yield stable and nontrivial differences in observables, if its abelian and quasi-fluidic regimes recover familiar gauge-theoretic behavior in a more unified way, and if its computational predictions continue to align with known constraints on expressiveness, verifiability, and reversibility. It would be weakened, or potentially falsified, if the proposed geometric strata fail to produce mathematically meaningful invariants, if the claimed transitions between regimes do not correspond to genuine differences in computational behavior, or if the framework proves unable to recover the standard structures it is intended to subsume.

5.6 On the Classification of Geometric Field Theories

We can now classify *geometric computational field theories* by specifying three layers of structure and composing the canonical maps between them:

$$\begin{array}{c}\text{decorated state/transition groupoid}\\ \Big\downarrow{\scriptstyle \text{higher rep.}}\\ \text{fully dualisable object in a symmetric monoidal } (\infty, n)\text{-category}\\ \Big\downarrow{\scriptstyle \text{Cobordism Hyp.}}\\ \text{fully extended } (\infty, n)\text{-TFT}\end{array}$$

Existence–uniqueness (Baez–Dolan [61]/Lurie [82]) identifies the last arrow once a fully dualisable object is supplied; the first arrow is where GCT lives, turning helix–decorated paths and bordisms into higher representations with explicit operational semantics.

Operational layer (constructive evaluator). As in section 5.5, *types* are boundary data (3–geometries with gauge labels), *terms* are 4–dimensional histories (cobordisms), and *composition* is gluing. Linearity tracks conserved resources; totality guarantees normalization of evaluations in the core; coarse–graining/renormalization are treated as algebraic *effects with handlers.* For every regime below, the interface is the same: registers are Wilson loops/surfaces, readouts are gauge–invariant class functions of holonomy (e.g. traces), and evaluation is symmetric monoidal and local.

Regime Ladder Overview.

(a) Symbolic. *Input symmetry/labels:* decorated fundamental groupoid with finite abelian labels $(V_4, (C_2)^3)$. *Higher rep target:* finite-group *2-representations* (objects are twist sectors, 1-morphisms are path homotopies). *Dualisability:* finite semisimple ⇒ left/right duals exist. *Predicted TFT:* 2D Dijkgraaf–Witten. *Ops:* commuting joins; characters χ_g as readouts.

(b) Quasi-fluidic. *Input:* $(C_2)^4$ with 1- and 2-form labels (higher-form flux). *Higher rep: 3-representations*/crossed-module representations. *Dualisability:* finite higher group ⇒ rigid. *Predicted TFT:* 3D higher-form DW/Yetter quantum double. *Ops:* loops and surfaces as registers; higher Wilsons; coarse–graining as 2-sheafification.

(c) Fully fluidic. *Input:* symmetry of a THM cell $D_4 \ltimes (C_2)^4$ acting on a decorated Π_1. *Higher rep: 4-representations*/2-connections on principal 2-bundles (toy non-abelian sector). *Dualisability:* restrict to *compact* structure groups; finite-depth truncations for constructivity. *Predicted TFT:* 4D *BF* (topological) as rigorous core; YM-term added as an *effect* in the evaluator. *Ops:* noncommuting joins; class functions of holonomy as readouts; renormalization as a certified handler.

(d) Emergent gauge. *Input:* bundle of simple Lie data (e.g. $SU(2)$) on helix-adapted charts. *Higher rep:* bundles of fully dualisable objects in Alg_∞ (e.g. E_2-algebras of Wilson lines). *Dualisability:* compactness/rigidity (fusion) or factorization criteria, depending on dimension. *Predicted TFT:* 3D Chern–Simons (fully extended); 4D topological sector (BF/CS-like); YM handled as an *effect*. *Ops:* same interface; node constraints become intertwiners in the representation category.

Lemma 5.2 (Evaluator Factors Through the Classified TFT)**.** *Let $\mathcal{D}$ be a symmetric monoidal (∞, n)-category and $X \in \mathcal{D}$ a fully dualisable object obtained as a higher representation of the decorated groupoid/bordism data. The constructive evaluator of section 5.5 (which is symmetric monoidal, local, and sends start/stop to coevaluation/evaluation) factors uniquely through the fully extended TFT Z_X classified by X.*

Proof sketch. By locality and monoidality, the evaluator is a symmetric monoidal functor on the bordism (∞, n)-category. Its values on units and caps/cups agree with the duality data of X by construction, hence it is equivalent to Z_X by the Cobordism Hypothesis. □

Table 11: Constructive classification: regimes, dualisable inputs, and predicted TFTs.

Group Type	**Fully dualisable object supplied**	**Predicted (n)-TFT**	**Comments**	**Dualisability guardrails**
Abelian	Finite-group *2-representation* of the decorated groupoid (V_4)	2D Dijkgraaf-Witten	Factors through finite sets → Vect; commuting joins	Finite semisimple category; duals automatic
Maximally abelian	Discrete *3-representation* of $(C_2)^4$ with higher gauge 2-forms	3D DW / Yetter double	$\mathbb{Z}_2$ higher-form lattice models; surfaces as registers	Finite higher group; rigid monoidal structure
Non-abelian	Smooth *4-representation* of $D_4 \ltimes (C_2)^4$ inside decorated Π_1	4D BF; Yang-Mills via effect	Noncommuting holonomies; renorm as certified handler	Compact structure groups; depth truncations
Simple Lie	Bundle of dualisable objects in Alg_∞ (e.g. E_2 of $SU(2)$ lines)	Chern-Simons / (∞, n)-TFT; YM via effect	Matches the Cobordism Hypothesis with decorated invariants	Compactness or factorization-algebra criteria

Guardrails in 4D: Topological Core vs. Metric Effects. In four dimensions, the Cobordism Hypothesis classifies *fully extended* topological field theories by *fully dualisable* objects in an appropriate symmetric monoidal (∞, n)-category. This classification is inherently *topological*: it is driven by local-to-global gluing, duals/evaluations, and functoriality under bordism composition, rather than by metric-dependent dynamics. [83]

Pure Yang–Mills is therefore not an acceptable *core* object for dualisability statements: the Yang–Mills action depends on a choice of metric through the Hodge operator $*$, and it carries propagating local degrees of freedom. To keep the 4D semantics fully compositional, we isolate a metric-free *BF topological core* [84] and treat the metric-sensitive Yang–Mills term as an *effect* handled by the constructive evaluator. Concretely, one may regard Yang–Mills in first-order form as a deformation of BF theory:

$$S_{\mathrm{BFYM}}(A, B) \;=\; \int_M \mathrm{tr}(B \wedge F_A) \;+\; \frac{g^2}{2} \int_M \mathrm{tr}(B \wedge *B),$$

where A is a connection with curvature F_A and B is an adjoint-valued 2-form. The BF term is topological (metric-free), while the second term is precisely where metric/scale enters via $*$. In the formal "topological limit" $g \to 0$, one recovers BF theory; integrating out B recovers the usual second-order Yang–Mills action.

This mirrors the Plebanski [85] paradigm: four-dimensional gravity can be presented as *constrained BF theory*, [86] where a BF-type action is supplemented by *simplicity constraints* (or, equivalently, constraint potentials) to recover metric geometry and propagating modes. In our setting, the analogy is operational:

- **Core (dualisable).** BF-type topological semantics, used for the Cobordism Hypothesis classification layer.
- **Effects (non-topological).** Metric-dependent dynamics (YM terms, constraint potentials, coarse-graining/RG) implemented as controlled effect-handlers in the evaluator.
- **Soundness criterion.** Effect-handlers must preserve symmetric monoidality (gluing/parallel composition) and must respect boundary gauge-invariant observables (e.g., class functions of holonomy such as Wilson loop traces).

All dualisability statements in this paper pertain to the BF core; the physically relevant Yang–Mills sector is recovered by applying the declared metric effects in a way that is compositional on boundaries.

Effects, semantic data, and QFT obligations. In the tradition of lattice gauge theory, we can interpret the constructive tooling as a *regulator with admissibility built in.* Rather than beginning with an unrestricted formal path integral over all field histories and then imposing gauge, locality, or regularity conditions afterward, we attempt to restrict the space of admissible histories at the level of geometric formation rules. Boundary data, gluing constraints, transport data, and regime-dependent readouts are not external decorations; they are part of the syntax of the theory. In this sense, GCFT aims to make the space of possible histories smaller before analytic or quantum-field-theoretic reconstruction begins.

This should not be misunderstood as avoiding the usual obligations of quantum field theory. A constrained geometric semantics must still account for the structures that make a quantum field theory physical: a measure or weighting on admissible histories, positivity conditions sufficient for Hilbert-space reconstruction, a nontrivial continuum limit, a spectrum of excitations, anomaly and global-symmetry data, and a well-defined algebra of observables. These are not optional add-ons. They are the tests by which a proposed geometric semantics becomes a physical theory rather than merely a kinematic model.

It is therefore important to distinguish several roles that are sometimes conflated:

Structure	Role in GCFT
Measure	semantic weighting/counting of admissible histories
Positivity	consistency condition for quantum reconstruction
Continuum limit	scaling/refinement theorem
Spectrum	output of reconstructed dynamics
Anomalies	obstruction/global consistency data
Observables	readout algebra or semantic target
Effects	controlled changes of regime or presentation

In particular, the term *effect* is not meant as a catch-all for every analytic or quantum-field-theoretic obligation. Effects are controlled operations internal to the semantic framework: examples include Wick rotation, coarse-graining, measurement-induced boundary updates, quotient/readout projections, renormalization-like scale changes, and the introduction of metric-sensitive Yang–Mills dynamics over a more topological compositional core. These operations may transform the semantic data of the theory, but they do not by themselves replace that data.

Wick rotation. For example, Wick rotation may be regarded as an effect because it changes the presentation of a geometric-computational history from a Lorentzian dynamical regime to a Euclidean or moduli-sensitive regime in which instanton-like sectors, compactness phenomena, and tunneling configurations become more visible. But the legitimacy of this operation is not automatic. A Wick-rotated presentation must still satisfy the relevant reconstruction conditions, such as reflection positivity, if it is to define a physical Lorentzian quantum theory. Thus Wick rotation may be an effect, while reflection positivity is a constraint on that effect.

Measurement, continuum limit and anomalies. Similarly, measurement may be treated as an effect in the sense that it changes boundary conditions, admissible continuations, or observer-accessible readout data. But the probabilities associated with measurement outcomes require a measure or weighting on the relevant space of admissible histories or moduli. The measure is not merely an effect; it is part of the semantic data of the theory. Likewise, the observable algebra is not an effect, but the target through which the theory exposes gauge-invariant or regime-invariant information: holonomy traces, Wilson-loop-type quantities, defect charges, boundary operators, spectral data, or moduli invariants.

The continuum limit also has a different status. Coarse-graining or renormalization may be represented by effect-like operations, but the existence of a nontrivial continuum limit is a theorem to be proved, not an operation to be postulated. A finite or regulated helical system may exhibit gaps, stable sectors, or finite-dimensional readouts, but the physical question is whether these structures survive refinement. In the Yang–Mills setting, for instance, one must eventually show that the relevant observables converge and that any proposed spectral gap remains uniformly bounded below in the continuum limit.

Anomalies occupy yet another role. In the present language, an anomaly should be viewed as an obstruction to making local symmetry, gauge, or gluing data globally consistent without additional boundary or higher-categorical structure. Thus, anomaly data are naturally related to failed descent, defect terms, or nontrivial cocycles in the global organization of the theory. They are not errors to be ignored; they are part of the global consistency structure that any serious gauge semantics must reproduce.

The intended architecture is therefore as follows:

(a) **Core geometry.** Helical local normal forms, admissible gluing, transport, and defects.

(b) **Semantic data.** Measures, observable algebras, boundary categories, and holonomy readouts.

(c) **Effects.** Wick rotation, measurement, coarse-graining, quotient/readout projection.

(d) **Consistency conditions.** Gauge invariance, positivity, anomaly cancellation, or inflow, functoriality.

(e) **Outputs.** Continuum limits, reconstructed Hilbert spaces, spectra, gaps, invariants.

This separation is essential. The effect structure organizes changes of presentation and regime, but it does not excuse the theory from reconstructing the physical content normally supplied by quantum field theory. Rather, the goal is to reorganize where these obligations enter. GCFT seeks to build admissibility into the geometric syntax of histories, then recover measures, observables, spectra, and continuum behavior from the resulting constrained semantic structure. If successful, the framework would not eliminate the analytic burdens of quantum field theory, but would relocate them to a setting in which the allowed histories, boundary data, and gauge-compatible readouts are already geometrically disciplined.

On topological Lagrangians. A useful historical precedent for our viewpoint is the Atiyah–Jeffrey [87] reinterpretation of Donaldson–Witten theory via the Mathai–Quillen formalism. Their perspective shows that, at least in the cohomological/topological setting, a gauge-theoretic action need not be understood merely as a dynamical law or a formal infinite-dimensional integral; it can instead be read as a geometric localization device that extracts invariant content from the moduli space of solutions. In modern terms, the action behaves like an infinite-dimensional Thom form or Euler-class representative, concentrating the theory onto gauge-invariant data rather than raw field configurations. This is closely aligned with the present setup: we likewise distinguish a topological/core semantics from the surrounding metric-dependent machinery, and we treat evaluation as the extraction of stable geometric information rather than as brute-force functional integration. Atiyah–Jeffrey does not provide the full constructive apparatus sought here, nor does it resolve the genuinely dynamical problems of four-dimensional Yang–Mills. Still, it offers a precise and respected prototype for the claim that gauge actions can be reinterpreted geometrically as evaluators of invariant content.

Why Lagrangians appear: local scoring, gluing, and invariance. Given the above comparison, a recurring question from the physics side is why then our dynamics are presented (in the fluidic/gauge regimes) via an "action principle" in the first place. In the present framework, this is not an aesthetic choice but a structural one. Once one insists that (a) evolution is *local*, (b) composite processes are formed by *gluing*, and (c) the description is *coordinate/gauge invariant*, the natural notion of "cost" for a geometric process is forced into the Lagrangian form.

Local scoring and additivity under gluing. Let M be a spacetime region (or computational bordism) and let ϕ denote the relevant geometric field data (e.g. a connection, framing field, or helix-state field) subject to prescribed boundary conditions on ∂M. We seek a functional

$$S_M[\phi] \in \mathbb{R}$$

interpreted as the score/cost of realizing the process ϕ on M. The minimal structural requirements are:

(a) **Locality.** $S_M[\phi]$ depends on ϕ through its local jets (a finite number of derivatives), so that modifications supported in a small region only affect the score locally.

(b) **Gluing additivity.** If $M = M_1 \cup_\Sigma M_2$ is obtained by gluing along an interface Σ and ϕ restricts compatibly, then

$$S_M[\phi] \;=\; S_{M_1}[\phi|_{M_1}] \;+\; S_{M_2}[\phi|_{M_2}],$$

up to boundary/interface terms that depend only on the glued data on Σ. [88]

(c) **Invariant symmetry.** S is invariant under the relevant redundancy group: diffeomorphisms (when applicable) and gauge transformations for connection-valued data.

Under these hypotheses, the functional must be representable (up to boundary terms) as an integral of a *local density*,

$$S_M[\phi] \;=\; \int_M \mathcal{L}(\phi, \nabla\phi, \dots)\,,$$

where $\mathcal{L}$ is built from coordinate- and gauge-invariant contractions of the available geometric tensors (metric, volume form, curvature, covariant derivatives, etc.). This is the precise sense in which a Lagrangian is the canonical "local score" compatible with gluing and invariance.

Stationarity, not optimization. The variational condition

$$\delta S_M[\phi] = 0$$

does not assert that Nature "minimizes" S in a global sense; rather, it singles out histories that are *stable under first-order perturbations* with fixed boundary data, yielding Euler–Lagrange equations [89]. This stationarity principle is the natural compatibility condition between local scoring and compositional evolution, and it is the mechanism by which continuous field descriptions produce constraint equations and conservation laws (via Noether-type arguments) [90].

Topological core versus evaluator. In the GCT viewpoint, an action decomposes conceptually into a *compositional core* that depends only on topological/geometric structure needed for gluing and typing of interfaces, and an *evaluator term* that penalizes "difficulty" (curvature/strain/instability) and selects physically realizable histories. This mirrors our BF/Yang–Mills split. BF-type terms are first-order, local, and naturally compatible with gluing; they encode kinematic constraints (flatness, defect charges, coupling to sources). Yang–Mills-type terms introduce a metric-sensitive cost for curvature and thereby define an effective dynamics/regularization. Concretely, for a connection A with curvature F_A, one has the paradigmatic pair

$$S_{\mathrm{BF}}(A, B) = \int_M \mathrm{tr}(B \wedge F_A), \qquad S_{\mathrm{YM}}(A) = \int_M \|F_A\|^2 \,\mathrm{vol}_g,$$

where S_{BF} supports gluing-amenable constraint semantics and S_{YM} acts as an evaluator (a resource/energy functional) selecting stable configurations. In our setting, the "fully fluidic" regime corresponds to the existence of such a local invariant evaluator; lower regimes (quasi-fluidic, symbolic) arise when only coarser readouts of the geometry are verifiable or when dissipation/coarse-graining dominates.

Dissipation and arrows. Finally, many effective evolutions used in modeling (including ours in certain regimes) are not strictly Hamiltonian. This is compatible with the above: one may pass to Euclidean signature or introduce a dissipation functional so that evolution follows a gradient-like flow for the evaluator (a Lyapunov monotone). In that case, the Lagrangian/action continues to serve as the correct local score compatible with gluing. At the same time, the *effective* time evolution becomes a semigroup on the verifiable state space.

Summary. Lagrangians arise because they are the unique local, gluing-additive, symmetry-respecting scoring rules on geometric processes, and the variational principle expresses stability of admissible histories under perturbation. In our language, they are the evaluator layer that sits atop a compositional (often topological) core.

Asymmetry of regimes. Recall again table 7. The four regimes split into "inner" (Symbolic, Fluidic) and "outer" (Quasi–fluidic, Gauge) types as detected by the join commutator (5.1):

(a) **Quasi–fluidic (outer).** $[J_1, J_2] = e$ and the local symmetry on phases is abelian. Here *independence is a coordinate choice* via decoupling if need be.

(b) **Symbolic (inner).** $[J_1, J_2] = e$ by construction; helices are *intentionally glued* through a shared discrete update table.

(c) **Fluidic (inner).** generically $[J_1, J_2] \neq e$ couplings are path–dependent via non–abelian holonomy and helices are *dynamically glued.*

(d) **Gauge (outer).** typically $[J_1, J_2] \neq e$, but there is gauge redundancy; after gauge–fixing, field lines behave as independent modules at the representation level, so *independence is a gauge choice* as the dynamics remains non–commutative.

Gluing and the sectors of ω–logic. There is a curious "outer vs. inner" asymmetry in the chart of n–logics. In the symbolic and fluidic bands, helices are defined as glued; in the quasi–fluidic and gauge bands, this need not hold. Under a C_2^4 embedding inside QF, joins commute yet are not always necessary (decoupling by normal form), whereas in the gauge band, joins play a different role altogether (gauge transformations). Thus, the observed breakdown of compulsory gluing toward the extreme ends of ω–logic should not be read as a loss of global compositional structure. Rather, it indicates that *local* gluing is no longer primitive there: in the quasi–fluidic band it becomes coordinate-dependent through decoupling and holonomy, while in the gauge band it becomes representation-dependent through gauge redundancy. What remains invariant is the global closure law governing admissible transport, transition, and regime change.

Question. *But why does nature even bother with gluing helices?* A single helix's axial $U(1)$ phase renders a gauge; with a twin, the relative phase $\Delta\theta \in S^1$ is gauge-invariant and robust. This yields an abelian "phase logic" realizing the $C_2 \times C_2$ actions used in the symbolic regime. Likewise, gluing balances torque and stabilizes the framing, thereby making the interfaces implementable. The most commonly cited continuous symmetry of the helix (i.e., rotation about the screw axis) is not amenable to computation.

For a complete overview of the Computational Cobordism Hypothesis, see appendix A.7.

5.7 Psi-Geometric

In light of our results, how can the existence of GCMs *help inform us* about potential interpretations of quantum mechanics?

It is plausible that the quantum state is an *ontic geometric structure*, a twin helix bundle with connection and density, while the usual wavefunction ψ is a *coordinate description* of that structure; probabilities arise from equivariant transport of density, and measurement is bordism–induced branching within the same geometry. This interprets ψ–ontic in the sense of PBR [91] (up to gauge), with an epistemic role for ψ as a representation.

Kinematic ontology. Fix a spacetime region M (nonrelativistic: $M \subset \mathbb{R}^3$ with time parameter; relativistic: a globally hyperbolic spacetime). The *beables* are:

(O1) A twin helix fiber bundle $\pi : \mathcal{H} \to M$ with fiber carrying two oriented strands (chirality data), equipped with a $U(1)$ connection A (phase transport) and a smooth positive density ρ on M (helix–flux density).

(O2) Global topological data: linking/holonomy classes of sections of $\mathcal{H}$ ("join" structure), which encode entanglement and AB–type effects.

Choosing a local trivialization of $\mathcal{H}$ yields fields (R, S) with

$$\psi(x) \;=\; R(x)e^{\frac{i}{\hbar}S(x)}, \qquad R = \sqrt{\rho}, \qquad \mathbf{v} = \frac{1}{m}\big(\nabla S - q\mathbf{A}\big),$$

and (relativistically) a conserved current J^μ with $\nabla_\mu J^\mu = 0$ and $J^\mu \sim \rho\, v^\mu$.

Dynamical Ontology.

(D1) *Madelung/Schrödinger form.* Writing $\psi = Re^{iS/\hbar}$ gives the continuity and Hamilton–Jacobi pair

$$\partial_t \rho + \nabla\cdot(\rho\,\mathbf{v}) = 0, \qquad \partial_t S + \frac{(\nabla S - q\mathbf{A})^2}{2m} + V + Q = 0, \quad Q = -\frac{\hbar^2}{2m}\frac{\nabla^2 R}{R}.$$

(D2) *Helix–geometric form.* Q is a curvature/torsion scalar of the amplitude sheet induced by π and A; $\mathbf{v}$ is the horizontal lift direction set by the connection. In relativistic settings, we use Tomonaga–Schwinger evolution $\Psi[\Sigma]$ on Cauchy surfaces Σ and define J^μ covariantly from the helix connection (spinorial or scalar sector as appropriate) [92].

A measurement is a *bordism surgery* in the coupled system–apparatus helix bundle: new, boundary/join conditions modify the allowed horizontal lifts. Dynamically, this selects an attracting sector (branch) of the global helix geometry. The post–measurement ψ is the restriction/re-expression of the same ontic object to the surviving branch; no stochastic primitive is added.

Probabilities and the Born rule. The density $\rho = |\psi|^2$ is *equivariant* under the guidance flow:

$$\partial_t \rho + \nabla\cdot(\rho\,\mathbf{v}) = 0.$$

Given typical initial ensembles (Liouville/mixing with respect to the helix dynamics), frequencies in repeated experiments converge to $\int_\Omega \rho\, dx$ on any detector region Ω. Thus, probabilities are *epistemic-dynamical* (ignorance over initial micro-geometry) while the underlying evolution remains deterministic.

Multi-particle states are single glued geometric objects: linked/braided helix fibers with shared holonomy. Bell nonlocality arises from this *shared geometry* (one global guidance field) rather than superluminal signals; operational no-signalling is preserved. AB shifts are the holonomy of A on a multiply connected base; interference channels move with $\Delta\varphi = (q/\hbar)\Phi$ while local forces vanish.

We use a covariant current J^μ and surface-based Tomonaga–Schwinger dynamics to avoid a preferred frame. Worldlines are integral curves of $v^\mu = J^\mu/\sqrt{-J^\nu J_\nu}$; statistics on any spacelike Σ agree by current conservation.

PBR and the Status of ψ. Let Λ denote the ontic state space of GCT beables:

$$\Lambda \;=\; \big\{(\mathcal{H}\to M,\; A,\; \rho;\; \text{holonomy/linking})\big\}/\text{gauge}.$$

(a) ψ–**ontic (structural).** If ψ_1 and ψ_2 are not related by a gauge transformation and global holonomy equivalence, then the induced pairs $(\rho_1, \nabla S_1 - q\mathbf{A})$ and $(\rho_2, \nabla S_2 - q\mathbf{A})$ define *disjoint* subsets of Λ. Hence, distinct quantum states correspond to distinct ontic geometries—compatible with PBR.

(b) ψ–**epistemic (representational).** A particular ψ is a *coordinate* on the gauge-equivalence class $[\lambda] \in \Lambda$ (choice of section/trivialization and coarse-graining). Different observers may use different ψ–charts for the same $[\lambda]$ without altering beables.

A Summary of Postulates.

P1. Ontology. Physical systems are twin helix bundles $(\mathcal{H}, A, \rho)$ over spacetime with specified holonomy/linking data. These structures are real (ontic).

P2. Dynamics. Evolution is deterministic guidance of defects/flux along the horizontal distribution set by A; in coordinates, it reproduces Schrödinger/Dirac (Madelung) with Q a geometric scalar.

P3. Equivariance. $\rho = |\psi|^2$ is transported by the guidance field, yielding Born statistics for typical ensembles.

P4. Measurement. Coupling to an apparatus induces bordism surgery that selects an attracting branch; "collapse" is the re-expression of the same ontic geometry on that branch.

P5. Nonlocality. Entanglement is shared holonomy/bordism in one global object; no-signalling holds.

P6. Relativity. The law is tensorial (via J^μ) and surface-local (Tomonaga–Schwinger), avoiding a preferred frame.

This gives a ψ–ontic, computable, and geometric ontology (no configuration-space fluid, a concrete origin for the "quantum potential" Q, the shape operator of the helix amplitude sheet), and a unified account of AB phases, interference, and entanglement via holonomy and join structure. What remains open and falsifiable is (a) a complete relativistic foliation-independence proof for worldlines in interacting QFT; (b) a first-principles determination of abelian coupling (fine-structure constant) from helix micro-geometry, and (c) loop-level matching of abelian β–function from helix excitations.

Summary. In GCT, *ψ is ontic as geometry and epistemic as description.* The "real" thing is the helix bundle with its connection, density, and holonomy; ψ is a chart on that object. Probabilities arise from deterministic equivariant flow, and measurement is geometric branching, giving a PBR-compatible, computable interpretation of quantum mechanics.

Closing Remarks on THM_4 and the role of V_4. We do not identify THM_4 with particular Standard Model particle families, nor do we claim that it directly realizes continuous gauge groups. Its significance is more primitive. At the kinematic level, the twin helix substrate carries two independent involutive operations:

$$\chi : (\text{chirality flip}), \qquad \omega : (\text{orientation flip}),$$

with

$$\chi^2 = \omega^2 = \mathrm{id}, \qquad \chi\omega = \omega\chi.$$

Hence, the substrate supports the minimal nontrivial abelian group generated by two commuting involutions,

$$\langle \chi, \omega \rangle \;\cong\; \mathbb{Z}_2 \times \mathbb{Z}_2 \;\cong\; V_4.$$

Equivalently, if one labels the four helical sectors by

$$\mathcal{H}_4 = \{THM_4^{(\epsilon,\delta)} : \epsilon, \delta \in \{\pm 1\}\},$$

where ϵ records chirality and δ records orientation, then $\mathcal{H}_4$ is naturally a V_4–torsor: the group acts freely and transitively by toggling these two binary invariants. In this sense, V_4 should be understood not as an incidental symmetry, but as the minimal algebraic skeleton of the twin helix substrate.

This point is strengthened by the fact that the same algebra arises in several canonical settings. First, the one-qubit Pauli group modulo its scalar center satisfies

$$\mathcal{P}_1/\{\pm 1, \pm i\} \;\cong\; \{I, X, Y, Z\}/\{\pm 1, \pm i\} \;\cong\; V_4,$$

so that V_4 is precisely the phase-stripped algebra of two independent binary flips. Second, the four Bell states form a natural V_4–torsor under local Pauli operations: starting from any one Bell state, the action of two commuting $\mathbb{Z}_2$ generators traverses the remaining three. Third, on a torus, one has

$$H^1(T^2; \mathbb{Z}_2) \;\cong\; \mathbb{Z}_2 \times \mathbb{Z}_2,$$

corresponding to the four spin structures determined by periodic/antiperiodic boundary conditions along the two independent cycles. In each case, the same abstract pattern appears: two commuting binary distinctions generate a four-sector structure. In our setting, the relevance of V_4 is not that it encodes the exact P or T symmetries of the Standard Model, but that it supplies the smallest nontrivial algebra in which independent binary geometric distinctions can be composed, inverted, and transported.

This perspective also clarifies the relation to quantum nonlocality. Historically, Einstein, Podolsky, and Rosen [93] identified a genuine incompleteness in any theory satisfying locality and separability; Bell [94] later showed that this package is incompatible with the observed quantum correlations. A deterministic but globally constrained theory preserves no-signalling while relaxing measurement independence, so the EPR impulse toward a deeper completion survives, even though the specific locality/separability package would, in the spirit of Bohm [95], yet again, be abandoned.

Our universe may be built from the smallest groups, not just the richest.

A Appendix

A.1 "Hidden" Variables?

It should be noted that two canonical join patterns for twin helices lead to distinct gauge–geometric behaviors.

Definition A.1 (Apex Twin Helix)**.** An *apex–joined* configuration is obtained by collapsing a boundary circle to a point, producing a conical join $C(S^1)$ (a stratified space with a point singularity). Equivalently, remove the apex and view the result as $M \setminus \{p\}$ equipped with a connection whose curvature may carry a localized charge at p (e.g. a Dirac monopole in the abelian case). Such joins model sources/sinks or defect cores, such as optical vortices, but most prominently **Dirac cones**.

By contrast, our THM as in definition 1.8 are the objects used for deterministic logic flow, as in the **Schrödinger equation**. The operational distinction is *holonomic*:

Proposition A.1 (Base vs. Apex Holonomy)**.** *Let A be a connection on M.*

(a) *(*Base–joined*) If F is smooth near the join annulus and B is simply connected there, then for sufficiently small loops γ linking the join, the holonomy class $[\,\mathrm{Hol}_\gamma(A)\,]$ is trivial; memory arises only from the distributed curvature, not from a singular core.*

(b) *(*Apex–joined*) If the join collapses to p and A has curvature with total flux through a small linking 2–sphere S^2_p equal to a quantized charge, then small loops γ encircling p acquire nontrivial holonomy classes. In $U(1)$,*

$$\mathrm{Hol}_\gamma(A) = \exp\Big(i\int_{S^2_p} F\Big) = e^{2\pi i k} \quad (k \in \mathbb{Z}),$$

and in the nonabelian case the conjugacy class of $\mathrm{Hol}_\gamma(A)$ *is determined by the localized topological data (e.g. degree of a normalized frame map, or the G–bundle class on S^2_p).*

Resolution by blow-up. An apex singularity can be *resolved* by the real blow-up of M at p, replacing p with the unit normal S^2_p. The connection extends over the exceptional sphere with a (possibly nontrivial) bundle; its Chern/characteristic class on S^2_p is exactly the stored charge. In this sense, apex–joined and base–joined pictures are two presentations of the same data: defects as boundary classes vs. smooth transport with an extra S^2 boundary component carrying flux.

Feature	**Twin Helical Manifolds**	**The Apex Twin Helix**
Join point	Smooth base join	Singular apex join
Topology	Globally smooth manifold	Manifold with point singularity
Flow	Bidirectional transport along an annulus	Radial (source/sink) around a defect core
Chirality	Controlled by local geometry	Emerges from defect topology
Singularities	Avoided (non-singular)	Essential (e.g. Dirac monopoles)
Field role	Symbolic/geometric logic (ordered joins)	Physical spinor dynamics
Purpose	Analog computational flow	Topological charge carriers

Summary. The "duality" is a clean dichotomy of join conditions: *smooth* joins emphasize programmable transport (deterministic logic), while *singular* joins emphasize conserved charges (defect logic). Hence, helicity plays two separate roles in physics. Holonomy conjugacy classes and curvature flux rigorously control both.

Likewise, as illustrated in fig. 5b, a complex plane wave

$$\Psi(x,t) = Ae^{i(kx-\omega t)}, \qquad \Re\Psi = A\cos(kx-\omega t), \quad \Im\Psi = A\sin(kx-\omega t),$$

traces a Type 1 THM in space: the real/imaginary components are $\pi/2$ out of phase and form two counterwinding strands whose joints occur at equal-phase points. In Maxwell's $U(1)$ sector (massless, no self-interaction), the physical photon inherits exactly this geometry: a pair of phase-locked, coaxial, counterwinding helices—the "helical orb."

Question. *But why the difference? Why do photons appear as Type 1 while Yang–Mills instantons (fig. 5a) present as Type 3 dual-cone configurations?*

Definition A.2 (Helical Normal Forms)**.** We say a localized geometric computational configuration on the twin helix substrate is:

- **Type 1 (Helical Orb Normal Form).** A coaxial pair of counterwinding helices as an abelian wave with a common propagation axis and phase-locked transverse frame; joints are equal-phase loci.
- **Type 3 (Symmetric Flow Event Normal Form).** A compact, oriented cobordism in spacetime whose incoming/outgoing faces carry oppositely oriented helical frames, visualized as two opposed cones joined along a finite 3-volume.

Proposition A.2 (Symmetry–Energetics–Boundary Selection Principle)**.** *Given a gauge sector G and boundary data:*

(a) ***Abelian, propagating waves*** *(e.g. photons in $U(1)$) minimize the geometric action by* pure transverse frame rotation *along an open worldline, yielding the* Type 1 *normal form.*

(b) ***Nonabelian, (anti)self-dual finite-action events*** *(instantons in $SU(2)/SU(3)$) minimize by a compact, topologically charged 4D configuration interpolating between vacua with different $k \propto \int \mathrm{tr}(F \wedge F)$. The oriented in/out faces enforce opposite frame orientations, producing the* Type 3 *dual-cone normal form.*

Proof sketch. (1) In the abelian case, holonomy abelianizes to a single phase; with no self-interaction or topological charge, the energetic minimizer is a traveling wave: equal-amplitude quadrature components assemble into a coaxial, phase-locked twin helix (Type 1). There are no junctions, cones, or domain walls to pin the frame. (2) For nonabelian (anti)self-dual solutions, the relevant invariant is:

$$k = \frac{1}{8\pi^2} \int \mathrm{tr}(F \wedge F)$$

so finite action requires compact support in Euclidean spacetime and yields a vacuum-to-vacuum *cobordism.* The in/out orientations of the action density impose opposite normals on the two temporal faces of the event, and the lifted spin/transverse frame rotates accordingly, giving the dual, opposed cones (Type 3). These are *events* (tunneling) rather than asymptotic quanta. □

Summary. Photons are asymptotic Type 1 helical waves in an abelian phase channel; instantons are nonabelian, topological *events*—Type 3 dual-cone cobordisms—not particles. The distinct pictures are fixed by (a) symmetry (abelian vs. nonabelian), (b) energetics (wave transport vs. self-dual soliton), and (c) boundary class (open propagation vs. vacuum-to-vacuum interpolation).

Question. *But what about the Type 2 and Type 4 twin helix configurations? What might they represent?*

Proposition A.3 (Composite Normal Forms for Type 2 and Type 4)**.** *Unlike Type 1 and Type 3, which arise naturally as asymptotic wave and instanton events normal forms, Type 2 and Type 4 are best interpreted as* composite *or* constrained *channels of gauge-theoretic flow.*

(a) ***Type 2*** *(Twisted pair dual carrier form) is the natural normal form for an* opposite-helicity paired channel*: two transverse spin-*1 *excitations propagating along a common axis with vanishing net helicity. In QCD, this is the structural regime of scalar and pseudoscalar two-gluon composites, i.e. the* 0^{++} *and* 0^{-+} *glueball channels.*

(b) ***Type 4*** *(Co-propagating twins dual carrier form) is the natural normal form for a* same-helicity co-propagating channel*: two aligned vector excitations whose twist data reinforce rather than cancel. In gauge theory, this appears most directly in helical-string/flux-tube models of QCD, and structurally it matches tensorial channels built from same-helicity vector data.*

Heuristic justification. For Type 2, the geometric obstruction of lemma 1.3 already indicates that this configuration is not the generic shape of a free propagating mode in $\mathbb{R}^3$, but rather a constrained or resolved pairing. This is precisely what one expects of helicity-cancelling two-gluon sectors: the individual vector modes remain transverse, but the composite object has scalar or pseudoscalar quantum numbers. The relevant gauge-invariant operators are schematically

$$\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}), \qquad \mathrm{tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu}),$$

and the two-gluon helicity formalism reproduces the observed low-lying glueball J^{PC} assignments without introducing spurious states associated with naive spin-1 constituent gluons [96, 97].

For Type 4, by contrast, chirality and orientation reinforce one another. The resulting configuration is not cancellation but coherent co-propagation. This is the regime most naturally associated with a twisted flux carrier: the QCD string itself can be modeled as helix-shaped, with physically meaningful winding density and azimuthal ordering (cf. Todorova-Nova [98].) At the purely representation-theoretic level, the same-helicity pairing of spin-1 modes is also the canonical route to a spin-2 object: for a massless graviton, the helicity-± 2 polarization tensor factorizes as

$$h^{(\pm 2)}_{\mu\nu} = \varepsilon^{\pm}_{\mu}\varepsilon^{\pm}_{\nu},$$

and in double-copy language, the positive-helicity graviton arises from positive-helicity gluonic data on both sides [99]. We therefore regard Type 4 not as a new elementary particle, but as the correct helical normal form for coherent same-handed flux and tensorial composite channels. □

Summary. The four THM normal forms separate naturally into two classes. Type 1 and Type 3 are *single-carrier* normal forms: abelian propagating waves and nonabelian topological events. Type 2 and Type 4 are instead *dual-carrier* normal forms. Type 2 is cancellation-dominated and therefore appears most naturally in scalar/pseudoscalar composite sectors; Type 4 is reinforcement-dominated and therefore appears in helical flux tubes, tensor channels, and same-helicity pairings. In this sense, the relative scarcity of Type 2 and Type 4 in textbook particle imagery is not an objection to the THM classification, but a consequence of the fact that these geometries belong primarily to bound, confined, or effective sectors rather than to free asymptotic quanta.

Sector	Object	Dominant functional	Boundary class	Form
$U(1)$	photon	wave (no self-int.)	open, asymptotic	Type 1
$SU(2), (3)$	instanton	(anti)self-dual soliton	vacuum → vacuum ($k \neq 0$)	Type 3
$SU(3)$	(pseudo)scalar glueball	opposite-helicity pairing	confined, composite	Type 2
$SU(3)$	helical flux tube	coherent same-helicity flux	confined, stringlike/effective	Type 4

Table 12: Selection of helical normal forms by symmetry, energetics, and boundary data.

Disclaimer regarding the SM and GUTs. Though we do not execute such a derivation in this work, we propose that the observed Standard Model gauge group splitting

$$\big(U(1) \times SU(2)\big) \times SU(3)$$

may reflect distinct computational primitives of the substrate: the ω-logic (binary orientation plus phase rotation) underlying the electroweak sector, and the χ-logic (triality/twist structure) underlying color. We offer this only as an organizing hypothesis for how different gauge behaviors might arise from distinct geometric-computational regimes. It does *not* constitute a derivation of the observed fermion content, Yukawa sector, mass hierarchies, mixing parameters, anomaly cancellation pattern, or measured coupling constants of the Standard Model. Nor does it forbid the possibility that a simple-group unification (e.g. $SU(5)$ or a related structure) may emerge at higher energies or in a more complete algebraic formulation. The claim here is therefore modest: that the low-energy factorization of the Standard Model may admit a natural interpretation in terms of distinct substrate-level computational primitives, even if the full particle-physics story lies well beyond the scope of the present paper.

Question. *Proof of Yang–Mills instantons being composed of holonomic helices has so far not been confirmed. How then can we safely verify their existence?*

The issue is that the helical structure we posit resides in the *connection* A (parallel transport in the internal bundle). In contrast, direct observables are gauge-invariant functionals of the *curvature*

$F = dA + A \wedge A$ or of holonomies (Wilson loops). A gauge transformation can erase any purely local feature of A; what survives to observation are scalar densities such as $\mathrm{tr}\, F_{\mu\nu}F^{\mu\nu}$ and $\mathrm{tr}\, F \wedge F$, which appear as lumps, rings, or cone-like envelopes in lattice plots and experiments. The micro–handedness of a helical A thus projects to gauge–invariant *envelopes* rather than visible corkscrews.

Helical winding is a property of the *fiber* (the G internal directions) over spacetime. Detectors and standard lattice diagnostics live on the base manifold and access only fiber–invariant combinations. Projecting fiber twist onto base scalars discards orientation data but retains energy and topological charge, yielding the observed dual-cone profiles for instanton-like configurations. Real instruments and lattice operators average over space, time, and ensembles. Without a phase-locked reference to the internal connection, linear twist contributions cancel, leaving only quadratic or topological signals. Hence, the helical microstructure is washed out, while $\mathrm{tr}\, F^2$ and $F \wedge F$ remain.

Many diagnostic solutions (instantons, calorons) are Euclidean saddles. The helix lives in the Euclidean/complexified connection; in real time, one measures its *consequences* (selection rules, rates, envelopes), not the helical motion itself. The appropriate observables are *nonlocal* and gauge sensitive:

- **Wilson–loop tomography.** Families of loops $\mathcal{W}(\gamma) = \mathrm{tr}\, \mathcal{P} \exp \int_\gamma A$ detect non–abelian holonomy; loop–shape/systematic studies can reconstruct local twist consistent with $\mathrm{tr}\, F^2$ envelopes.
- **Gauge–fixed diagnostics.** In a fixed gauge one may visualize the Chern–Simons 3–form K_μ to track handedness (useful for diagnosis, though not itself gauge invariant).
- **Analog platforms.** Phase–locked probes in $SU(2)$ optical media, cold atoms with synthetic gauge fields, or superconducting circuits access non–abelian holonomy directly, providing experimental surrogates for helical transport [100, 101].

Summary. The helical substrate is "invisible" for the same reason the vector potential is: nature reveals only gauge–invariant aggregates locally and holonomy nonlocally. Helices manifest as the *geometry of transport* (path–ordered phases), while energy/topology plots display their *envelopes.* To expose the helices, one must either reconstruct in a fixed gauge or measure holonomy via nonlocal probes.

Psi–geometric semantics and "quantum quandaries". Our program does not attempt to explain *why* helices in nature behave as they do; that work is best left to our colleagues in adjacent departments. Rather, it takes seriously what their behavior appears to implement: something simultaneously ordinary and mildly unsettling—they enforce global consistency in a world that, operationally, only permits local access.

A useful way to read this is through the categorical perspective advocated by Baez [102]: many familiar "quantum quandaries" are not additional physical mysteries, but structural consequences of working in a non-cartesian, process-theoretic setting (closer in spirit to **nCob** than to **Set**). In such a setting, composition is *gluing* rather than pairing; one should not expect a canonical diagonal ("copying") nor a cartesian product that cleanly models independent subsystems. From this viewpoint, interference and AB-type phenomena are the generic signatures of phase transport and holonomy: the relevant compatibility data is global (encoded in connections and their path-dependence), while local descriptions are related by gauge. Entanglement, likewise, is best understood as the failure of factorization of the underlying glued state: two distant "systems" may simply be two boundary ports of a single global geometry prepared in their common past. No superluminal signalling is implied; the observed correlations reflect shared constraints on how boundary data can be capped, compared, and recombined.

A.2 Higher Dimensional Algebra and Geometric Computational Field Theory

To our knowledge, no single algebraic framework presently captures computation in GCMs across the full symbolic–to–fluidic range. By this we mean a framework that simultaneously (a) composes systems under joins—both commuting and non-commuting, (b) handles local–to–global propagation through interfaces, and (c) couples continuous transport (flows/phases) with discrete readout. Lie groupoids (and stacky variants) currently provide the most serviceable language for local symmetry, defects, and gluing. However, a general categorical semantics that bridges a GCM's geometry with its computability is still lacking.

As one passes from discrete to fluidic regimes, finite algebraic invariants rapidly lose discriminative power: phase freedoms and continuous deformations overwhelm purely combinatorial structure. This suggests a hybrid treatment in which dynamical systems methods provide the evolution laws, while higher algebraic machinery provides the compositional spine. Accordingly, we use "algebra" here in a programmatic sense: the aim is a language that unifies discrete composition, continuous flow, and topological joining within a single, compositional setting.

What is needed? A viable semantics for GCMs must be simultaneously symbolic and geometric: it must (a) compose discrete states and interfaces (information–theoretic layer), (b) transport phases and flows continuously (geometric layer), (c) express gluing along boundaries and track joins, and (d) specify how update laws transform the underlying geometry (chirality/orientation/topological data).

The key obstacle is nonlocal constraint propagation across helices: interface compatibility is not decided on a single patch, and path–dependence (holonomy) persists under coarse discretization. Global states are therefore not simple products of local states but solutions of descent/compatibility conditions along the interface groupoid (as in definition 4.1). Consequently, naive local linearization or discretization erases the very invariants that govern computation.

A workable mathematical language for GCMs must support four capabilities and tie each to a concrete compositional device.

(a) **Continuous transformation.** Represent smooth deformations (tube radius, phase, twist) and their flows.

(b) **Local-to-global behavior.** Track how interface–local updates assemble to global states and constraints.

(c) **Structural invariance and laws.** Express conservation and controlled change of chirality, orientation, and topological charges.

(d) **Rich morphism structure.** Compose fragments along boundaries; allow deformations as higher morphisms.

Definition A.3 (Minimal Compositional Spine for GCMs)**.** Let $\mathbb{G}$ be a symmetric monoidal double category of open GCMs, defined as follows:

- **Objects** (0-cells). Interfaces/boundary components equipped with the declared readout structure.
- **Horizontal 1-cells**. Open GCM fragments with specified input/output interfaces, i.e., gluable bordisms equipped with interface rules $\mathcal{I}$.
- **Vertical 1-cells**. Structure-preserving maps of interfaces (refinements, relabelings, or symmetry-compatible identifications).
- **2-cells**. Commuting squares of GCM data, interpreted as compatible deformations/flows/homotopies between fragments.
- **Monoidal product**. Disjoint union $\otimes$ (parallel composition), with horizontal composition given by gluing along matching interfaces.

Decorate $\mathbb{G}$ by a stack (or higher stack) $\mathcal{S}$ of local state objects on the site of open GCMs, valued in the geometric ambient category $\mathcal{E}_{\mathrm{GCM}}$. Thus, each open region U is assigned a local state object $\mathcal{S}(U)$ together with descent data along overlaps and interfaces. Dynamics consists of:

- **Pulses.** Distinguished endo-2-cells representing discrete update steps.
- **Flows.** Connections or transport laws on $\mathcal{S}$ representing continuous evolution.

The point of definition A.3 is to separate the *compositional skeleton* from the choice of local state model. The double-category structure records how GCM fragments glue and deform; the stack $\mathcal{S}$ records what state data live on those fragments. The minimal semantics above should be read as a workable core rather than a final synthesis. A fuller algebraic account of GCM behavior appears to radiate outward in several tightly related directions. We group by function:

Local-to-global semantics. The first extension concerns how local computational states glue into global geometric configurations. Here, the natural language is that of *stacks*, *descent*, and *higher gauge theory* [103]. One should assign to each open set U a higher groupoid of local states and transports, then recover global configurations as descent data rather than as naive products of local coordinates. In the abelian/quasi-fluidic band, this collapses to familiar local systems (our $B/T/Z$-type readouts, torus phases, and integer charges), but in the fluidic and gauge regimes, the gluing problem becomes genuinely nonabelian: one must track connections, holonomies, and coherent homotopies of gauge data under refinement and recombination. This is exactly the setting in which higher gauge theory becomes the minimal bookkeeping required for nonabelian transport, defect fusion, and feedback under gluing. From this perspective, global GCM states are not tuples of local values but *glued objects in a differential cohesive geometric semantics.*

Differential dynamics and update laws. The second extension concerns the internal mechanics of updates. Here *synthetic differential geometry*, *jet bundles*, and the *Cartan–Spencer formalism* [104, 105] provide the natural language for finite-order control, prolongation, and integrability. Working internally to a differential cohesive ∞-topos (cf. A.7) allows vector fields, flows, infinitesimals, and differential modalities (e.g. $\int$, $\flat$, $\sharp$) to be treated as first-class objects rather than external analytic artifacts. In parallel, a future *bordism type theory* could package the same ideas proof-theoretically: types as stacks of decorated manifolds, terms as global sections or configured geometries, identity types as homotopies/deformations, and Π– and Σ–types as homotopy right/left Kan extensions along fibrations modeling parameterized families and gluing. Followed by a sequent calculus whose cut/elimination rules match horizontal/monoidal composition in the double category of open GCMs. Our aim is not to multiply formalisms for their own sake, but to isolate a setting in which update laws, prolongations, phase slips, and defect motion can be represented as executable geometric rules rather than described informally.

Observables, defects, and stable invariants. The third extension concerns observables in the genuinely higher regimes. Once one allows defects, interfaces, and nontrivial operator insertions, ordinary bundle language quickly becomes too coarse. Here *stratified topological field theory*, *factorization homology*, *K-theory*, and *noncommutative geometry* [106, 107, 108] enter naturally. Stratified/extended TFT organizes computations by defect type and gluing pattern; factorization homology provides the corresponding local-to-global integration of observables; (possibly twisted) K-theory classifies stable sectors and conserved charges; and noncommutative geometry becomes relevant when quotienting by holonomy/gauge data destroys any useful ordinary space of states. In the abelian subtheories, these tools reduce to comparatively tame flux and phase bookkeeping, but in the fluidic/gauge band, they are the natural repositories of the data that survive after passing to moduli and quotienting by symmetry. In other words, once the geometry is sufficiently expressive, the correct outputs cease to be "numbers at points" and become stable invariants of operator-valued or defect-decorated structures.

Equivariant and Floer-type refinements. The final extension concerns the onset of genuinely equivariant and cohomological phenomena. Our base symmetry $V_4 \simeq C_2 \times C_2$ embeds in $O(2)$, but in the regimes sensitive to spinorial or orientation-reversing data one should lift along $\mathrm{Pin}(2) \to O(2)$; the corresponding preimage of V_4 is the quaternion group $Q_8 \subset \mathrm{Pin}(2) \subset SU(2)$. Thus, chirality and orientation reversal are not cosmetic labels but genuine equivariant structure, aligning the fluidic/-gauge band with the $\mathrm{Pin}(2)$-equivariant viewpoint that has become central in modern Floer theory, particularly in Manolescu's work [109]. Practically, the abelian/quasi-fluidic analysis still sees only the V_4 projection, while the higher gauge regime naturally asks for the $\mathrm{Pin}(2)$ lift. This is one reason that Floer-type and Donaldson-type refinements appear not as optional embellishments, but as the likely endpoint of any serious attempt to extract stable computational invariants from the highest regime.

Remark A.1. There are, of course, many further directions, but these four themes already indicate the scale of the task. A full algebraic description of GCM behavior likely requires a synthesis beyond the reach of any single mathematician—something closer to a coordinated language than a solitary breakthrough. Grothendieck famously spoke of a rising sea in which difficult problems might eventually be submerged and simplified. Yet when nature insists on phenomena such as quantum mechanics, sometimes no ocean is deep enough. To counter this, we may need to carve a new path, one with a river of flow.

A.3 Implications for Reversible Computing

GCT opens an avenue for *reversible computation.* This matters for physics because microscopic physical laws are, to an excellent approximation, information-preserving: classical Hamiltonian dynamics preserve phase-space volume (Liouville) and quantum dynamics evolve by unitary (or antiunitary) transformations. In both cases, the evolution can, in principle, be inverted. A computational model that respects this structure aligns directly with how physical systems propagate.

Let Conf be the space of configurations of a computing device. Then for:

(a) **Finite/discrete models.** A step of reversible computation is a bijection $F : \mathsf{Conf} \to \mathsf{Conf}$. Composition gives a group $\mathrm{Bij}(\mathsf{Conf})$ of updates; gates such as Toffoli/Fredkin generate subgroups used in practice (see [110] for an overview).

(b) **Classical continuum models.** Let (M, ω) be a symplectic phase space. An evolution $\{\phi_t\}_{t\in\mathbb{R}}$ is reversible when each ϕ_t is a *symplectomorphism* and $\phi_{t+s} = \phi_t \circ \phi_s$, $\phi_0 = \mathrm{id}$, hence $\phi_{-t} = \phi_t^{-1}$. Hamiltonian flows are the canonical example.

(c) **Quantum models.** On a Hilbert space $\mathcal{H}$, closed-system time evolution is implemented by a one-parameter unitary group $U_t = e^{-iHt}$, so $U_{-t} = U_t^\dagger$. More generally, any unitary (or antiunitary, for time-reversal) map is logically reversible; non-unitary channels (measurement, dissipation) are not.

Reversible computation can be expressed as *isomorphisms* in a category of states and processes. If $\mathcal{C}$ is a category of configurations, a reversible step is an isomorphism $f : X \to Y$ with inverse $f^{-1} : Y \to X$ and $f \circ f^{-1} = \mathrm{id}_Y$, $f^{-1} \circ f = \mathrm{id}_X$. Collecting objects and isomorphisms yields a groupoid of configurations and reversible transitions. Equivalently, a reversible dynamics is an autoequivalence $F : \mathcal{C} \to \mathcal{C}$ with a quasi-inverse F^{-1}, i.e. $F \circ F^{-1} \cong \mathrm{id}_{\mathcal{C}}$ and $F^{-1} \circ F \cong \mathrm{id}_{\mathcal{C}}$. This captures "no information loss" at the structural level.

Example A.1 (Flows and Time-reversal)*.* Let $\phi_t : M \to M$ denote a flow on a manifold M. If every ϕ_t is a diffeomorphism and $\phi_{-t} = \phi_t^{-1}$ for all t, the evolution is reversible. In classical mechanics cs this is strengthened by symplecticity $\phi_t^\omega = \omega$, while mechanics conservation of phase-space volume and the existence of a well-posed inverse flow. In quantum theory, the analogous statement is $U_{-t} = U_t^\dagger$ for unitary U_t.

Reversibility is not merely an aesthetic constraint on computation; it mirrors the fundamental invertibility of closed-system physics.

(a) **Information conservation.** Reversible updates preserve *all* fine-grained information. Logical irreversibility (many-to-one maps) necessarily discards information and, when implemented physically, has thermodynamic cost.

(b) **Microscopic reversibility vs macroscopic irreversibility.** Hamiltonian systems are still reversible (invertible maps); apparent irreversibility arises from coarse-graining, noise, and coupling to environments. Computationally, introducing erasure, measurement, or stochastic resets corresponds to non-invertible maps and semigroup dynamics.

(c) **Design principles.** Algorithms and architectures that factor into reversible cores (bijections/symplectomorphisms/unitaries) with controlled, explicit points of irreversibility (enable tighter accounting of energy and error, and interface cleanly with physics-based models.

Modeling computation by invertible transformations (bijections, symplectomorphisms, unitaries) places the theory squarely inside the same structural class as classical and quantum dynamics, with logical irreversibility reserved for explicit coarse-grainings or measurements.

A.4 Topological Kleene Field Theories and Computational Geometric Bordisms

Miranda, Peralta, and Gonzales [111] have recently introduced *Topological Kleene Field Theory* (TKFT) as a smooth-topological realization of classical computability. In their framework, computation is encoded by a *dynamical bordism*: a manifold with boundary equipped with a suitably well-behaved vector field, together with embeddings of discrete input and output data into distinguished boundary components. The associated reaching map recovers a partial recursive function. In this sense, TKFT should be read as a model of computability in the Kleene–Turing sense, not merely as an analogy with it.

What is especially relevant for the present work is that, in TKFT, the topology of the bordism is not ornamental. Branching and iteration are reflected in the topology of the underlying manifold: pair-of-pants type decompositions model conditional branching, while nontrivial handles encode iterative or looping behavior. TKFT, therefore, offers a precise way to view program structure through smooth topology and dynamical flow.

Our use of bordism is related, but conceptually distinct. In TKFT, the bordism is primarily the ambient geometric stage through which a computation is routed between designated boundary components. In GCT, by contrast, the computational content is intended to be carried more intrinsically by the decorated geometry itself: by the manifold, its local interfaces, its admissible deformations, and the algebraic or gauge-theoretic structures placed upon it. This motivates the following definition.

Definition A.4 (Computational Geometric Bordism)**.** Let $\mathcal{S}$ be a class of computational structures on smooth manifolds (for example, principal G-bundles with connection, decoration functors, or other geometric state data). An n-dimensional GCM is a pair (M, σ), where M is an oriented smooth n-manifold, possibly with boundary, and $\sigma \in \mathcal{S}(M)$.

A *computational geometric bordism* from (M_0, σ_0) to (M_1, σ_1) is a compact oriented smooth $(n+1)$-manifold W with boundary, equipped with a collar-preserving identification

$$\partial W \;\cong\; \overline{M_0} \;\sqcup\; M_1,$$

together with a structure $\Sigma \in \mathcal{S}(W)$ whose restrictions along the collars recover the endpoint data:

$$\Sigma|_{M_0} = \sigma_0, \qquad \Sigma|_{M_1} = \sigma_1.$$

Two such bordisms are regarded as equivalent if they are related by a boundary-fixing diffeomorphism together with an isomorphism in $\mathcal{S}$.

Example A.2 (A Helical Cylinder as Bordism). Let

$$C := S^1 \times I, \qquad I = [0, 1],$$

so that

$$\partial C = S^1 \times \{0\} \;\sqcup\; S^1 \times \{1\}.$$

For any fixed parameter $\lambda \in \mathbb{R}$, the map

$$F_\lambda(\theta, t) = \big(\cos(\theta + \lambda t),\ \sin(\theta + \lambda t),\ t\big), \qquad (\theta, t) \in S^1 \times I,$$

is an embedding into $\mathbb{R}^3$. Its image is a helically twisted cylinder joining two copies of S^1. Thus, the underlying bordism is still the cylinder $S^1 \times I$, but now equipped with an explicitly chiral geometric realization. In the present setting, the point is not that the bordism class changes, but that geometric structure carried by the bordism can itself become computationally meaningful.

TKFT provides a smooth-topological language in which classical computable functions may be organized by dynamical bordisms; GCT seeks a more intrinsic implementation in which computational state is carried directly by decorated geometry. The two frameworks should therefore be viewed as complementary rather than competing. Heuristically, one may prototype a computation in the TKFT setting and then seek a geometric implementation of the same computational content within a GCFT, where the relevant data are borne by the manifold itself rather than by an external symbolic encoding alone.

A.5 Relationship to the Blum-Shub-Smale Model

The Blum–Shub–Smale (BSS) model [112] extends classical computability from $\mathbb{N}$ to $\mathbb{R}$ by postulating a register machine whose instructions are field operations and branches on real values. It yields a rich theory of decision/complexity over $\mathbb{R}$, but with an *algebraic* semantics predicated on unit-cost real arithmetic.

Our framework is inspired but orthogonal. We retain smooth reasoning internally (synthetic/smooth reals) and execute externally over effective number systems (computable reals/algebraics), while moving the *computational substrate* from algebra to geometry: We therefore use BSS only as *context*: it shows that enlarging the base field can change computability classes. In contrast, GCT changes the *operational layer*: computation is controlled by the evolution of geometric state, with gauge invariants as outputs and coarse-graining/renormalization modeled as algebraic effects (cf. section 5.5). This preserves smooth calculus for reasoning, avoids hidden oracles in execution, and matches the holonomy-based semantics used throughout.

Definition A.5 (Geometric E_∞-algebra)**.** Let $\mathcal{C}_\infty$ denote the symmetric monoidal ∞-category of smooth (derived) manifolds and smooth maps, equipped with the Cartesian product $\times$ as its tensor product. A GCM lattice is specified by

$$X \in \mathcal{C}_\infty, \qquad \{\mu_n : X^{\times n} \longrightarrow X\}_{n\geq 0},$$

where the operations μ_n encode parallel composition (disjoint gluing) of n unit helix components. The family (μ_n) is required to satisfy the coherence relations of the E_∞-operad in $\mathcal{C}_\infty$, making $(X, \mu_\bullet)$ a commutative E_∞-algebra object is as follows:

(a) The unit map $\mu_0 : * \to X$ corresponds to the empty (null) computation.

(b) The binary product $\mu_2 : X \times X \to X$ realizes the geometric join of two independent lattice components.

(c) Higher μ_n encodes n-way concurrent composition, commutativity holds up to coherent homotopy, giving the required flexibility for fluidic deformations.

In this language, a classical BSS machine is recovered by restricting to the full sub-∞-category of finite–dimensional real vector spaces with polynomial maps, whose objects form discrete E_∞-algebras (strict commutative monoids). The GCM lattice generalizes this: it is still an E_∞-monoid, but now geometry (i.e., smooth structure, group actions, bordisms) replaces uncountable real registers as the primitive data store.

$$\text{BSS (discrete commutative monoid)} \quad \longrightarrow \quad \text{GL (geometric } E_\infty\text{-algebra)}.$$

See Lurie [113] for the general theory of E_∞-algebras in symmetric monoidal ∞-categories. This synthesis of higher category theory and geometry moves us closer to a physical theory of computation.

A.6 On Causal Fermion Systems

Causal Fermion Systems (CFS), developed by Prof. Felix Finster and collaborators [114], is, from the GCT perspective, the current leading attempt, bar none, to rewrite the foundations of physics in a way that is *geometric-first* and *variational-first*, while treating standard quantum mechanics as a *limiting* (and in a sense, *effective*) description rather than an irreducible axiom. A brief treatment is now possible, as our journey through the computational landscape is complete.

A CFS starts from an operator-theoretic object (a Hilbert space and a measure on a space of operators), and declares that space-time itself is not given a priori: it is reconstructed as the *support* of the measure. Dynamics is not postulated in the usual "Hamiltonian on a background manifold" way, but instead emerges from minimizing a nonlinear action functional (the *causal action*). In the appropriate continuum limit, one recovers familiar structures: Lorentzian geometry, classical gauge fields, and (in further approximations) the quantum-mechanical and QFT toolkits as effective formalisms.

The core data of a causal fermion system are often packaged as a triple

$$(\mathcal{H}, \mathcal{F}, \rho),$$

where:

- $\mathcal{H}$ is a Hilbert space of "physical" spinorial states (think: a space of Dirac-type solutions, but *not* necessarily tied to a pre-existing manifold in the fundamental formulation);
- $\mathcal{F}$ is a set of (typically finite-rank) self-adjoint operators on $\mathcal{H}$, with an imposed signature constraint (this is part of how "spin dimension" enters);

- ρ is a positive measure on $\mathcal{F}$ (the *universal measure*).

The conclusion is then:

$$M := \mathrm{supp}(\rho)$$

is defined to be space-time. One then extracts causal relations, metric information, and spin structure from spectral data of operator products xy for $x, y \in M$. The dynamics are: vary ρ (subject to constraints) to minimize a positive action functional constructed from the eigenvalues of the products xy. The theory is therefore "equations of motion as an optimization principle on measures," not "equations of motion as PDEs on a background manifold."

A Fisherman of the Dirac Sea? In our regime language, CFS is best read as living predominantly in the gauge band with an essential side channel into the fully fluidic band. The most faithful placement is that CFS is a gauge theory of computation: non-commutative dynamics with a considerable quotient by redundancy, such that apparent modular independence is a gauge choice rather than a literal factorization. This is precisely the hallmark of the gauge regime in our classification: joins do not commute in general, but one regains a notion of "independence" after gauge-fixing; the state is stored in gauge-invariant aggregates (holonomies/curvature/spectral invariants) rather than in raw potentials.

CFS is often perceived as intimidating because it is broad, and in addition, Finster is, by training, also a mathematician. The same broadness could be said for our own program, but as a result, the overlap is clean at the conceptual level:

(a) **Spacetime is derived, not assumed.** In GCT, we treat the manifold structure as constructed from computational geometry. CFS likewise treats spacetime as emergent (support of the universal measure) rather than a fixed stage.

(b) **Physics is "action and constraints" at the fundamental layer.** In GCT, we repeatedly re-encounter the principle that what matters are invariants that are stable under the allowed deformations (gluing, transport, holonomy, charge). CFS makes this explicit by encoding dynamics as a variational principle on the measure itself.

(c) **Gauge invariance is not a cosmetic symmetry: it is the computational quotient.** Our gauge regime is precisely the regime where the machine's *actual* state is not the raw configuration, but the equivalence class under gauge transformations. CFS is philosophically the same move, except made at the operator level: local unitary freedom is built in, and only gauge-invariant spectral aggregates are physically meaningful.

(d) **Quantum formalism as a readout/effective theory.** CFS is often read as "de-quantizing" quantum mechanics: the linear Hilbert-space formalism is not the fundamental law, but a limiting description arising from coarse-graining and continuum approximations. This resonates strongly with the GCT posture that many "quantum" features are emergent bookkeeping for geometric microstructures.

CFS Feature	**GCT Interpretation**
Universal measure ρ on operator space	Microstate distribution/ensemble over admissible geometric configurations (a "state-selection" layer)
Spacetime $M = \mathrm{supp}(\rho)$	Working manifold (or effective base) as an emergent support of stable computational configurations
Spectral data of xy defines causal structure	Causality/communication constraints extracted from transport/holonomy invariants
Causal action principle (minimize action)	"Physics as compilation": admissible computations selected by global consistency/extremal principles
Local unitary freedom (gauge)	Gauge regime: independence after gauge-fixing; observables are gauge-invariant aggregates
Continuum limit yields classical gauge fields + gravity	Gauge regime semantics: computation stored in curvature/holonomy classes; classical fields as macroscopic limit of microstructure

We need not adopt the full CFS formalism to benefit from it. What we all can take away is the *structural lesson*:

> If one treats quantum theory as an effective linearization of a deeper geometric object, then the correct primitive is not "wavefunctions on spacetime" but rather a *geometric substrate* whose macroscopic shadows look like Hilbert-space quantum mechanics.

In this sense, CFS is not a competitor but a strong sanity-check: it shows that a long-running research program exists which is already comfortable with (a) emergent spacetime, (b) gauge-as-quotient, and (c) quantum mechanics as a derived language rather than a first principle. All three are compatible with the GCT regime taxonomy, and CFS can be read as a sophisticated inhabitant of the gauge band.

Summary. CFS is best understood here as an operator-level realization of the gauge regime: non-commutative dynamics with ample redundancy, where the physically meaningful information is the gauge-invariant residue. This makes it a natural "bridge theory" between our regime picture and conventional QFT language.

A.7 The Computational Cobordism Hypothesis

The algebraic enrichment required by this framework demands categories that encode both geometric and computational structure. To this end, we introduce the category $\mathcal{ESM}^+$ ("Embedded Smooth Manifolds with Enrichments") as the natural habitat for all GCMs.

Definition A.6 (Embedded Smooth Manifolds with Enrichments)**.** Let **Man** denote the category of smooth manifolds and smooth maps. The *master category* $\mathcal{ESM}^+$ is a symmetric monoidal category whose:

(a) **Objects.** Pairs (M, str) where M is a smooth (possibly stratified) manifold and str is a finite tuple of auxiliary geometric/computational structures over M (e.g. an orientation double cover, specified principal bundles and connections, distinguished vector fields/flows, discrete labels, etc.)

(b) **Morphisms.** $(M, \mathrm{str}) \to (N, \mathrm{str}')$ are smooth maps $f : M \to N$ that preserve the declared structure (pull back covers/bundles, intertwine flows, respect discrete labels.)

(c) **Monoidal product.** Disjoint union:

$$(M, \mathrm{str}) \otimes (N, \mathrm{str}') := (M \sqcup N,\ \mathrm{str} \sqcup \mathrm{str}'), \qquad 1 := (\varnothing, \varnothing).$$

There is a faithful forgetful functor $U : \mathcal{ESM}^+ \to \mathbf{Man}$. We suppress set-theoretic size issues (all categories are essentially small).

Remark A.2. The choice $\otimes = \sqcup$ models parallel composition of independent systems; other monoidal products (e.g. fibered sums under boundary matching) can be introduced when needed, but are not required for the present development.

We separate the geometric substrate from its semantic evaluation target.

Definition A.7 (Differentially Cohesive Topos for GCMs)**.** To reason natively about bundles, connections, curvature, and holonomy while retaining an executable semantics, we separate *geometry* from *execution* and connect them by a left–exact bridge. Let $\mathcal{E}_{\mathrm{GCM}}$ be a (higher) differentially cohesive topos [115] in the sense of cohesive/hypoelliptic geometry (e.g. a smooth ∞–topos of sheaves on Cartesian spaces). It comes equipped with the standard modalities

$$\Pi \dashv \flat \dashv \mathrm{id} \dashv \sharp \qquad \text{and} \qquad \mathrm{dR} : \mathcal{E}_{\mathrm{GCM}} \to \mathcal{E}_{\mathrm{GCM}}$$

so that one may speak internally of principal bundles, connections, differential characters, and holonomy. Decorated groupoids and bordisms land in $\mathcal{E}_{\mathrm{GCM}}$ as objects carrying these structures.

Definition A.8 (Execution Ambient and Extraction)**.** Let $\mathcal{D}_{\text{exec}}$ be a symmetric monoidal category of effective observables and executable transformations, whose numeric objects are effectively representable (e.g. rationals, algebraics, or computable reals). A *left–exact, symmetric monoidal extraction*

$$[\![-]\!] \;:\; \mathcal{E}_{\text{GCM}} \longrightarrow \mathcal{D}_{\text{exec}}$$

preserves finite limits and the declared monoidal structure, and sends gauge–invariant functionals (for example, class functions of holonomy or differential–cohomological observables) to computable readouts.

The overall pipeline is thus:

$$\text{decorated bordisms } \mathbf{Bord}_n^{\mathcal{S}} \xrightarrow{\text{assignment}} \mathcal{E}_{\text{GCM}} \xrightarrow[{[\![-]\!]}]{} \mathcal{D}_{\text{exec}}$$

Remark A.3 (Modalities as semantics)*.* The cohesive modalities organize the readouts:

- Π ("shape") extracts homotopy–level/topological content (e.g. conjugacy classes of holonomy).
- $\flat$ ("flat/discrete") forgets curvature, yielding symbolic data for verification.
- dR produces differential–cohomological observables (differential characters).
- $\sharp$ ("codiscrete") forgets topology when analytic control is required.

Operationally, "path integration is replaced by sheafification": global states are computed as (homotopy) sheafified sections in $\mathcal{E}_{\text{GCM}}$, then evaluated by $[\![-]\!]$.

Remark A.4 (Effects and Handled Evaluation)*.* Coarse-graining, renormalization, metric insertion, and related scale-changing operations live at the level of $\mathcal{D}_{\text{exec}}$ as *handled effects*: explicit transformations of geometric terms together with a stated observational equivalence on boundary readouts. Geometry itself remains smooth and local in $\mathcal{E}_{\text{GCM}}$; execution records only the effective observables extracted from that geometry. This separation keeps the constructive core focused on compositional gauge semantics while ensuring that all externally visible outputs remain computable.

We now pass from the state/transition layer (decorated groupoids) to a control–flow layer where computations are realized as decorated bordisms.

Definition A.9 (Decorated Bordism Category)**.** Fix a dimension $n \geq 0$, a decoration monoid/group $\mathcal{S}$, and a class of $(n{+}1)$–dimensional compact smooth manifolds with boundary and $\mathcal{S}$–decoration data compatible with composition (gluing). The *decorated n–bordism category* Bord_n^{S} has:

(a) **Objects.** Closed (n)–manifolds with chosen $\mathcal{S}$–decorations on their state/transition groupoids (e.g. $d : \Pi_1^*(\,\cdot\,) \to \mathbf{B}S$).

(b) **Morphisms.** Compact n–manifolds W with boundary $\partial W \cong \overline{M_0} \sqcup M_1$ carrying compatible $\mathcal{S}$–decorations, composed by gluing along collars.

(c) **Symmetric monoidal product.** Is disjoint union.

When a geometric substrate $(M, \mathrm{str}) \in \mathcal{ESM}^+$ is fixed, we write $\mathsf{Bord}_n^{S}(M, \mathrm{str})$ for the full subcategory of bordisms that embed in $M \times I$ and whose decorations agree with str on the boundary.

Conjecture A.4 (The Computational Cobordism Hypothesis (CCH))**.** *Let $\mathcal{ESM}^+$ be the ambient category, and let $\mathcal{E}_{\text{GCM}}$ be a topos. There is a classification principle such that symmetric monoidal functors*

$$Z \;:\; \mathsf{Bord}_n^{S}(M, \mathrm{str}) \longrightarrow \mathcal{E}_{\text{GCM}}$$

*(*geometric field theories*) are in natural bijection with choices of* fully dualizable *decorated objects associated to* (M, str) *in a suitable symmetric monoidal* $(\infty, 1)$*–completion of* $\mathcal{ESM}^+$*. Moreover, when $\mathcal{S}$ is trivial, and decorations factor through Π_1, this reduces to the corresponding case of the (once-extended) Cobordism Hypothesis.*

Three–line operational sketch.

(a) **States/transitions.** Encodes local updates by an $\mathcal{S}$–decorated (groupoid or action groupoid) $(\mathcal{G}, d)$ over boundary cuts.

(b) **Control–flow.** Models a computation as a decorated bordism $W \in \mathsf{Bord}_n^S(M, \mathrm{str})$; composition is concatenation of phases.

(c) **Evaluation.** A symmetric monoidal functor Z sends boundary data to objects of $\mathcal{E}_{\mathrm{GCM}}$ and sends W to a morphism in $\mathcal{E}_{\mathrm{GCM}}$, yielding denotational semantics for the computation.

Remark A.5 (Reduction to Classical Cases). If the decoration d factors through Π_1 and $\mathcal{S}$ is discrete/finite, then Z specializes to a finite–group TQFT. If $\mathcal{S}$ carries continuous abelian data (e.g. $S^1 \times C_2$), Z gives an abelian "gauge–like" semantics; non-abelian $\mathcal{S}$ interpolates to gauge–type models.

We now transition from the state/transition layer (decorated groupoids) to a control–flow layer, where computations are realized as decorated bordisms.

Example A.3 (Helical Bordism Computation). Let $M \cong S^1_\theta \times D^2_{(r,\varphi)}$ carry a chirality label $\chi \in C_2$ and an orientation phase $\theta \in S^1$. Set

$$G \;=\; C_2 \times U(1) \times \mathbb{R},$$

acting on core states by

$$(\varepsilon, R_\alpha, t) \cdot (\chi, \theta, \phi, \dot{\phi}) \;=\; (\,\varepsilon\chi,\ \theta + \alpha,\ \Phi_t(\phi, \dot{\phi})\,),$$

where R_α is a rotation of the θ-coordinate and Φ_t is the time-t flow on internal variables.

Use the gauge-invariant monoidal target

$$\mathcal{S} \;:=\; \mathrm{Cl}(U(1)) \times C_2 \;\cong\; U(1) \times C_2, \qquad (z, \eta) \cdot (z', \eta') \;:=\; (zz',\, \eta \oplus \eta'),$$

where $\mathrm{Cl}(U(1))$ denotes conjugacy classes (for $U(1)$ this is $U(1)$ itself). Fix a $U(1)$ connection A representing a chosen 1-cocycle in $H^1(M; U(1))$ and define the (symmetric monoidal) *decoration*

$$d:\ \Pi_1(M) \longrightarrow \mathcal{S}, \qquad d([\gamma]) \;=\; \big(\mathrm{Hol}_\gamma(A),\, \chi|_\gamma\big).$$

Let G act on $\mathcal{S}$ by

$$(\varepsilon, R_\alpha, t) \cdot (z, \eta) \;:=\; (z,\, \varepsilon\eta),$$

so that d is G-equivariant: $d\big((\varepsilon, R_\alpha, t) \cdot [\gamma]\big) = (\varepsilon, R_\alpha, t) \cdot d([\gamma])$. Functoriality holds: $d([\gamma_1] \cdot [\gamma_2]) = d([\gamma_1]) \cdot d([\gamma_2])$, $d([\gamma]^{-1}) = d([\gamma])^{-1}$.

Consider the trivial bordism (cylinder) $W = S^1 \times I \subset M \times I$ with helical embedding

$$(\theta, t) \longmapsto \big(\cos(\theta + \lambda t),\ \sin(\theta + \lambda t),\ t\big), \quad (\theta, t) \in S^1 \times I,$$

whose boundary components are $S^1 \times \{0\}$ and $S^1 \times \{1\}$. Along a loop at fixed radius r, define a local angular rate $\Omega(r, \theta)$ and the one–turn phase gate

$$U(r) \;=\; \exp\Big(i \int_0^{2\pi} \Omega(r, \theta)\, d\theta\Big) \;\in\; U(1).$$

Then the decoration assigns

$$d([\gamma_r]) \;=\; \big(U(r),\, \chi\big).$$

Two loops with the same homotopy class but different radii $r_1 \neq r_2$ may yield $U(r_1) \neq U(r_2)$; hence the computational outcome depends on geometric control (not only on the bare π_1 class), while composition in $\mathcal{S}$ remains abelian and compatible with the groupoid structure and with the bordism functoriality of W.

A *geometric field theory* $Z : \mathsf{Bord}_1^S(M, \mathrm{str}) \to \mathcal{E}_{\mathrm{GCM}}$ sends the boundary circle to an object $Z(S^1) \in \mathcal{E}_{\mathrm{GCM}}$ and the decorated cylinder W to a morphism $Z(W)$ which, in this example, acts by multiplication by $U(r)$ on the appropriate component. When Ω is constant and χ is trivial, d factors through Π_1 and Z reduces to a discrete finite–group semantics.

A.8 Fourier Mode Calculations

Here, we examine kinematic selection effects in the idealized computational flow on the L^2 field of a trivial bundle. Let the cross-section be a disk D^2 in the xy-plane. The velocity field $\vec{v}(r,\theta)$ at each cross-section can be written in polar coordinates as

$$\vec{v}(r,\theta) = v_r(r,\theta)\,\hat{r} + v_\theta(r,\theta)\,\hat{\theta},$$

where v_r and v_θ are the radial and angular components, respectively.

Suppose the flow is smooth and incompressible, and consider the angular dependence of v_r and v_θ. Each component admits a Fourier expansion:

$$v_r(r,\theta) = \sum_{k=-\infty}^{\infty} a_k(r)\,e^{ik\theta}, \qquad v_\theta(r,\theta) = \sum_{k=-\infty}^{\infty} b_k(r)\,e^{ik\theta}.$$

Imposing rotational symmetry of order n means requiring invariance under rotation by $2\pi/n$:

$$v_r(r,\theta + 2\pi/n) = v_r(r,\theta) \qquad \forall\,\theta.$$

This forces all Fourier coefficients $a_k(r)$ and $b_k(r)$ with $k \notin n\mathbb{Z}$ to vanish. Thus, only harmonics with k divisible by n remain:

$$v_r(r,\theta) = \sum_{m=-\infty}^{\infty} a_{nm}(r)\,e^{inm\theta}.$$

In the absence of symmetry ($n = 1$), all $2K + 1$ modes with $|k| \leq K$ are permitted. Under n-fold symmetry, only those with $k = 0, \pm n, \pm 2n, \ldots, \pm Mn$, with $M = \lfloor K/n \rfloor$, survive—yielding $2M + 1$ allowed modes.

We define the *relative expressiveness* as the fraction of permitted angular degrees of freedom:

$$E(n) := \frac{\text{Number of allowed modes for symmetry } n}{\text{Number for } n = 1} = \frac{2M + 1}{2K + 1}.$$

For example, with $n = 4$ and $K = 8$, we have $M = 2$ (i.e., $k = 0, \pm 4, \pm 8$), so $E(4) = 5/17 \approx 0.29$.

As $n \to \infty$, $M \to 0$ and $E(n) \to 0$; i.e., only the constant mode survives. The only flow invariant under arbitrary rotation is uniform, and all angular structure is lost. This directly parallels the earlier result that the only smooth closed 1-manifold with infinite rotational symmetry is the circle.

Thus, increasing symmetry necessarily reduces the space of physically expressible patterns. The tradeoff is exact, quantitative, and determined by the order of symmetry imposed on the system.

Remark A.6. If a field $F(r,\theta)$ on an annulus is invariant under all rotations $\theta \mapsto \theta + \phi$ ($U(1)$ symmetry), then in its Fourier expansion $F(r,\theta) = \sum_{m\in\mathbb{Z}} F_m(r)e^{im\theta}$ one has $F_m \equiv 0$ for $m \neq 0$. Equivalently, only the θ-independent ($m = 0$) mode survives; axisymmetric azimuthal components may remain, but no angular structure ($|m| \geq 1$) does.

A.9 Disclaimers

Regarding Scope and Fundamental Assumptions

This paper does not challenge the empirical validity of quantum field theory, Quantum Chromodynamics, Yang–Mills theory, or the Standard Model. Its purpose is instead to investigate whether some of the geometric and computational structures underlying gauge theory can be reformulated in terms of prequantum geometry. Our work here is best viewed as a candidate semantics for organizing known physical structures, rather than as a competing phenomenological theory.

Glossary and Scope of Terminology

To avoid ambiguity, we distinguish three related but non-identical layers of the present program.

(a) **Geometric Computability Theory (GCT).** The broad research program concerns computation realized through geometric, topological, and gauge-theoretic structures, as well as the logical, complexity-theoretic, and physical consequences of such realizations. In this sense, GCT is the umbrella field of study. It includes, for example, the classification of computational regimes, the role of helicoidal substrates, the EVT hypothesis, and *possible* future connections to number theory/arithmetic geometry (e.g., Berry-Keating), physics, and pedagogy.

(b) **Geometric Field Theory (GFT).** The field-theoretic sector of GCT in which fields are treated primarily as geometric objects: manifolds, bundles, connections, holonomy, curvature, and their associated algebraic or categorical invariants. A GFT need not, by itself, be constructive or computational in the strong sense; rather, it provides the geometric language in which field-theoretic structure is expressed. Topological quantum field theories may be viewed as special cases of GFT in which metric dependence is absent or suppressed.

(c) **Geometric Computational Field Theory (GCFT).** The constructive (total) formalism (5.5) developed within GCT, in which geometric histories are interpreted as composable computational processes. In a GCFT, boundaries carry typed state data, gluing corresponds to evaluation or composition, gauge equivalence is handled at the level of observable output, and physical observables arise as gauge-invariant, coarse-grained results of computation. GCFT is thus the most structured and formal layer of the program: it is to GCT what a functorial field theory is to the broader study of field-theoretic phenomena.

These notions satisfy the schematic inclusion

$$\mathrm{GCFT} \subset \mathrm{GFT} \subset \mathrm{GCT},$$

where GCT denotes the broad foundational program, GFT its geometric field-theoretic sector, and GCFT the constructive compositional formalism proposed in this work.

Author's Note on Naming Conventions

If the twin helix manifold (THM) described here ultimately proves to be of lasting foundational significance, the author would prefer that it not be given an eponymous name. This preference is not intended as a rejection of historical naming conventions. Still, it reflects the view that *geometric* objects of genuinely fundamental scope are often better served by descriptive terminology that emphasizes mathematical content rather than personal attribution. This is in contrast to *theorems*, *conjectures/hypotheses*, and *lemmas*, which I would not object to.

Still, the emphasis of this paper is therefore not on personal legacy, but on the development of Geometric Computability Theory and its possible mathematical, physical, and pedagogical consequences. The work is written from the perspective of the computational sciences, and readers are encouraged to interpret its terminology in that spirit.

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Conflicts of Interest

None.

Data Availability

There is no associated data with this manuscript.

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Likewise, the AI was tasked with designing the experiments above to detect the helices, as this is simply beyond my ability.

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